MGSC 706: Management Research Statistics

Lecture Notes

PhD Core Course in Management Research Statistics

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These notes cover the course lecture material for the PhD core course in management research statistics.

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Contents

MGSC 706 is a core doctoral course in management research statistics. Its scope proceeds from probability to sampling, statistical inference, hypothesis testing, regression, and model building. Although formulas are essential, the course emphasizes business applications and interpretation. Our plan is to demonstrate Python in a cloud environment, while allowing students to use other statistical platforms such as R or SPSS when appropriate.

The pedagogical principle is learning by doing. Manual calculations are useful for understanding definitions, but real research datasets require software. The deeper goal is to learn how a statistical quantity maps to a research question, how to select the correct comparison group, and how to interpret output rather than merely generate it.

Assessment note. There will be four assignments that are worth 10% each, the midterm exam is worth 20%, the final exam is worth 30%, and participation is worth 10%.

1 | Lecture 1: Probability

Sample spaces • Conditional probability • Simpson's paradox • Independence • Covariance • Bayes' rule

1.1 Executive summary

Probability is the language that connects raw data to statistical reasoning, and this lecture builds that language from the ground up. The methodological message is as important as any formula: begin with transparent, model-free summaries of the data; use those summaries to identify patterns and candidate mechanisms; and only then turn to formal inference to judge whether the pattern is stable or merely an artifact of sampling variation. Mathematical machinery is not a substitute for research intuition, and intuition is not a substitute for rigorous analysis — strong empirical work moves back and forth between the two.

Four technical pillars organize the lecture. First, *empirical probability* is estimated by relative frequency in observed data. Second, *conditional probability* restricts attention to a subgroup and therefore changes the denominator used in the calculation. Third, *independence and covariance* describe whether two events co-occur more or less often than their marginal probabilities alone would predict. Fourth, *Bayes' rule* reverses the direction of conditioning, turning a prior belief into a posterior belief once new evidence arrives.

Throughout, a single applied case — graduate admissions by gender and department — carries the running example, while a set of smaller vignettes (a fair die, masks and COVID, online ad clicks, and defective parts from two suppliers) isolate individual ideas. A closing theme, echoed in several places, is that probability is also the mathematical core of modern generative AI: a language model is, at every step, estimating a conditional probability distribution over its next token.

1.1.1 Learning objectives

1. Distinguish a sample space, an event, a subgroup, and a random variable, and recognize different variable types.
2. Compute empirical, joint, marginal, and conditional probabilities from counts, including under counting rules for uniform outcomes.
3. Explain why the conditioning event determines the denominator in a conditional probability.
4. Decompose an aggregate rate into subgroup-specific rates and subgroup weights.

5. Diagnose Simpson's paradox and separate within-group comparisons from compositional effects.
6. State the chain rule, the multiplication criterion for independence, and the covariance of binary indicators.
7. Apply Bayes' rule and the law of total probability to update a prior probability after observing evidence.
8. Separate descriptive association, statistical significance, and causal interpretation.
9. Connect conditional probability, independence, and the chain rule to how generative AI models — especially language models — represent and update uncertainty.

A practical research workflow runs through the lecture's examples: inspect the raw records and understand what each variable represents; compute simple counts, proportions, and subgroup comparisons before fitting any model; form a provisional explanation for the observed pattern; use standard errors, confidence intervals, hypothesis tests, or regression to assess uncertainty and competing explanations; return to the data when formal analysis contradicts the initial intuition, since research is iterative rather than strictly linear; and report contradictory evidence and unsuccessful replications rather than selecting only the result that supports the preferred story.

1.2 Probability Models, Sample Spaces, and Events

1.2.1 Defining a probability

A probability is a normalized number measuring how likely an event is. A probability of zero means impossibility within the model, a probability of one means certainty, and intermediate values quantify uncertainty:

$$0 \leq P(A) \leq 1.$$

1.2.2 Sample spaces, events, and variable types

A probability model has three ingredients: an experiment, a sample space, and probabilities assigned to outcomes. The sample space, written Ω, contains every elementary outcome the model recognizes. An event A is any subset of Ω and may contain one or many elementary outcomes:

$$A \subseteq \Omega, \qquad \sum_{\omega \in \Omega} P(\omega) = 1.$$

This normalization axiom means that increasing the probability of one outcome requires decreasing probability elsewhere — the same constraint enforced by a softmax layer in a neural network, discussed below.

Before assigning probabilities, it helps to classify the variables involved. *Nominal categorical* variables (gender, department, admission decision) carry distinct labels with no natural order. *Ordinal* variables (a Likert rating, class rank) have ordered levels whose distances need not be equal. *Quantitative* variables (GPA, age, sales, waiting time) carry numerical magnitude, so arithmetic on them is meaningful, subject to measurement assumptions. Table 1.1 summarizes the distinction and the summaries each type supports.

Table 1.1: Measurement types and their mathematical implications

Type	**Defining feature**	**Examples**	**Typical summaries**
Nominal categorical	Distinct labels; no natural order	Gender; department; admission decision	Counts, proportions, mode
Ordinal	Ordered levels; distances need not be equal	Likert rating; class rank	Median, quantiles, rank methods
Quantitative	Numerical magnitude; arithmetic is meaningful	GPA, age, sales, waiting time	Mean, variance, regression

1.2.3 Empirical probability as relative frequency

When outcomes are drawn from data rather than assumed, probability is estimated by relative frequency. Let Ω be the set of all units under consideration and let $E \subseteq \Omega$ be an event with $n(E)$ satisfying records:

$$\widehat{P}(E) = \frac{n(E)}{N}. \tag{1.1}$$

An empirical probability lies in $[0, 1]$; multiplying by 100 expresses it as a percentage. In repeated random samples of size m, the expected number of observations satisfying E is approximately $mP(E)$ — for $m = 100$ and $P(E) = 0.459$, the expected count is 45.9, or about 46.

The running case for this lecture is a graduate-admissions dataset of $N = 2{,}844$ applicant records, each carrying an applicant identifier, gender, department, and admission outcome. Of these applicants, 1,306 were admitted, so the overall empirical acceptance probability is

$$\widehat{P}(A) = \frac{1306}{2844} = 0.459 = 45.9\%. \tag{1.2}$$

This number supports three compatible readings: 45.9% of the observed applicants were accepted; a randomly selected record from this dataset has empirical acceptance probability 0.459; and a random sample of 100 comparable applicants would contain about 46 accepted applicants on average, an interpretation that depends on random sampling from the relevant population.

1.2.4 Large language models as a motivating example

Conditional probability estimation is also the mathematical core of a large language model. Treat a token as an elementary outcome and the vocabulary as the sample space for the next-token decision; the model estimates a probability distribution over that vocabulary. A marginal probability such as $P(\text{school})$ ignores context, while a language model instead estimates the conditional probability that the next token is "school" given everything that precedes it. Every additional prompt token changes the conditioning information, and the model recomputes its distribution over the next token accordingly — search autocomplete is a simpler, everyday version of the same idea.

Simple experiments make the same structure visible at smaller scale. One die has six elementary outcomes; two dice have thirty-six ordered outcomes; a quality test may have only success and failure; a digital image has an enormous structured state space determined by pixel positions and color intensities. Generative modeling is difficult largely because real sample spaces are huge and their probabilities are far from uniform.

Normalizing arbitrary model scores into a valid probability distribution is typically done with *softmax*. Given scores $z_1, \dots, z_K$,

$$\mathrm{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}. \tag{1.3}$$

Every output is positive and the outputs sum to one, satisfying the normalization axiom above. Softmax guarantees only the mathematical *form* of a categorical distribution, not that the resulting probabilities are well calibrated or correct; the training data and loss function determine whether they become useful.

1.3 Counting, Combining, and Structuring Outcomes

1.3.1 Counting with and without replacement

Counting rules supply probabilities only once a symmetry assumption is justified. If all elementary outcomes are equally likely,

$$P(A) = \frac{|A|}{|\Omega|} \qquad \text{(under uniformity)}. \tag{1.4}$$

For two sequential draws from three colors *with replacement*, the ordered sample space has $3 \times 3 = 9$ outcomes, since each position independently has three possibilities; if all colors and draws are symmetric, each outcome has probability $1/9$. *Without replacement*, and with one marble of each color available, only $3 \times 2 = 6$ ordered pairs remain. The lesson is that a counting formula should never be applied before the experimental protocol — inventory, replacement, and whether order matters — has been specified; most probability errors begin with an ambiguous sample space rather than difficult arithmetic.

1.3.2 Elementary outcomes versus events

An elementary outcome is indivisible at the model's chosen resolution; an event is any collection of elementary outcomes. The same real-world description can generate different sample spaces depending on the question asked. If the experiment is "draw one marble and record its color," the outcomes are red, green, and yellow. If the question is instead "how many red marbles appear in two draws," the outcomes are 0, 1, and 2, even though the same underlying ordered color pairs are being observed at finer resolution.

1.3.3 Union, intersection, and the complement rule

An event can combine several elementary outcomes. In the two-draws-with-replacement example, the event "same color twice" contains red-red, green-green, and yellow-yellow, so by additivity for mutually exclusive outcomes,

$$P(\text{same color}) = \tfrac{1}{9} + \tfrac{1}{9} + \tfrac{1}{9} = \tfrac{1}{3}.$$

More generally, the union of two events needs a correction whenever they overlap:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B). \tag{1.5}$$

Let A be "at least one red" and B be "two marbles of the same color" in the marble example. Then $P(A) = 5/9$, $P(B) = 3/9$, and $A \cap B$ is red-red with probability $1/9$, so $P(A \cup B) = 7/9$. Simply adding $5/9$ and $3/9$ would double-count red-red. The complement rule follows from the same normalization axiom:

$$P(A^c) = 1 - P(A). \tag{1.6}$$

For disjoint events $A_1, A_2, \ldots$, countable additivity gives

$$P\left(\bigcup_i A_i\right) = \sum_i P(A_i), \qquad \text{when the } A_i \text{ are pairwise disjoint.} \tag{1.7}$$

Splitting $A \cup B$ into three disjoint regions — only A, only B, and the intersection — recovers Equation (1.5); for three or more overlapping events the same idea extends to inclusion–exclusion, which alternates sums and subtractions of intersections.

A *joint-probability table*, with rows for the first draw and columns for the second, is a convenient representation for two categorical variables: each cell is an elementary joint outcome, row and column sums are marginal probabilities, and selected blocks or diagonals represent events. Making overlap visible in a table is a simple, reliable way to avoid double counting.

1.3.4 From uniform assumptions to empirical estimation

Uniform distributions are a useful starting model, but observed systems are rarely uniform: a biased die, unequal marble counts, language frequencies, and image structure all produce unequal probabilities. Section 1.2.3's empirical estimator, $\widehat{P}(A) = n(A)/N$, replaces the uniformity assumption with observed relative frequency, and the estimate can be refined as more data accumulate. A generative model performs a far more structured version of this same task, sharing statistical strength across contexts rather than independently counting every possible sentence or image.

1.4 Conditional Probability

1.4.1 Definition and the changing denominator

Conditional probability restricts attention to cases in which another event is already known to have occurred. The event B becomes the new reference population, so the intersection of A and B must be renormalized by $P(B)$ rather than by the size of the full sample space:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B)}{n(B)}, \qquad P(B) > 0. \tag{1.8}$$

For a fair die, let $A = \{1, 2, 3\}$ (a result less than four) and $B = \{1, 3, 5\}$ (an odd result). The intersection is $\{1, 3\}$, so

$$P(A \mid B) = \frac{2/6}{3/6} = \frac{2}{3}, \qquad P(A^c \mid B) = \frac{1}{3},$$

and the two conditional probabilities correctly sum to one. This illustrates why the denominator is required: it renormalizes probabilities after the sample space has been restricted to B. Conditioning can also be pictured as filtering and rescaling — first discard outcomes outside B, then keep the portion of B that also belongs to A, and finally divide by the total mass of B.

1.4.2 Applying conditioning: acceptance by gender

Returning to the admissions data, the raw acceptance rates by gender are shown in Table 1.2.

Table 1.2: Aggregate acceptance rate by gender

Group	**Applicants** $n(g)$	**Accepted** $n(A \cap g)$	**Acceptance** $P(A \mid g)$
Men	1,963	967	967/1,963 = 49.3%
Women	881	339	339/881 = 38.5%
All applicants	2,844	1,306	1,306/2,844 = 45.9%

The raw difference is $38.5\% - 49.3\% = -10.8$ percentage points for women relative to men. At this stage the gap is purely descriptive: it does not by itself establish discrimination, statistical significance, or causality, since department choice may be associated with both gender and department selectivity — the subject of the next section.

1.4.3 The mask-and-COVID example

A second illustrative case, used repeatedly below, considers 100 people cross-classified by consistent mask use and whether they contracted COVID (Table 1.3).

Table 1.3: Mask use and COVID outcome (percent of full sample)

	COVID	**No COVID**	**Total**
Mask	15	45	60
No mask	15	25	40
Total	30	70	100

The marginal probabilities are $P(\text{COVID}) = 0.30$ and $P(\text{mask}) = 0.60$. Conditioning on mask use,

$$P(\text{COVID} \mid \text{mask}) = \frac{0.15}{0.60} = 0.25, \qquad P(\text{COVID} \mid \text{mask}^c) = \frac{0.15}{0.40} = 0.375. \tag{1.9}$$

Observing mask use moves the estimated COVID probability from a 30% prior (marginal) probability down to a 25% posterior probability — a 12.5-percentage-point gap between the two conditional rates, and a five-percentage-point drop from the unconditional rate. This is an association visible in the table; whether mask use *caused* the reduction depends on study design, measurement, compliance, and possible confounding, and the table itself is a teaching example rather than evidence for a real-world causal estimate.

1.5 Simpson's Paradox and Aggregation

1.5.1 Department-specific results

A finer, post-hoc analysis conditions jointly on gender and department, examining $P(A \mid g, D = d)$ instead of $P(A \mid g)$ alone. Table 1.4 reports acceptance rates within each department.

Table 1.4: Acceptance rates within department

Department	**Men acc. / n**	**Men rate**	**Women acc. / n**	**Women rate**	**W−M**
Management	496 / 616	80.5%	155 / 175	88.6%	+8.1 pp
Law	53 / 272	19.5%	94 / 481	19.5%	+0.1 pp
Engineering	418 / 1,075	38.9%	90 / 225	40.0%	+1.1 pp
Aggregate	967 / 1,963	49.3%	339 / 881	38.5%	−10.8 pp

Within every single department, women's acceptance rate is equal to or higher than men's; yet the aggregate comparison reverses to favor men by 10.8 percentage points. This reversal is a textbook instance of **Simpson's paradox**: an association that holds in every subgroup disappears, or even flips, once the subgroups are pooled. Figure 1.1 illustrates the reversal.

1.5.2 The mechanism: different weights, not different truths

An aggregate conditional rate is a weighted average of department-specific rates, with weights given by how each gender's applications are distributed across departments:

$$P(A \mid g) = \sum_{d \in \mathcal{D}} P(A \mid g, D = d)\, P(D = d \mid g). \tag{1.10}$$

Table 1.5 shows the mix driving the reversal: men are far more concentrated in Management, the *least* selective department, while women are far more concentrated in Law, the *most* selective department.

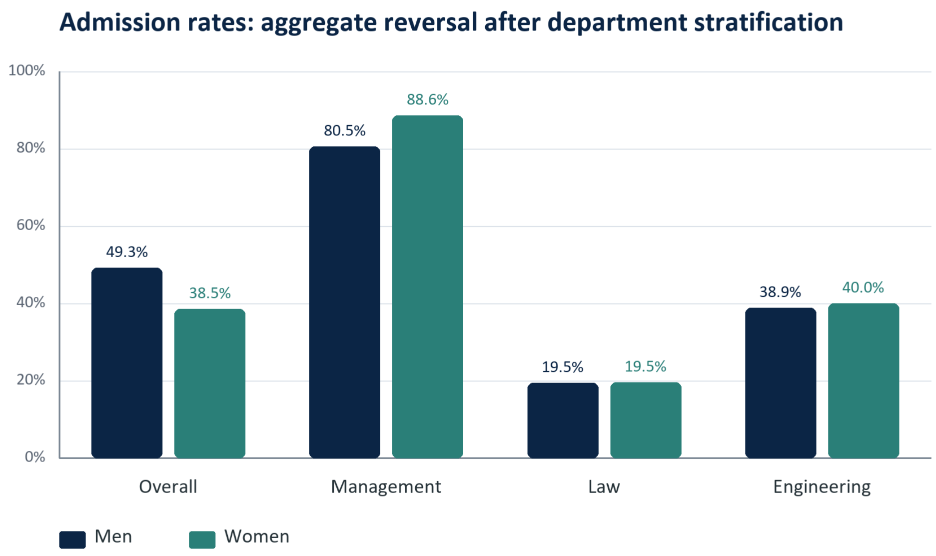

Figure 1.1: The aggregate gender comparison reverses after conditioning on department.

Table 1.5: Application mix by gender and department selectivity

Department	$P(D = d \mid \textbf{Men})$	$P(D = d \mid \textbf{Women})$	**Selectivity** $P(A \mid D = d)$
Management	616/1,963 = 31.4%	175/881 = 19.9%	651/791 = 82.3%
Law	272/1,963 = 13.9%	481/881 = 54.6%	147/753 = 19.5%
Engineering	1,075/1,963 = 54.8%	225/881 = 25.5%	508/1,300 = 39.1%

Applying Equation (1.10) explicitly,

$$P(A \mid M) = 0.805(0.314) + 0.195(0.139) + 0.389(0.548) = 0.493, \tag{1.11}$$
$$P(A \mid F) = 0.886(0.199) + 0.195(0.546) + 0.400(0.255) = 0.385. \tag{1.12}$$

Nearly equal within-department rates, combined with very different weights, reproduce the aggregate rates from Table 1.2. Simpson's paradox is not a mathematical contradiction: the aggregate and stratified quantities condition on different information and apply different weights, so a responsible analysis states exactly which estimand is being interpreted and why it matches the research question.

One descriptive way to remove the compositional difference is *standardization*: apply a common department mix — here the overall proportions of 27.8% Management, 26.5% Law, and 45.7% Engineering — to both genders. This produces standardized rates of 45.3% for men and 48.1% for women, a swing to +2.8 percentage points in favor of women. Standardization is an adjusted descriptive comparison, not by itself a causal estimate.

1.5.3 From descriptive pattern to formal inference

A visible difference in a finite dataset is evidence of a descriptive pattern, not yet proof of a population-level difference. Formal inference asks whether the observed gap could plausibly arise from sampling variation, using machinery covered later in the course: standard errors, confidence intervals, hypothesis tests, and regression. The large-sample confidence interval for a single proportion is

$$\widehat{p} \pm z_{1-\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}, \tag{1.13}$$

and for two independent proportions the standard error of the difference depends on both group sizes and both estimated rates. A hypothesis test would formalize a null statement such as $H_0 : p_F = p_M$, while a stratified test or regression would account for department directly. Model-free analysis generates the question; formal inference quantifies the uncertainty around the answer.

1.6 Joint Probability, the Chain Rule, and Independence

1.6.1 Joint probability and the chain rule

The joint event $A \cap B$ occurs when both A and B occur; marginal probabilities refer to each event separately. Rearranging the conditional-probability definition (Equation (1.8)) gives the **chain rule** (equivalently, the multiplication rule), valid whenever the conditioning probability is positive:

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A). \tag{1.14}$$

1.6.2 Independence

Events A and B are **independent** when learning that one occurred does not change the probability of the other. The following statements are equivalent whenever the relevant probabilities are nonzero:

$$A \perp\!\!\!\perp B \iff P(A \mid B) = P(A) \iff P(A \cap B) = P(A)P(B). \tag{1.15}$$

A related diagnostic in binary settings compares $P(A \mid B)$ with $P(A \mid B^c)$: if the two rates differ, B carries information about A. Independence is symmetric — if A is independent of B, then B is independent of A, A is independent of B^c, and A^c is independent of B — and a joint table for independent events factors into the product of its row and column marginals. Independence should not be confused with mutual exclusivity: two mutually exclusive events with positive probability cannot be independent, since the occurrence of one makes the other impossible. Independence is also a property of the joint distribution, not a claim that two events are causally unrelated in every scientific sense.

1.6.3 Covariance of binary event indicators

Let I_A equal 1 when event A occurs and 0 otherwise. For two event indicators, covariance has a direct probability interpretation:

$$\mathrm{Cov}(I_A, I_B) = P(A \cap B) - P(A)P(B). \tag{1.16}$$

Applying this to the mask-and-COVID table (Section 1.4.3),

$$\mathrm{Cov}(I_C, I_M) = 0.15 - (0.30)(0.60) = -0.03.$$

Under independence, the expected joint proportion would be $0.30 \times 0.60 = 0.18$; the observed joint proportion of 0.15 is three percentage points lower, so mask use and COVID occurrence co-occur *less* often than independence would predict in this dataset. A technical caveat is worth stating precisely: zero covariance means *uncorrelated*, not generally *independent*. For two Bernoulli indicators specifically, $\mathrm{Cov}(I_A, I_B) = 0$ is equivalent to $P(A \cap B) = P(A)P(B)$ and so does imply independence of those two binary events; for general random variables, however, nonlinear dependence can persist even when covariance is zero. Refer to the counter example provided in §2.7.5.

1.6.4 Independence of a collection of events

The definition of independence extends naturally from two events to a whole collection. For three events A_1, A_2, and A_3, independence means satisfying all four of the conditions

$$P(A_1 \cap A_2) = P(A_1)P(A_2), \tag{1.17}$$
$$P(A_1 \cap A_3) = P(A_1)P(A_3), \tag{1.18}$$
$$P(A_2 \cap A_3) = P(A_2)P(A_3), \tag{1.19}$$
$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3). \tag{1.20}$$

The first three conditions simply assert that every pair of events is independent, a property called **pairwise independence**. The fourth condition is a separate requirement, and — perhaps surprisingly — it does not follow from the first three: a collection of events can be pairwise independent while still failing the joint condition in Equation (1.20). The converse also fails: satisfying Equation (1.20) alone does not guarantee that the three pairwise conditions in Equations (1.17)–(1.19) hold. The two examples that follow illustrate each direction of this gap, which is why a model that assumes several events are **mutually independent** must state and check all four conditions — pairwise independence checked one pair at a time is not enough.

1.6.5 Pairwise independence does not imply independence

Independence is a genuinely joint property of a collection of events, not something that can be checked one pair at a time. The following example shows three events that are pairwise independent — every pair satisfies the multiplication rule — while the full collection is not independent.

Consider two independent fair coin tosses, and the following events:

$$H_1 = \{\text{1st toss is a head}\}, \qquad H_2 = \{\text{2nd toss is a head}\}$$

and

$$D = \{\text{the two tosses have different results}\}.$$

The events H_1 and H_2 are independent by definition, since the two tosses are independent. To see that H_1 and D are also independent, condition on H_1:

$$P(D \mid H_1) = \frac{P(H_1 \cap D)}{P(H_1)} = \frac{1/4}{1/2} = \frac{1}{2} = P(D).$$

By the same argument, H_2 and D are independent as well. Every pair among H_1, H_2, D therefore satisfies the independence criterion from Equation (1.15).

Yet the three events together are not independent. If H_1 and H_2 both occur, the two tosses agree, so D cannot occur:

$$P(H_1 \cap H_2 \cap D) = 0 \qquad \text{while} \qquad P(H_1)P(H_2)P(D) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

Since $0 \neq 1/8$, the three-way multiplication rule fails, even though it holds for every pair separately. **Pairwise independence, checked two events at a time, is strictly weaker than mutual independence of the whole collection** — a distinction that matters whenever a model assumes several events or random variables are jointly independent rather than merely uncorrelated in pairs.

1.6.6 The triple-intersection condition alone is not enough

The above example showed that pairwise independence does not imply the joint condition $P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3)$. The converse gap is just as real: satisfying that single joint condition does not imply any of the three pairwise conditions.

Consider two independent rolls of a fair die, and the following events:

$$A = \{\text{1st roll is } 1, 2, \text{ or } 3\}, \qquad B = \{\text{1st roll is } 3, 4, \text{ or } 5\}, \qquad C = \{\text{the sum of the two rolls is } 9\}.$$

We have

$$\begin{aligned}
P(A \cap B) &= \frac{1}{6} \neq \frac{1}{2} \cdot \frac{1}{2} = P(A)P(B), \\
P(A \cap C) &= \frac{1}{36} \neq \frac{1}{2} \cdot \frac{4}{36} = P(A)P(C), \\
P(B \cap C) &= \frac{1}{12} \neq \frac{1}{2} \cdot \frac{4}{36} = P(B)P(C).
\end{aligned}$$

Thus the three events A, B, and C are not independent, and indeed *no two* of these events are independent. On the other hand,

$$P(A \cap B \cap C) = \frac{1}{36} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{4}{36} = P(A)P(B)P(C).$$

The triple-intersection condition holds exactly, even though every pairwise condition fails.

The intuition behind the independence of a collection of events is analogous to the two-event case: independence means that the occurrence or non-occurrence of any number of the events in the collection carries no information about the remaining events or their complements. For example, if the events A_1, A_2, A_3, A_4 are independent, one obtains relations such as

$$P(A_1 \cup A_2 \mid A_3 \cap A_4) = P(A_1 \cup A_2)$$

or

$$P(A_1 \cup A_2^c \mid A_3^c \cap A_4) = P(A_1 \cup A_2^c).$$

1.6.7 Design matters: the advertisement-and-purchase example

In a stylized group of 100 online consumers, 30 click a displayed advertisement and 12 of those clickers go on to purchase. The critical missing quantity is the number x of purchases among the 70 non-clickers — without that comparison group, no effect or association involving ad engagement can be assessed. The clicker purchase rate is $12/30 = 0.40$; independence requires the same rate among non-clickers, so

$$\frac{x}{70} = \frac{12}{30} = 0.40 \implies x = 28, \tag{1.21}$$

giving 40 total purchasers under the independent benchmark, with 12 among clickers and 28 among non-clickers; equivalently, $P(\text{Purchase} \cap \text{Ad}) = 0.12 = (0.40)(0.30) = P(\text{Purchase})P(\text{Ad})$. The broader design principle: a rate observed only among exposed or engaged users is never enough to establish an effect. A valid comparison needs an appropriate unexposed group and, for a causal claim, a design that addresses selection into exposure — people who click ads may differ systematically from those who do not.

1.6.8 Conditional independence

Two events that are marginally dependent may become independent once a third event is conditioned on:

$$A \perp B \mid C \quad \iff \quad P(A \mid B, C) = P(A \mid C). \tag{1.22}$$

Conditional independence is central to larger probabilistic models: graphical models, naive Bayes, hidden Markov models, and many sequence models all exploit conditional-independence structure to avoid estimating an impossibly large, unrestricted joint distribution over all variables at once.

1.7 Bayes' Rule and Probabilistic Updating

1.7.1 From the chain rule to Bayes' rule

Equating the two right-hand sides of the chain rule (Equation (1.14)) and solving for $P(A \mid B)$ produces **Bayes' rule**:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}. \tag{1.23}$$

Bayes' rule carries two readings at once: it reverses the direction of conditioning, deriving $P(A \mid B)$ from $P(B \mid A)$; and it updates a *prior* belief $P(A)$ into a *posterior* belief $P(A \mid B)$ using the *likelihood* $P(B \mid A)$ and the total *evidence* $P(B)$. The denominator ensures posterior probabilities across competing hypotheses still sum to one. When A is a hypothesis and its complement A^c is the only alternative, the denominator expands by the **law of total probability**:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B \mid A)P(A) + P(B \mid A^c)P(A^c)}. \tag{1.24}$$

More generally, if $B_1, \ldots, B_k$ form a mutually exclusive and exhaustive partition of Ω, the law of total probability reconstructs a marginal probability from conditional pieces,

$$P(A) = \sum_{i=1}^{k} P(A \mid B_i)P(B_i), \tag{1.25}$$

and the same sum, computed over all hypotheses, replaces the two-term denominator of Equation (1.24) for a partition with more than two pieces. In a diagnostic setting A might be a positive test and the B_i disease states; in a forecasting setting A might be a positive signal and the partition good versus bad economic conditions.

Returning to the mask example, $P(\text{mask} \mid \text{COVID}) = 0.5$, since half of the COVID group wears masks. Bayes' rule recovers the same posterior found directly from the table in Equation (1.9):

$$P(\text{COVID} \mid \text{mask}) = \frac{0.50 \times 0.30}{0.60} = 0.25.$$

1.7.2 Base rates and posterior odds

Bayes' rule is most informative — and most often misapplied — when base rates matter. A high $P(B \mid A)$ does not automatically imply a high $P(A \mid B)$; the posterior combines the likelihood with the prior prevalence and divides by the overall evidence, and ignoring $P(A)$ is the classic *base-rate fallacy*. An equivalent technical form separates prior belief from evidential strength using odds:

$$\text{Posterior odds of } A \text{ vs. } A^c = \text{prior odds} \times \text{likelihood ratio}, \qquad \text{likelihood ratio} = \frac{P(B \mid A)}{P(B \mid A^c)}. \tag{1.26}$$

A likelihood ratio above one favors A; below one favors A^c; exactly one carries no information about A at all. In the mask table, this comparison is purely descriptive: conditioning identifies an association, not a cause, and a causal conclusion would require stronger design assumptions about confounders, selection, and measurement.

1.7.3 Worked example: the supplier-quality problem

A company receives 60% of its parts from Montreal (M) and 40% from Toronto (T); the good-part rates are 96% for Montreal and 98% for Toronto. A part is observed to be good (G), but its supplier label is missing, and the question is $P(T \mid G)$.

Table 1.6: Prior supplier shares and quality likelihoods

Supplier	**Prior** $P(\textbf{Supplier})$	**Good rate** $P(G \mid \textbf{Supplier})$	**Joint** $P(G \cap \textbf{Supplier})$
Montreal (M)	0.60	0.96	$0.60 \times 0.96 = 0.576$
Toronto (T)	0.40	0.98	$0.40 \times 0.98 = 0.392$
Total good	1.00	—	$0.576 + 0.392 = 0.968$

$$P(T \mid G) = \frac{(0.40)(0.98)}{(0.40)(0.98) + (0.60)(0.96)} = \frac{0.392}{0.968} = 0.405. \tag{1.27}$$

The prior probability of Toronto was 0.400; observing a good part raises it only slightly, to 0.405, because Toronto's good-part rate is just barely higher than Montreal's. Evidence changes a prior substantially only when it discriminates strongly among the competing hypotheses — here the likelihoods 0.98 and 0.96 are close, so the update is small. As an instructive contrast, observing a *defective* part moves the probability the other way:

$P(T \mid \text{defective}) = (0.40 \times 0.02)/[(0.40 \times 0.02) + (0.60 \times 0.04)] = 0.25$, since a defect is relatively more compatible with Montreal, whose defect rate is twice Toronto's.

Table 1.7 organizes the four moving pieces of any Bayesian update using this example.

Table 1.7: Components of a Bayesian update

Component	**Notation**	**Meaning in the supplier example**
Prior	$P(T)$	Supplier probability before seeing the part: 0.40
Likelihood	$P(G \mid T)$	Probability of a good part if from Toronto: 0.98
Evidence	$P(G)$	Overall probability of a good part: 0.968
Posterior	$P(T \mid G)$	Updated supplier probability after seeing a good part: 0.405

The same logic underlies adaptive prediction systems: as a user supplies new information, a model updates its probabilities over possible next actions or outcomes. Sequential learning from evidence is the general principle, though real production systems may combine Bayesian updating with other, non-Bayesian algorithms.

1.8 Association, Significance, and Causality

Every example in this lecture — the admissions gap, the mask-and-COVID table, the ad-and-purchase design, the supplier data — raises three distinct questions that require different kinds of evidence to answer, summarized in Table 1.8.

Table 1.8: Three levels of empirical interpretation

Question	**Typical quantity**	**What it establishes**	**What it does not establish**
Descriptive association	Rate difference, covariance, correlation	Variables move together in the observed data	Population stability or causal direction
Statistical significance	Standard error, confidence interval, p-value	Observed association is difficult to explain by a specified sampling model under H_0	Practical importance or causality
Causal effect	Randomized contrast or identified causal estimand	Outcome change attributable to an intervention under assumptions	Universal validity beyond the design and target population

A nonzero correlation or covariance can arise because A affects B, B affects A, a third variable affects both, the sample is selected, variables are measured imperfectly, or the association is simply due to chance. Causal claims always require research design and assumptions beyond a nonzero covariance. This caution extends to research practice itself: when an observational dataset suggests an interesting pattern but a costly follow-up experiment fails to reproduce it, the second result must not be discarded merely because it is inconvenient. Observational evidence may be confounded, and randomized evidence may reveal that an initial relationship was never causal; a credible research report documents the full sequence of evidence, its limitations, and the differences in design.

1.9 Sequential Probability and Generative AI

The tools of this lecture — conditioning, the chain rule, and conditional independence — are exactly the machinery behind sequence generation in modern AI. For language, the chain rule decomposes a sentence's probability into a product of next-token conditionals:

$$P(w_1, \ldots, w_T) = \prod_{t=1}^{T} P(w_t \mid w_1, \ldots, w_{t-1}). \tag{1.28}$$

A language model estimates each of these factors, normalized into a valid distribution by softmax (Equation (1.3)); generating text then means sampling or selecting from each next-token distribution and appending the chosen token to the conditioning context for the next step. Just as conditional independence lets graphical models and hidden Markov models avoid an intractable full joint distribution (Section 1.6), a language model's contextual conditioning lets it avoid separately estimating the probability of every possible sentence. The probability module of this course therefore connects directly to the computational behavior of the generative systems built on top of it.

1.10 Compact Formula Reference

Table 1.9: Core probability formulas

Concept	Formula	Interpretation
Empirical probability	$\widehat{P}(E) = n(E)/N$	Fraction of observed units satisfying E
Complement	$P(A^c) = 1 - P(A)$	Probability that A does not occur
Union	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Correct for double-counted overlap
Conditional probability	$P(A \mid B) = P(A \cap B)/P(B)$	Probability of A within subgroup B
Chain rule	$P(A \cap B) = P(A \mid B)P(B)$	Build a joint probability in stages
Total probability	$P(B) = \sum_i P(B \mid H_i)P(H_i)$	Add mutually exclusive routes to B
Independence	$P(A \cap B) = P(A)P(B)$	The joint equals the product of marginals
Binary covariance	$\mathrm{Cov}(I_A, I_B) = P(A \cap B) - P(A)P(B)$	Observed joint minus independent benchmark
Bayes' rule	$P(A \mid B) = P(B \mid A)P(A)/P(B)$	Reverse conditioning and update a prior

Table 1.9: Core probability formulas

Concept	Formula	Interpretation
Aggregation	$P(A \mid g) = \sum_d P(A \mid g, d)P(d \mid g)$	Overall rate is a weighted subgroup average
Softmax	$\text{softmax}(z_i) = e^{z_i} / \sum_j e^{z_j}$	Normalize scores into a categorical distribution

1.11 Practice Problems with Concise Solutions

1.11.1 Random-sample interpretation

If 500 comparable applicants are sampled randomly and the empirical acceptance probability is 0.459, what is the expected number accepted?

Solution. $E[X] = np = 500(0.459) = 229.5$, so approximately 230 accepted applicants.

1.11.2 Conditional denominator

Why is 339/2,844 *not* the probability of acceptance given woman?

Solution. The condition "woman" restricts the sample space to 881 women, so $P(A \mid F) = 339/881 = 0.385$. The quantity 339/2,844 is instead the joint probability $P(A \cap F)$.

1.11.3 Aggregate reversal

Women have an equal or higher acceptance rate in every department but a lower overall rate. Identify the mathematical mechanism.

Solution. The aggregate rates use different weights $P(D = d \mid \text{gender})$. Women are concentrated in Law, whose overall acceptance rate is 19.5%, while men are more concentrated in Management, whose rate is 82.3%. Different weighted averages can reverse the within-stratum comparison — Simpson's paradox.

1.11.4 Independence threshold

Thirty of 100 users click an ad, and 12 clickers purchase. How many of the 70 non-clickers must purchase for click status and purchase to be independent?

Solution. Set $x/70 = 12/30 = 0.40$, giving $x = 28$. Then both conditional purchase rates equal 0.40, and the joint probability 0.12 equals 0.30×0.40.

1.11.5 Bayes update

Using the supplier data, compute the probability that a defective part came from Toronto.

Solution. $P(T \mid \text{defective}) = [0.40(0.02)]/[0.40(0.02) + 0.60(0.04)] = 0.008/0.032 = 0.25$.

1.11.6 Covariance interpretation

In the mask example, why is the covariance -0.03?

Solution. The observed joint probability $P(C \cap M)$ is 0.15. Independence predicts $P(C)P(M) = 0.30(0.60) = 0.18$. Their difference, $0.15 - 0.18 = -0.03$, shows less co-occurrence than the independent benchmark.

Appendix: Reconstructed Admissions Counts

Table 1.10: Complete count reconciliation

Department	Men total	Men accepted	Women total	Women accepted	Total applicant
Management	616	496	175	155	791
Law	272	53	481	94	753
Engineering	1,075	418	225	90	1,300
Total	1,963	967	881	339	2,844

Every row and column reconciles to the totals used throughout the lecture.

2 | Lecture 2: Random Variables

Distributions • Expectation • Variance • Standard deviation • The normal distribution

2.1 Executive summary

This lecture moves from probability over events to probability over numerical quantities. A random variable assigns a number to every outcome in a sample space; once that mapping is fixed, a probability distribution describes how mass or density is allocated across its possible values. Expected value locates the distribution's center; variance and standard deviation quantify its spread; and, when a normal model is appropriate, standardization converts any such distribution onto a single universal reference scale.

A handful of running examples carry the ideas. MBA salaries and automobile fuel efficiency motivate empirical distributions, binning, and histograms. A ranking problem shows how the chain rule generates a discrete probability mass function from first principles. A three-dice casino game demonstrates that a seemingly balanced payoff can still carry a negative expected return, because a small class of outcomes — triples — quietly transfers probability to the house. Two idealized lotteries with equal means but very different spreads motivate why variance and covariance matter, and simple uniform and triangular densities make the mechanics of continuous probability concrete before the normal distribution is introduced. Throughout, generative AI supplies a running motivation: a language model's next token is a random variable, softmax is its probability mass function, and decoding is sampling.

Core principle. A mean alone is not a complete description. Two populations or decisions can have the same expected value but very different variability. Statistical comparisons therefore require both a measure of center and a measure of uncertainty or spread.

2.1.1 Learning objectives

1. Define a random variable as a function from outcomes to real numbers, and distinguish discrete from continuous random variables.
2. Construct probability mass functions, probability density functions, and cumulative distribution functions, and distinguish density from probability.
3. Build empirical probability masses from counts and explain how bin choices affect a histogram.
4. Compute expected values as probability-weighted averages, including for transformations of a random variable.

5. Use the probability chain rule to derive a discrete distribution from first principles.
6. Calculate the expected return and house edge of a finite game, and the expected value and variance of simple lotteries.
7. Compute variance and standard deviation, interpret their units, and apply the transformation rules for affine functions of a random variable.
8. Distinguish independence from zero covariance, and compute the variance of a sum of dependent random variables.
9. Standardize a normal random variable and compute probabilities using the standard normal table.
10. Distinguish population parameters from the sample estimators used in later statistical inference.
11. Explain the empirical-distribution (plug-in) principle behind non-parametric estimation, and why it requires a genuinely random sample.

2.2 Random Variables as Numerical Mappings

2.2.1 Operational and formal definitions

An event is a subset of a sample space; a random variable goes one step further and assigns a numerical value to every elementary outcome. Formally, if Ω is the sample space, a random variable X is a function

$$X : \Omega \to \mathbb{R}, \qquad \omega \mapsto X(\omega). \tag{2.1}$$

The word *random* describes the outcome ω selected from Ω, not the function itself: once ω is known, $X(\omega)$ is determined. For a coin toss, one convenient coding is $X(H) = 1$ and $X(T) = 0$; for height, an outcome is a person and $X(\omega)$ is that person's height in centimeters; for gender or another category, numeric codes are simply labels, and assigning 0 and 1 does not by itself create an ordering, a distance, or a causal interpretation. A chosen encoding must be applied consistently once selected, even though another valid encoding of the same experiment could exist.

Technical extension. In a fully formal probability model $(\Omega, \mathcal{F}, P)$, X must be *measurable*: for every Borel set B on the real line, the preimage $\{\omega : X(\omega) \in B\}$ belongs to $\mathcal{F}$. This condition is what makes statements such as $P(X \leq x)$ well-defined probabilities.

2.2.2 Discrete and continuous variables

Table 2.1 summarizes the two principal classes of random variable.

Table 2.1: Two principal classes of random variables

Class	**Possible values**	**Examples**	**Probability representation**
Discrete	Finite or countably infinite	Coin indicator, coded category, rank, casino payoff, number of children, a language token's integer identifier	Probability mass function $p_X(x)$
Continuous	An interval or other uncountable set	Height, salary as measured, fuel efficiency, time, temperature, a financial return, a normalized pixel intensity	Density $f_X(x)$ and distribution function $F_X(x)$

Measured continuous variables are often stored with finite precision, but the conceptual model may still be continuous; conversely, counts such as the number of purchases are discrete even when they take many values. The mathematical representation should follow the data-generating quantity, not merely the spreadsheet format.

2.2.3 A generative-AI motivating example

Text-generating models convert text into tokens, associate each token with a discrete numerical identifier, and generate a conditional probability distribution over the next token — a genuinely discrete random variable, however large its support. Image, audio, and video systems, by contrast, often represent data as high-dimensional numerical arrays; although stored pixels are quantized on a computer, mathematical models frequently treat latent variables or normalized intensities as continuous. For an RGB image with height H and width W, a simple representation contains three values per pixel, so

$$\begin{aligned} \text{dimension of an } H \times W \text{ RGB image} &= 3HW; \\ \text{for } H = W = 256,\ \text{dimension} &= 196{,}608. \end{aligned} \tag{2.2}$$

The resulting joint random vector is enormously higher-dimensional than a single coin toss, which helps explain why training and inference on such models are computationally expensive.

2.3 Probability Mass, Density, and Cumulative Probability

2.3.1 Definitions

Every probability model must be nonnegative and normalized. For a discrete random variable, the *probability mass function* (PMF) assigns a probability to each attainable value, and the masses must sum to one:

$$p_X(x) = P(X = x), \qquad p_X(x) \geq 0, \qquad \sum_{x \in S_X} p_X(x) = 1. \tag{2.3}$$

For a continuous random variable, the *probability density function* (PDF) f_X is also nonnegative, but normalization is expressed as an integral rather than a sum, and probability is obtained by integrating the density over an interval:

$$\begin{gathered} f_X(x) \geq 0, \qquad \int_{-\infty}^{\infty} f_X(x)\,dx = 1, \\ P(a \leq X \leq b) = \int_a^b f_X(x)\,dx. \end{gathered} \tag{2.4}$$

A central technical distinction is that *density is not probability*: a density may exceed one when its support is narrow, whereas probabilities are areas under the density curve and must lie between zero and one. For a genuinely continuous variable, the probability of any exact point is zero, even when the density at that point is positive: $P(X = x_0) = 0$.

The *cumulative distribution function* (CDF) works for either discrete or continuous variables and answers a directly interpretable question — what is the probability that X does not exceed x:

$$F_X(x) = P(X \leq x). \tag{2.5}$$

A related terminological distinction is worth making precise. The word *likelihood* is sometimes used loosely for the height of a density, but density and likelihood answer different questions. A density $f(x \mid \theta)$ is a function of possible data x when the parameter θ is held fixed; a likelihood $L(\theta \mid x)$ treats the same mathematical expression as a function of the unknown parameter θ, with the data held fixed. Density describes curve height as data vary; likelihood is reserved for parameter inference.

2.3.2 The uniform distribution

The simplest continuous example is the uniform distribution. If X is uniform on the interval from a to b, the density is constant inside the interval and zero outside it; because the enclosed rectangle must have area one, its height is the reciprocal of the interval length:

$$\begin{gathered} X \sim \text{Uniform}(a, b) \implies f_X(x) = \frac{1}{b-a} \text{ for } a \leq x \leq b, \\ \text{and } f_X(x) = 0 \text{ otherwise.} \end{gathered} \tag{2.6}$$

For $\text{Uniform}(1, 10)$, the support has length nine, so the density is $1/9$; a proposed constant height of five would be positive but invalid, since its area would be $5 \times 9 = 45 \neq 1$. The probability of landing in the narrow interval from 7 to 7.1 is the area of a thin rectangle:

$$P(7 \leq X \leq 7.1) = \frac{7.1 - 7}{10 - 1} = \frac{0.1}{9} = \frac{1}{90} \approx 0.0111.$$

A narrower uniform distribution can have density greater than one without any contradiction. $\text{Uniform}(1, 1.1)$ has width 0.1 and density 10; this exceeds one but is still valid because $10 \times 0.1 = 1$, and, for instance, $P(1.05 \leq X \leq 1.06) = 10(0.01) = 0.10$.

2.3.3 Empirical distributions and histograms

With observed values $x_1, \ldots, x_N$, divide the measurement axis into bins $B_j = [b_j, b_{j+1})$. The count n_j records how many observations fall in bin j, and dividing by N converts counts into an empirical probability mass:

$$n_j = \sum_{i=1}^{N} \mathbf{1}\{x_i \in B_j\}, \qquad \widehat{p}_j = \frac{n_j}{N}. \tag{2.7}$$

For equal-width bins, bar heights may display counts or proportions directly. For unequal widths, a proper density histogram should use height $n_j/(Nw_j)$, where w_j is the bin width, so that bar *area* — not height alone — equals probability. A histogram is therefore an estimate or grouped representation of a distribution, not literally the underlying probability density function.

Two applied datasets illustrate the construction: an MBA salary dataset of 2,597 anonymized, inflation-adjusted records, and an automobile dataset of 392 vehicle models produced from roughly 1970 to 1982. Table 2.2 contrasts the distinct questions that different summaries of such a distribution answer, and Table 2.3 previews what a standard descriptive-statistics call reports before the two datasets are worked through in code below.

Table 2.2: Distinct questions answered by common summaries

Summary	**Question answered**	**Sensitivity**
Exact mode	Which recorded value occurs most often?	Sensitive to rounding and repeated values
Modal bin	Which interval contains the most observations?	Sensitive to bin origin and width
Median	Where is the 50th percentile?	Robust to extreme values
Mean	Where is the probability-weighted balance point?	Pulled toward extreme (e.g. very high salary) values

Table 2.3: Typical descriptive-statistics output

Output	**Mathematical meaning**	**Interpretive use**
count	Number of nonmissing observations	Data completeness
mean	Arithmetic sample average	Estimated center
std	Sample standard deviation	Estimated spread
min / max	Observed extremes	Range and possible anomalies
25% / 50% / 75%	Sample quartiles	Robust location and spread

2.3.3.0.1 The MBA salary dataset in Python. Loading the file and calling `describe()` gives the summary in Table 2.4:

```
import numpy as np
import pandas as pd

salary = pd.read_excel("M2 Salary Data.xlsx", sheet_name="DataTable")
salary.describe()
```

Table 2.4: `salary.describe()` output

Statistic	**Salary ($)**
count	2,597
mean	111,208.37
std	47,515.30
min	33,000.00
25%	73,930.80
50% (median)	99,684.00
75%	140,000.00
max	336,966.00

The distribution is right-skewed: the mean, $111,208, exceeds the median, $99,684, by more than $11,000 — exactly the asymmetry a right-skewed monetary variable produces, since every value enters the mean in proportion to its magnitude while the median only counts rank. A symmetric normal model should never be assumed merely because a histogram happens to have one central peak.

Binning the salary variable into $20,000 intervals and normalizing the counts gives the empirical PMF in Table 2.5:

```
bins = range(30000, 350000, 20000)
salary["Bins"] = pd.cut(salary["Salary"], bins)
salary["Bins"].value_counts(normalize=True).sort_index()
```

Table 2.5: Empirical PMF of binned MBA salary

Bin	$\widehat{p}_j$
(30,000, 50,000]	2.7%
(50,000, 70,000]	18.3%
(70,000, 90,000]	20.8%
(90,000, 110,000]	16.4%
(110,000, 130,000]	12.0%
(130,000, 150,000]	9.0%
(150,000, 170,000]	6.6%
(170,000, 190,000]	7.3%
(190,000, 210,000]	4.1%
(210,000, 230,000]	1.4%
(230,000, 250,000]	0.5%
(250,000, 270,000]	0.4%
(270,000, 290,000]	0.2%
(290,000, 310,000]	0.2%

Table 2.5: Empirical PMF of binned MBA salary

Bin	$\widehat{p}_j$
(310,000, 330,000]	0.1%
(330,000, 350,000]	0.0%

The most populated interval, \$70,000–\$90,000, holds about 20.8% of the sample — the "about one-fifth" figure quoted informally in lecture. Note also that `pandas`' default binning convention, used above, constructs *right-closed* intervals $(b_j, b_{j+1}]$, the opposite of the left-closed convention $[b_j, b_{j+1})$ used in Equation (2.7); either convention is valid so long as it is documented, since a value sitting exactly on a boundary is assigned differently under each rule. Because the `describe()` output above shows the salary range running from \$33,000 to \$336,966, entirely inside the chosen bin range of \$30,000 to \$350,000, every observation lands in some bin here and none are silently dropped.

2.3.3.0.2 The automobile fuel-efficiency dataset in Python. The same workflow bins miles per gallon (MPG) into intervals of width five:

```
auto = pd.read_excel("M2 Auto_data.xlsx", sheet_name="Original Dataset")
bins = range(0, 50, 5)
auto["bins"] = pd.cut(auto["mpg"], bins)
auto["bins"].value_counts(normalize=True).sort_index()
```

Table 2.6: Empirical PMF of binned automobile fuel efficiency (MPG)

Bin	$\widehat{p}_j$
(0, 5]	0.0%
(5, 10]	0.8%
(10, 15]	16.9%
(15, 20]	23.3%
(20, 25]	19.4%
(25, 30]	18.7%
(30, 35]	12.8%
(35, 40]	6.4%
(40, 45]	1.8%

The distribution is unimodal, centered around 15–25 MPG, and the proportions sum to 100% up to rounding, confirming that every recorded MPG value between 0 and 45 was captured by the chosen bins. Section 2.6 returns to this table to compute an approximate grouped-data mean directly from these bin proportions.

Choosing bins responsibly. Both examples above avoided trouble only because their bin edges happened to cover the full observed range — a fact confirmed here by checking, not assumed. In general: always check the minimum and maximum against the chosen edges (Python's `range` and `arange` both exclude the stop value, a common source of a silently truncated top bin); keep widths equal when comparing bar heights directly as probability masses; test whether a conclusion changes under reasonable alternative widths or starting points; report the sample size and bin definitions so the figure can be reproduced; and never treat a smooth, bell-like appearance as proof of normality.

2.4 Sampling, Empirical Frequency, and Reproducibility

A *sample* is a realized value from a random variable; a dataset of size n contains n such draws, which can be generated in Excel or Python from a specified distribution. If the draws are independent and identically distributed, the empirical proportion of observations falling in an event A estimates its model probability:

$$\widehat{P}_n(A) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i \in A\}. \tag{2.8}$$

In one set of 100 simulated draws from Uniform$(1, 1.1)$, seven fell between 1.05 and 1.06, giving an empirical proportion of 0.07 against the theoretical value 0.10 found above. This is not a contradiction: sampling error is expected in a finite dataset, and under the **law of large numbers** the empirical proportion converges to the true probability as n grows,

$$\widehat{P}_n(A) \to P(A) \quad \text{as } n \to \infty,$$

under standard independent-sampling assumptions. A *pseudorandom seed* initializes a deterministic number generator; reusing the same seed with the same generator and software implementation reproduces the same sequence, which is valuable for validation and debugging, though cross-platform results may differ when implementations differ. A seed does not make observations more random — it makes a pseudorandom experiment reproducible.

This machinery is closely connected to AI inference. A generative model produces outputs by sampling from learned conditional distributions, though the analogy should not be taken too literally: a modern language model does not merely transform one scalar uniform distribution into English. It learns a high-dimensional parameterized network, computes conditional token probabilities with a temperature-scaled softmax layer, and applies a decoding rule such as greedy selection, temperature sampling, top-k sampling, or nucleus sampling:

$$P(\text{next token} = j \mid \text{context}) = \frac{\exp(z_j/T)}{\sum_k \exp(z_k/T)}. \tag{2.9}$$

Here z_j is the model's score for token j and T is the temperature. Lower temperature concentrates probability on high-scoring tokens; higher temperature flattens the distribution and increases output diversity. This is the operational link between probability distributions, sampling, and why the same prompt can produce different responses on different runs.

2.4.1 The empirical distribution and non-parametric (plug-in) estimation

The estimator $\widehat{P}_n(A)$ in Equation (2.8) is a special case of a much more general idea, the **plug-in** or **non-parametric** estimation principle: whenever a population quantity can be written as an expectation under the population distribution $P(x)$, and $P(x)$ itself is unknown, replace $P(x)$ by the *empirical distribution* that assigns weight $1/n$ to each of the n sampled observations. No parametric family — normal, uniform, or otherwise — needs to be assumed for this to work; the estimator is built directly from the data.

Concretely, the population expectation $E[X] = \sum_x x\, P(x)$ becomes, after substituting the empirical weights $1/n$ for $P(x)$, exactly the sample mean:

$$\overline{X} = \sum_{i=1}^{n} x_i \cdot \frac{1}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i. \tag{2.10}$$

The same substitution applied to a variance, a proportion, or any other functional of the distribution produces the corresponding sample statistic. This is why ordinary sample means and sample variances are special cases of the broader **Monte Carlo principle**: random sampling lets us approximate an expectation under $P(x)$ without ever writing $P(x)$ down explicitly.

For an MBA salary population of more than 2,000 observations with population mean about \$111,208, a random sample of $n = 100$ produced a sample mean of about \$111,568. The two numbers are not identical, and should not be: the empirical distribution built from any one finite sample only approximates the population distribution, and a different sample of the same size would give a different sample mean. The law of large numbers guarantees convergence of this approximation as n grows, not exact agreement at any fixed n.

Why random sampling matters. The plug-in principle is only as good as the sample it is built on. If the sampling mechanism systematically favors part of the population — surveying only certain neighborhoods, or observations that happen to be easiest to collect — the empirical distribution can converge to the wrong target no matter how large n becomes: a large biased sample can yield a very precise estimate of the wrong quantity. Before applying any estimator in this chapter, it is worth asking how the data were actually collected: were observations selected independently, and does observing one unit change the chance that another is observed?

2.5 Transformations of Random Variables

2.5.1 Affine transformations

A transformation constructs a new random variable from an existing one: if $Y = g(X)$, each realized X value is converted by the deterministic function g. The most important basic case is an **affine transformation**,

$$Y = aX + b.$$

The coefficient a changes scale — stretching the distribution if $|a| > 1$, compressing it if $0 < |a| < 1$, and reversing its orientation if $a < 0$ — while the constant b shifts every value by the same amount. If X has support from a lower bound L to an upper bound U and $a > 0$, then $Y \in [aL + b,\ aU + b]$.

As a worked example, map Uniform$(1, 10)$ onto Uniform$(4, 6)$ in three steps: subtract one (moving the support from 0 to 9), multiply by $2/9$ (compressing its length from 9 to

2), and add four (shifting the support to 4 through 6):

$$Y = 4 + \frac{2}{9}(X-1) = \frac{2}{9}X + \frac{34}{9}.$$

Checking endpoints verifies the mapping: $X = 1 \Rightarrow Y = 4$, and $X = 10 \Rightarrow Y = 6$. Because the transformation is linear and increasing and the input is uniform, the output is also uniform, with density $f_Y(y) = 1/2$ for $4 \le y \le 6$.

For a differentiable, one-to-one transformation $Y = g(X)$, density changes according to the Jacobian rule, which preserves total probability while horizontal distances are stretched or compressed:

$$f_Y(y) = f_X\Big(g^{-1}(y)\Big)\left|\frac{d}{dy}g^{-1}(y)\right|. \tag{2.11}$$

A valid transformation need not be one-to-one; $Y = X^2$, for instance, is perfectly valid. One-to-one behavior is required only for the simplest single-inverse change-of-variables formula above. If several X values map to the same Y, their density contributions must be summed over all inverse branches:

$$f_Y(y) = \sum_{x:\, g(x)=y} \frac{f_X(x)}{|g'(x)|}, \qquad \text{when the inverse branches are regular.}$$

2.5.2 Uniform and triangular densities

A density is valid whenever it is nonnegative and its total area is one — a criterion that extends naturally beyond rectangles. For a triangular density with base from 1 to 10 (base length nine) and peak height h, normalization requires

$$\tfrac{1}{2}(9)h = 1 \quad \Longrightarrow \quad h = \tfrac{2}{9}.$$

Moving the triangular peak changes where probability mass concentrates but not the height needed for normalization, as long as the base endpoints stay fixed at 1 and 10 and the triangle returns to zero at both ends. A peak near the left creates a right-skewed distribution with a long right tail; a peak near the right creates a left-skewed distribution; a peak at the midpoint 5.5 produces a symmetric triangular distribution, for which the mean and median coincide at 5.5. For a skewed triangular distribution the mode, median, and mean generally differ, since these three measures answer different questions and need not coincide: the mode is the location of the density peak, the median divides probability mass in half, and the mean is the balance point.

2.5.3 Quantiles and the discrete median convention

The median is a quantile: a continuous median m satisfies equal probability on either side when the density has no flat ambiguity, and more generally a median satisfies

$$P(X \le m) \ge 0.5 \qquad \text{and} \qquad P(X \ge m) \ge 0.5.$$

Using the CDF $F_X(x) = P(X \le x)$, the course convention for a *discrete* median is the smallest attainable value whose cumulative probability reaches or exceeds one half:

$$m = \inf\{x : F_X(x) \ge 0.5\}.$$

For a family-size distribution assigning probabilities 0.15, 0.20, 0.35, and 0.30 to 0, 1, 2, and 3 children, the cumulative probabilities are 0.15, 0.35, 0.70, and 1. Since $F(1) = 0.35 < 0.5$ while $F(2) = 0.70 \ge 0.5$, the median is 2 children.

2.6 Expected Value

2.6.1 Probability-weighted center

The expected value, also called the mean and written μ, is the probability-weighted average of a random variable's possible outcomes — a sum for a discrete variable, an integral against the density for a continuous one:

$$\begin{aligned} E[X] &= \sum_x x\, p_X(x) \quad \text{(discrete)}, \\ E[X] &= \int_{-\infty}^{\infty} x\, f_X(x)\, dx \quad \text{(continuous)}. \end{aligned} \tag{2.12}$$

A center-of-mass interpretation is useful: imagine each possible value placed on a balance beam with mass equal to its probability, and the expected value is the point at which the beam balances.

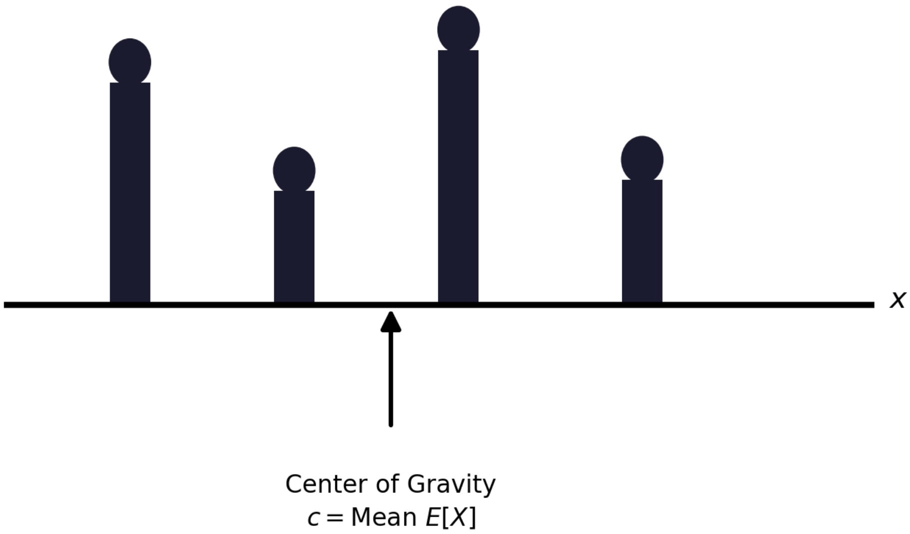

Figure 2.1: Interpretation of the mean as a center of gravity. Given a bar with a weight $p_X(x)$ placed at each point x with $p_X(x) > 0$, the center of gravity c is the point at which the sum of the torques from the weights to its left are equal to the sum of the torques from the weights to its right.

That is,

$$\sum_x (x - c)\, p_X(x) = 0, \qquad \text{or} \qquad c = \sum_x x\, p_X(x), \tag{2.13}$$

and the center of gravity is equal to the mean $E[X]$.

Unlike the discrete median, the expected value need not be an attainable outcome — a family cannot have 1.8 children, and no die shows 3.5, yet both are meaningful population averages. For the family-size distribution above,

$$E[X] = 0(0.15) + 1(0.20) + 2(0.35) + 3(0.30) = 1.80.$$

Many research quantities are transformations $g(X)$ rather than X itself — expected utility, squared error, and profit are examples. The *law of the unconscious statistician* computes such an expectation directly from the distribution of X, without first deriving the distribution of $g(X)$:

$$E[g(X)] = \sum_x g(x)\, p_X(x). \tag{2.14}$$

Expectation is linear even when variables are dependent, and constants move through the expectation operator in the natural way:

$$E[aX + b] = aE[X] + b, \qquad E[X_1 + \cdots + X_n] = E[X_1] + \cdots + E[X_n]. \tag{2.15}$$

For 1,000 families with the family-size distribution above, the expected total number of children is $1000(1.8) = 1800$; if each child is independently a girl with probability one half, the expected number of girls is 900. Linearity alone supplies the expected total — independence is not required for the addition of expectations, although the one-half conditional assumption is needed for the gender calculation.

We finally illustrate by example a common pitfall: unless $g(X)$ is a linear function, it is not generally true that $E[g(X)]$ is equal to $g\big(E[X]\big)$.

Example. Average Speed Versus Average Time. If the weather is good (which happens with probability 0.6), Alice walks the 2 miles to class at a speed of $V = 5$ miles per hour, and otherwise drives her motorcycle at a speed of $V = 30$ miles per hour. What is the mean of the time T to get to class?

The correct way to solve the problem is to first derive the PMF of T,

$$p_T(t) = \begin{cases} 0.6 & \text{if } t = 2/5 \text{ hours}, \\ 0.4 & \text{if } t = 2/30 \text{ hours}, \end{cases}$$

and then calculate its mean by

$$E[T] = 0.6 \cdot \frac{2}{5} + 0.4 \cdot \frac{2}{30} = \frac{4}{15} \text{ hours}.$$

However, it is wrong to calculate the mean of the speed V,

$$E[V] = 0.6 \cdot 5 + 0.4 \cdot 30 = 15 \text{ miles per hour},$$

and then claim that the mean of the time T is

$$\frac{2}{E[V]} = \frac{2}{15} \text{ hours}.$$

To summarize, in this example we have

$$T = \frac{2}{V}, \qquad \text{and} \qquad E[T] = E\left[\frac{2}{V}\right] \neq \frac{2}{E[V]}.$$

When only histogram bins are available, replace each observation in bin j by a representative value m_j, usually the midpoint, giving an approximate grouped-data mean:

$$\widehat{\mu} \approx \sum_j m_j\, \widehat{p}_j. \tag{2.16}$$

Practical caution. Using lower bin boundaries instead of midpoints systematically shifts this approximation downward. Whenever the original observations are available, calculate the mean from the raw values and reserve the histogram for visualization rather than estimation.

Applying Equation (2.16) to the automobile MPG bins of Table 2.6, using each bin's midpoint m_j (e.g. $m = 2.5$ for $(0, 5]$, $m = 7.5$ for $(5, 10]$, and so on up to $m = 42.5$ for $(40, 45]$) as the representative value:

$$\begin{aligned}\widehat{\mu} \approx\ & 2.5(0.000) + 7.5(0.008) + 12.5(0.169) + 17.5(0.233) + 22.5(0.194) \\ & + 27.5(0.187) + 32.5(0.128) + 37.5(0.064) + 42.5(0.018) \approx 23.1 \text{ MPG}.\end{aligned}$$

This grouped-data estimate is only approximate — it replaces every observation inside a bin with that bin's midpoint — but it requires nothing beyond the histogram itself, which is exactly the setting in which Equation (2.16) is used in practice.

2.6.2 Ranking example: deriving a PMF from the chain rule

Two men and two women receive distinct examination ranks, and every labeled ordering is equally likely. Let X be the best (smallest numerical) rank achieved by a woman; the possible values are 1, 2, and 3, since rank 4 is impossible (only two men can occupy positions ahead of the first woman). Table 2.7 derives the PMF using the chain rule.

Table 2.7: Distribution of the best rank achieved by a woman

x	**Required gender pattern**	**Probability calculation**	$P(X = x)$
1	W	2 women / 4 people	1/2
2	M, then W	(2/4)(2/3)	1/3
3	M, M, then W	(2/4)(1/3)(2/2)	1/6

For example, the chain rule gives the middle row explicitly:

$$P(X = 2) = P(G_1 = M)\, P(G_2 = W \mid G_1 = M) = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}.$$

The PMF sums correctly, $1/2 + 1/3 + 1/6 = 1$, and the expected best rank is

$$E[X] = 1\left(\tfrac{1}{2}\right) + 2\left(\tfrac{1}{3}\right) + 3\left(\tfrac{1}{6}\right) = \tfrac{5}{3} \approx 1.667,$$

a value that, like the family-size mean, is not itself an attainable rank but the average over repeated random rankings.

This calculation generalizes. With n people including w women, the set of positions occupied by women is uniformly distributed over the $\binom{n}{w}$ possible position sets. If X is the best female rank, then for feasible r the first woman occupies position r and the remaining $w - 1$ female positions are chosen from the $n - r$ later positions:

$$\begin{aligned} P(X = r) &= \frac{\binom{n-r}{w-1}}{\binom{n}{w}}, \qquad r = 1, \ldots, n - w + 1, \\ E[X] &= \frac{n+1}{w+1}. \end{aligned} \tag{2.17}$$

For $n = 4$ and $w = 2$, the general formula gives $(4+1)/(2+1) = 5/3$, matching the direct calculation above.

2.6.3 Three-dice casino game

Three fair six-sided dice produce $6^3 = 216$ equally likely ordered outcomes. A bet on "Small" wins \$100 when the sum is 4 through 10, *except* that any triple is a house win; it loses \$100 for a "Big" sum of 11 through 17 and for all triples. There are six triples — $(1,1,1), \ldots, (6,6,6)$ — so $P(\text{triple}) = 6/216 = 1/36$. Removing triples leaves 210 outcomes, and dice symmetry maps every nontriple Small outcome to a nontriple Big outcome, so each class contains 105 outcomes:

$$\begin{aligned} P(\text{Small}) &= \frac{105}{216} = \frac{35}{72} \approx 0.4861, \\ P(\text{Loss}) &= \frac{35}{72} + \frac{1}{36} = \frac{37}{72} \approx 0.5139. \end{aligned} \tag{2.18}$$

Table 2.8 lays out the full payoff distribution for a \$100 Small bet.

Table 2.8: Casino payoff distribution for a \$100 Small bet

Event	**Outcomes**	**Probability**	**Net payoff R**
Small, nontriple	105	$35/72 = 0.4861$	$+\$100$
Big, nontriple	105	$35/72 = 0.4861$	$-\$100$
Triple	6	$1/36 = 0.0278$	$-\$100$
Total	216	1.0000	—

The expected net return per \$100 bet is negative:

$$E[R] = 100\left(\frac{35}{72}\right) - 100\left(\frac{37}{72}\right) = -\frac{25}{9} \approx -\$2.78. \tag{2.19}$$

The player loses about \$2.78 per \$100 bet on average, so the house edge is approximately 2.78%; a single bet can still win, since expectation describes only the average over repeated play. By linearity, the expected cumulative return after n independent bets is

$$E\left[\sum_{i=1}^{n} R_i\right] = nE[R] = -\frac{25n}{9}. \tag{2.20}$$

2.7 Variance, Standard Deviation, and Risk

2.7.1 Why the mean is not enough

Two distributions can share an expected value while carrying very different risk. Consider lottery W, which pays -10 or $+10$ with equal probability, and lottery Y, which pays -100 or $+100$ with equal probability:

$$E[W] = (-10)(0.5) + (10)(0.5) = 0, \qquad E[Y] = (-100)(0.5) + (100)(0.5) = 0.$$

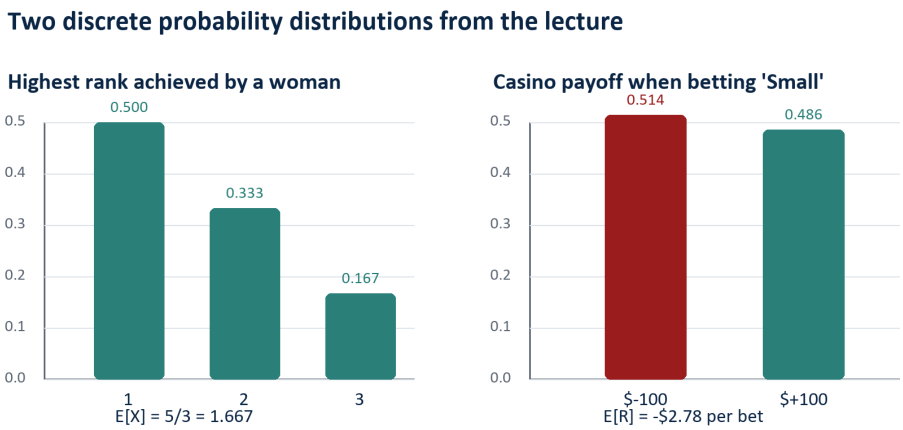

Figure 2.2: Probability mass determines both the center (expectation) and spread of a random variable.

Both means are zero, but Y exposes the decision maker to gains and losses ten times larger than W's. Preference between the two cannot be explained by expected value alone: a risk-averse person may prefer W, a risk-seeking person may prefer Y, and a complete decision model could apply expected utility, downside risk, value at risk, or another criterion,

$$\text{Expected utility} = \sum_x p(x)\, u(x), \qquad \text{where a concave } u \text{ represents risk aversion.}$$

2.7.2 Definitions

The expected deviation from the mean is always zero, $E[X-\mu] = E[X] - \mu = 0$, so positive and negative deviations cancel and a useful spread measure must remove their signs. **Variance** squares the deviations before averaging them:

$$\begin{aligned}\mathrm{Var}(X) = \sigma_X^2 &= E\Big[(X-\mu_X)^2\Big] \\ &= \sum_x (x-\mu_X)^2 p_X(x) \quad \text{or} \quad \int_{-\infty}^{\infty} (x-\mu_X)^2 f_X(x)\, dx.\end{aligned} \tag{2.21}$$

Variance carries squared units — if X is measured in dollars, variance is measured in dollars squared — so **standard deviation** returns to the original units and is easier to interpret:

$$SD(X) = \sigma_X = \sqrt{\mathrm{Var}(X)}. \tag{2.22}$$

Both variance and standard deviation are nonnegative and equal zero only when X is constant with probability one. Table 2.9 summarizes W and Y: both outcomes of W are ten units from its zero mean, giving variance 100 and standard deviation 10; both outcomes of Y are one hundred units from its zero mean, giving variance 10,000 and standard deviation 100, consistent with $Y = 10W$.

Table 2.9: Same expected value, different variability

Game	Distribution	E[Game]	Variance	SD
W	$+10$ or -10, each w.p. 1/2	0	100	10
Y	$+100$ or -100, each w.p. 1/2	0	10,000	100

$$SD(W) = \sqrt{\tfrac{1}{2}(10)^2 + \tfrac{1}{2}(-10)^2} = 10.$$

2.7.3 The computational identity and transformation rules

Expanding the square in Equation (2.21) gives an equivalent, often computationally convenient identity, although the centered formula is usually more numerically stable in software when the mean is very large relative to the spread:

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2. \tag{2.23}$$

Variance in Terms of Moments Expression

$$\mathrm{Var}(X) = E[X^2] - \big(E[X]\big)^2.$$

This expression is verified as follows:

$$\begin{aligned}
\mathrm{Var}(X) &= \sum_x \big(x - E[X]\big)^2 p_X(x) \\
&= \sum_x \Big(x^2 - 2xE[X] + \big(E[X]\big)^2\Big) p_X(x) \\
&= \sum_x x^2 p_X(x) \;-\; 2E[X]\sum_x x p_X(x) \;+\; \big(E[X]\big)^2 \sum_x p_X(x) \\
&= E[X^2] - 2\big(E[X]\big)^2 + \big(E[X]\big)^2 \\
&= E[X^2] - \big(E[X]\big)^2.
\end{aligned}$$

Example. Mean and Variance of the Bernoulli. Consider the experiment of tossing a biased coin, which comes up a head with probability p and a tail with probability $1-p$, and the Bernoulli random variable X with PMF

$$p_X(k) = \begin{cases} p & \text{if } k = 1, \\ 1-p & \text{if } k = 0. \end{cases}$$

Its mean, second moment, and variance are given by the following calculations:

$$\begin{aligned}
E[X] &= 1 \cdot p + 0 \cdot (1-p) = p, \\
E[X^2] &= 1^2 \cdot p + 0 \cdot (1-p) = p, \\
\mathrm{Var}(X) &= E[X^2] - \big(E[X]\big)^2 = p - p^2 = p(1-p).
\end{aligned}$$

Shifting every outcome by a constant changes the mean but not the variance; scaling every outcome by a multiplies variance by a^2 and standard deviation by $|a|$:

$$\begin{aligned} E[aX+b] &= aE[X]+b, \\ \operatorname{Var}(aX+b) &= a^2 \operatorname{Var}(X), \\ SD(aX+b) &= |a|\, SD(X). \end{aligned} \tag{2.24}$$

The absolute value matters because standard deviation cannot be negative: if $a=-2$, the distribution is reflected and stretched by two, so its standard deviation doubles rather than becoming negative.

As an illustration of the mean shifting under unequal probabilities, suppose the Y lottery is modified so that -100 occurs with probability 0.4 and $+100$ with probability 0.6. The mean shifts to 20, since the positive outcome is now more likely, and the variance falls to 9,600:

$$\begin{aligned} E[Y] &= (-100)(0.4)+(100)(0.6) = 20, \\ \operatorname{Var}(Y) &= (-100-20)^2(0.4)+(100-20)^2(0.6) = 9{,}600, \\ SD(Y) &\approx 97.98. \end{aligned}$$

The drop from variance 10,000 to 9,600 reflects the shifted mean and unequal masses: the $+100$ outcome is now both closer to the mean and more likely, modestly reducing the average squared deviation.

For the casino payoff R from Section 2.6, $R^2 = 10{,}000$ in both payoff states, so

$$\operatorname{Var}(R) = 10{,}000 - \left(-\tfrac{25}{9}\right)^2 \approx 9{,}992.28, \qquad SD(R) \approx \$99.96.$$

The expected loss is small relative to the volatility of a single bet, which helps explain why short winning streaks can coexist with a long-run house advantage.

2.7.4 Independence, covariance, and adding random variables

Variance is not linear the way expectation is. For *independent* random variables, the variance of a weighted sum is the sum of weighted variances, with each coefficient squared:

$$X_1 \perp X_2 \implies \operatorname{Var}(c_1X_1+c_2X_2) = c_1^2 \operatorname{Var}(X_1) + c_2^2 \operatorname{Var}(X_2). \tag{2.25}$$

For dependent variables, **covariance** captures how their deviations move together — positive covariance means above-average values of the two variables tend to occur together, negative covariance means one tends to be above average when the other is below:

$$\begin{aligned} \operatorname{Cov}(X_1,X_2) &= E[(X_1-\mu_1)(X_2-\mu_2)]\,, \\ \operatorname{Var}(c_1X_1+c_2X_2) &= c_1^2 \operatorname{Var}(X_1) + c_2^2 \operatorname{Var}(X_2) + 2c_1c_2 \operatorname{Cov}(X_1,X_2). \end{aligned} \tag{2.26}$$

Independence implies zero covariance when second moments exist, but zero covariance does not generally imply independence — a distinction that matters in regression, portfolio analysis, forecasting, and any system where multiple uncertain inputs move together.

The failure of the independent-sum formula is easy to see with $X_2 = 3X_1+5$: since X_2 is built directly from X_1, the two are perfectly dependent, and $\operatorname{Var}(X_2) = 9\operatorname{Var}(X_1)$. Adding them gives $X_1+X_2 = 4X_1+5$, so the exact variance is

$$\operatorname{Var}(X_1+X_2) = \operatorname{Var}(4X_1+5) = 16\operatorname{Var}(X_1),$$

sixteen times $\mathrm{Var}(X_1)$, not the ten times that the independent-sum formula would (incorrectly) suggest. The missing six units of variance come from twice the covariance: since $\mathrm{Cov}(X_1, 3X_1 + 5) = 3\,\mathrm{Var}(X_1)$,

$$2\,\mathrm{Cov}(X_1, X_2) = 2 \times 3\,\mathrm{Var}(X_1) = 6\,\mathrm{Var}(X_1).$$

Because covariance depends on measurement units, **correlation** is often used to express standardized dependence: dividing covariance by the two standard deviations produces a quantity between -1 and $+1$,

$$\mathrm{Corr}(X_1, X_2) = \frac{\mathrm{Cov}(X_1, X_2)}{SD(X_1)\,SD(X_2)}, \qquad -1 \leq \mathrm{Corr} \leq 1. \tag{2.27}$$

Values near $+1$ indicate strong positive linear co-movement, values near -1 indicate strong negative linear co-movement, and values near zero indicate little linear association. For $X_2 = 3X_1 + 5$ with X_1 having positive variance, $\mathrm{Corr}(X_1, X_2) = +1$, since X_2 is a perfectly increasing affine function of X_1; a negative coefficient would instead give correlation -1. This makes visible, on a standardized scale, exactly the dependence that the naive independent-sum formula missed. The broader modeling lesson: whether risks diversify depends on their dependence structure. Adding independent risks can smooth relative uncertainty, while adding highly positively correlated risks provides little of that diversification benefit.

2.7.5 A counterexample: zero covariance without independence

The caveat raised in Lecture 1 is not a technicality that only matters for exotic distributions — it is easy to construct an explicit random variable pair with zero covariance that is nevertheless strongly dependent, in the strongest possible sense: one variable is a deterministic function of the other.

2.7.5.0.1 A discrete example. Let X take the values $-1, 0, 1$, each with probability $1/3$, and define

$$Y = X^2.$$

Table 2.10 lays out the resulting joint distribution: since Y is a deterministic function of X, only three (x, y) pairs receive positive probability.

Table 2.10: Joint distribution of X and $Y = X^2$

x	$y = x^2$	$P(X = x, Y = y)$
-1	1	1/3
0	0	1/3
1	1	1/3

Zero covariance. By symmetry, $E[X] = \frac{1}{3}(-1) + \frac{1}{3}(0) + \frac{1}{3}(1) = 0$. Since $x^3 = x$ for each of $-1, 0, 1$ in this example, $E[XY] = E[X^3] = E[X] = 0$ as well, so

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0 - (0)\left(\tfrac{2}{3}\right) = 0.$$

Strong dependence. Despite this, Y is completely determined by X — knowing X leaves no uncertainty about Y at all, which is about as dependent as two random variables can be. The joint table confirms this directly: $P(X = 0, Y = 0) = 1/3$, but the marginals give $P(X = 0)P(Y = 0) = \left(\frac{1}{3}\right)\left(\frac{1}{3}\right) = 1/9 \neq 1/3$, so the independence factorization from Equation (1.15) fails. Covariance is blind to this dependence because it only measures *linear* co-movement: as X moves away from 0 in either direction, Y increases either way, and the positive and negative contributions to $E[XY]$ cancel exactly.

2.7.5.0.2 A continuous analogue. The same construction works with a continuous variable: let $X \sim \text{Uniform}(-1, 1)$ and again set $Y = X^2$. By the same odd/even symmetry argument, $E[X] = E[X^3] = 0$, so $\text{Cov}(X, Y) = E[X^3] - E[X]E[X^2] = 0$. Yet Y is again an exact deterministic function of X, so the two variables are as dependent as possible. In both cases, the lesson is the same one flagged in Lecture 1: $\text{Cov}(X, Y) = 0$ rules out *linear* association, not dependence in general — only for pairs of Bernoulli indicators does zero covariance upgrade to full independence.

2.7.6 Population parameters and sample estimators

The formulas above describe true, population-level moments; with observed data, analysts instead estimate them from a sample. Capital letters X_i denote random observations, lowercase x_i denote realized values, and the sample mean and (unbiased) sample variance are

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2. \tag{2.28}$$

The $n - 1$ denominator corrects the downward bias created by estimating μ with the same sample mean used inside the sum; this is why software may report slightly different standard deviations depending on whether a population or sample convention is selected. Table 2.11 summarizes the correspondence, including the standard error of the sample mean, which reappears when confidence intervals are introduced later in the course.

Table 2.11: Population parameters and sample estimators

Target	**Population quantity**	**Sample estimator**
Center	$\mu = E[X]$	$\bar{x} = (1/n)\sum_i x_i$
Variance	$\sigma^2 = E[(X - \mu)^2]$	$s^2 = [1/(n-1)]\sum_i (x_i - \bar{x})^2$
Standard deviation	$\sigma = \sqrt{\sigma^2}$	$s = \sqrt{s^2}$
Uncertainty of the sample mean	$SD(\bar{x}) = \sigma/\sqrt{n}$	Estimated by $s/\sqrt{n}$

Why the $n - 1$ correction, more precisely. If the population mean μ were known, variance could be estimated directly by averaging $(X_i - \mu)^2$ using all n observations freely. In practice μ is unknown and is replaced by $\bar{X}$, but this substitution is not free: once $\bar{X}$ is fixed, the n deviations $X_i - \bar{X}$ are constrained to sum to zero,

$$\sum_{i=1}^{n}(X_i - \bar{X}) = 0,$$

so only $n - 1$ of them can vary independently — the last is always determined by the other $n - 1$. One **degree of freedom** has been spent estimating $\bar{X}$, and dividing by

n instead of $n-1$ would systematically understate the population variance; the $n-1$ correction removes exactly that downward bias. The same bookkeeping recurs, in a more elaborate form, whenever a dataset is used to estimate several parameters that then enter a further calculation — for instance in regression, where each additional fitted coefficient consumes one more degree of freedom.

Diminishing returns to more data. Because $SD(\bar{X}) = \sigma/\sqrt{n}$, statistical error shrinks with sample size only at the rate $1/\sqrt{n}$. Quadrupling n halves the standard error; to quarter it requires sixteen times as much data. Equivalently, cutting a target standard error in half is never achieved by doubling the sample — it requires roughly four times as much data. This square-root law is why moving from a mediocre estimate to a good one is often cheap, while moving from a good estimate to an excellent one can demand disproportionately more data; the same pattern reappears in machine learning, where reducing model error further and further can require very large increases in training data.

2.8 The Normal Distribution, Standardization, and Z Probabilities

2.8.1 The normal density

Unlike the uniform examples above, which have finite lower and upper bounds, a normal random variable has support across the entire real line. Its density is symmetric, peaks at the mean μ, and decays rapidly as x moves away from μ; the parameter σ is the standard deviation and σ^2 is the variance:

$$\begin{aligned} &X \sim N(\mu, \sigma^2), \\ &f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \qquad -\infty < x < \infty. \end{aligned} \tag{2.29}$$

Squaring $x-\mu$ makes the curve symmetric around μ; the negative sign causes density to decrease as squared distance grows; and the factor $1/(\sigma\sqrt{2\pi})$ normalizes the total area to one. Changing μ shifts the entire curve without changing its shape; increasing σ spreads the same total probability over a wider range and lowers the peak, while decreasing σ concentrates probability and raises the peak. Symmetry implies that mean, median, and mode all coincide at μ for every normal distribution.

2.8.2 The empirical rule and tail behavior

Although normal distributions are unbounded, most probability mass lies within a few standard deviations of the mean. The canonical 68–95–99.7 empirical rule gives

$$P(\mu - \sigma \le X \le \mu + \sigma) \approx 0.6827, \qquad P(\mu - 2\sigma \le X \le \mu + 2\sigma) \approx 0.9545$$

and

$$P(\mu - 3\sigma \le X \le \mu + 3\sigma) \approx 0.9973.$$

The probability outside $\pm 3\sigma$ is therefore about 0.0027 in total, or roughly 0.00135 in each tail:

$$P(|X-\mu| > 3\sigma) \approx 1 - 0.9973 = 0.0027.$$

Rare does not mean impossible: the normal model still assigns positive probability to arbitrarily extreme finite observations.

2.8.3 Affine transformations and standardization

Normal distributions are closed under affine transformations: if $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, then Y is also normal, with its mean transformed linearly and its variance multiplied by a^2:

$$\begin{aligned} aX + b &\sim N(a\mu + b,\ a^2\sigma^2), \\ E[aX + b] &= a\mu + b, \\ SD(aX + b) &= |a|\sigma. \end{aligned} \tag{2.30}$$

A pure shift $X + b$ changes only the mean; a pure scale aX changes both mean and spread. **Standardization** converts any nondegenerate normal variable to the standard normal distribution by subtracting μ (shifting the mean to zero) and dividing by σ (rescaling the standard deviation to one):

$$Z = \frac{X - \mu}{\sigma}, \qquad Z \sim N(0, 1). \tag{2.31}$$

The standard normal cumulative distribution function, written Φ, returns the left-tail probability, and a Z table is simply a printed lookup table for Φ:

$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt, \qquad \Phi(-z) = 1 - \Phi(z).$$

2.8.4 Worked Z-table calculations

Suppose X is normal with mean 100 and standard deviation 50. To find $P(X \leq 110)$, standardize the cutoff:

$$z = \frac{110 - 100}{50} = 0.20, \qquad P(X \leq 110) = \Phi(0.20) \approx 0.57926,$$

so about 57.9% of this distribution lies at or below 110. For an interval probability, standardize both endpoints and subtract cumulative left-tail areas; the interval from 90 to 110 maps to -0.20 through $+0.20$:

$$P(90 \leq X \leq 110) = \Phi(0.20) - \Phi(-0.20) = 0.57926 - 0.42074 = 0.15852.$$

The same method handles any interval: for a right-tail probability, subtract a left-tail table value from one; for a two-sided probability, standardize both bounds and subtract:

$$\begin{aligned} P(X > c) &= 1 - \Phi\left(\frac{c - \mu}{\sigma}\right), \\ P(a \leq X \leq b) &= \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right). \end{aligned}$$

It is important to always state whether a normal distribution's second parameter denotes variance or standard deviation; these notes write $N(\mu, \sigma^2)$, so the standard deviation used in a Z score is σ. Standardization converts a problem with arbitrary units into a universal reference scale: once the distribution of a standardized statistic is known, table values or software translate distances from the mean into probabilities, tail areas, critical values, and confidence levels — exactly the machinery that confidence intervals and hypothesis tests build on in later chapters.

2.8.5 Why is the normal distribution important? The Central Limit Theorem

A natural question is why the normal distribution deserves this much attention in the first place, rather than the uniform, triangular, or any other shape already introduced. Part of the answer lies in the behavior of the plug-in estimator from Section 2.4.1: the sample mean $\overline{X}$.

Because $\overline{X}$ is built from a random sample, it is itself a random variable with its own probability distribution — the **sampling distribution** of the sample mean. This is a different object from the population distribution of a single observation X: the population distribution describes individual outcomes, while the sampling distribution describes how the statistic $\overline{X}$ would vary if the sampling procedure were repeated many times.

Two properties of this sampling distribution follow directly from the expectation and variance rules developed earlier in this chapter. Since $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ and expectation is linear,

$$E[\overline{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \frac{1}{n}\sum_{i=1}^{n} \mu = \mu, \tag{2.32}$$

so the sample mean is centered exactly at the population mean — the mathematical reason $\overline{X}$ is a sensible estimator of μ in the first place. Assuming the observations are independent, the variance of a sum is the sum of variances, and the constant $1/n$ enters squared:

$$\mathrm{Var}(\overline{X}) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{1}{n^2}\left(n\sigma^2\right) = \frac{\sigma^2}{n}, \qquad SD(\overline{X}) = \frac{\sigma}{\sqrt{n}}. \tag{2.33}$$

This last quantity is exactly the standard error of the mean already listed in Table 2.11.

Equations (2.32) and (2.33) pin down the center and spread of $\overline{X}$, but not its *shape*. The **Central Limit Theorem** supplies that missing piece: as the sample size n grows, the sampling distribution of $\overline{X}$ approaches a normal distribution,

$$\overline{X} \overset{\text{approx.}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right), \tag{2.34}$$

regardless of the shape of the original population distribution. The population being averaged may be uniform, sharply right-skewed like the MBA salary data, or any other shape entirely — the sample mean nevertheless comes to resemble a bell curve as n increases. This is why the normal distribution earns its central place in this chapter: it is not merely one convenient shape among many, but the shape that plug-in estimators such as $\overline{X}$ converge toward under very general conditions, which is what ultimately licenses the standardization and Z-table machinery developed in the rest of this section.

Standardizing $\overline{X}$ using Equation (2.34) gives a statistic with a known, tabulated distribution even when the population itself is unknown:

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \overset{\text{approx.}}{\sim} N(0, 1), \qquad \text{so} \qquad P\left(-1.96 \le \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \le 1.96\right) \approx 0.95. \tag{2.35}$$

This probability statement is the bridge from the Central Limit Theorem to confidence intervals: rearranging it so that the unknown μ, rather than the observed $\overline{X}$, sits in the middle of the inequality produces $\overline{X} \pm 1.96\,\sigma/\sqrt{n}$, a 95% confidence interval for the population mean — the subject taken up in later chapters.

2.9 Interpretation Cautions for Research

Table 2.12 lists several informal shortcuts that are common in practice, why each is incomplete, and the more rigorous correction.

Table 2.12: Common shortcuts and the rigorous correction

Informal shortcut	Why it is incomplete	Rigorous correction
A larger mean proves an effect.	Observed means vary across samples.	Evaluate the difference relative to its standard error and design.
The histogram is the probability density.	Bars depend on the sample and the chosen bins.	Treat it as an empirical estimate; report the bin construction.
A bell-shaped plot is normal.	Many distributions look approximately bell-shaped.	Use diagnostics and substantive modeling assumptions.
A numeric category code has quantitative meaning.	Codes may be arbitrary labels.	Model categories explicitly unless the ordering is substantive.
Equal means imply equivalent populations.	Spread and shape may differ strongly.	Compare variability, distributional shape, and the target estimand.

Statistical-significance clarification. A difference between treatment and control means is descriptive evidence, not automatically a significant effect. Significance depends on the estimated difference, group variability, sample sizes, the assignment mechanism, the dependence structure, and the null model used for inference.

2.10 Compact Formula Reference

Table 2.13: Core formulas from this lecture

Concept	Formula	Purpose
Random variable	$X : \Omega \to \mathbb{R}$	Map outcomes to numerical values
Discrete PMF	$p_X(x) = P(X = x)$; $\sum_x p_X(x) = 1$	Allocate mass across discrete values
Continuous PDF	$\int_a^b f_X(x)\,dx = P(a \leq X \leq b)$	Probability as area under a density
CDF	$F_X(x) = P(X \leq x)$	Accumulate probability through x
Uniform density	$f_X(x) = 1/(b-a)$ for $a \leq x \leq b$	Constant density on a bounded interval

Table 2.13: Core formulas from this lecture

Concept	Formula	Purpose
Empirical bin mass	$\widehat{p}_j = n_j/N$	Convert histogram counts to proportions
Discrete expectation	$E[X] = \sum_x x\, p_X(x)$	Probability-weighted center
Continuous expectation	$E[X] = \int x f_X(x)\, dx$	Density-weighted center
Linearity	$E[aX + b] = aE[X] + b$	Transform means without re-deriving distributions
Variance	$\text{Var}(X) = E[(X - \mu)^2]$	Average squared distance from the mean
Variance identity	$\text{Var}(X) = E[X^2] - (E[X])^2$	Efficient computation
Affine variance rule	$\text{Var}(aX + b) = a^2 \text{Var}(X)$	Rescale variance under a linear transform
Standard deviation	$SD(X) = \sqrt{\text{Var}(X)}$	Spread in original units
Covariance	$\text{Cov}(X_1, X_2) = E[(X_1 - \mu_1)(X_2 - \mu_2)]$	Co-movement of two variables
Variance of a sum	$\text{Var}(c_1X_1 + c_2X_2) = c_1^2 \text{Var}(X_1) + c_2^2 \text{Var}(X_2) + 2c_1c_2 \text{Cov}(X_1, X_2)$	Combine dependent risks
Correlation	$\text{Corr}(X_1, X_2) = \text{Cov}(X_1, X_2)/[SD(X_1)SD(X_2)]$	Standardized, unit-free dependence
Normal density	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp[-(x - \mu)^2/(2\sigma^2)]$	Bell-shaped continuous model
Standardization	$Z = (X - \mu)/\sigma$	Convert to the $N(0, 1)$ reference scale
Interval probability	$P(a \le X \le b) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$	Normal probabilities via the Z table
Sample mean	$\bar{X} = (1/n) \sum_i X_i$	Estimate the population mean
Sample variance	$s^2 = [1/(n - 1)] \sum_i (X_i - \bar{X})^2$	Estimate the population variance

2.11 Practice Problems with Concise Solutions

2.11.1 Coding versus meaning

A dataset codes departments as 0, 1, and 2. Is the resulting variable quantitative?

Solution. Not necessarily. If the codes only identify categories, arithmetic comparisons such as "department 2 is twice department 1" are meaningless. The variable is categorical even though it is stored numerically.

2.11.2 Empirical bin probability

A salary bin contains 519 of 2,597 observations. Estimate its probability mass.

Solution. $\widehat{p} = 519/2{,}597 \approx 0.200$, so the bin represents about 20.0% of the observed population.

2.11.3 Ranking distribution

Verify that the probabilities for X, the best female rank among two men and two women, form a valid PMF.

Solution. Each probability is nonnegative, and $1/2 + 1/3 + 1/6 = 3/6 + 2/6 + 1/6 = 1$.

2.11.4 Casino expectation

A Small bet wins \$100 with probability 35/72 and loses \$100 with probability 37/72. Find the expected payoff.

Solution. $E[R] = 100(35/72) - 100(37/72) = -200/72 = -25/9 \approx -\$2.78$.

2.11.5 Variance from a PMF

Let Z equal -2 with probability 0.25 and 2 with probability 0.75. Find $E[Z]$, $\text{Var}(Z)$, and $SD(Z)$.

Solution. $E[Z] = -2(0.25) + 2(0.75) = 1$. $E[Z^2] = 4$, so $\text{Var}(Z) = 4 - 1^2 = 3$ and $SD(Z) = \sqrt{3} \approx 1.732$.

2.11.6 Same mean, different risk

Why are the ± 10 and ± 100 games not equivalent even though both expected values are zero?

Solution. Their standard deviations are 10 and 100. The second game produces outcomes ten times farther from the common mean and therefore carries substantially greater risk.

2.12 Chapter Summary

The chapter follows a single pipeline: define a numerical random variable and its support; specify a normalized PMF or PDF; distinguish theoretical probabilities from finite-sample empirical frequencies; use transformations to connect variables and derive their means and variances, including for sums of dependent variables; and, when a normal model is appropriate, standardize to Z and calculate areas with the standard normal distribution.

$$\text{Model} \to \text{distribution} \to \text{sample} \to \text{summary measures} \to \text{standardized inference.}$$

The generative-AI examples fit the first half of this pipeline: model outputs are random variables, softmax supplies normalized conditional masses, and decoding samples from those masses. The statistical examples fit the second half: expectation summarizes center, variance summarizes spread, covariance captures dependence, and standardization makes normal probabilities comparable across measurement scales. Random variables are, in this sense, the bridge between a qualitative sample space and numerical statistical analysis: their distributions answer which values are possible and how likely those values are, histograms provide an empirical view of a distribution, expectation compresses location into a single balance point, and variance and standard deviation describe the dispersion that the mean alone cannot capture. These same population quantities become sample estimators, standard errors, and formal comparisons once the population distribution is unknown — the subject of the chapters that follow.

Rapid Review

- A random variable maps experimental outcomes to numbers; it may be discrete or continuous.
- A PMF sums to one; a PDF integrates to one, and probability is area under the density.
- For Uniform(a, b), density is $1/(b-a)$; a histogram is an empirical estimate of a distribution, not the distribution itself.
- An affine transformation $Y = aX + b$ shifts and rescales a distribution; the Jacobian rule extends this to nonlinear transformations.
- The mean is a probability-weighted balance point; expectation is linear, even for dependent variables.
- Variance is the expected squared distance from the mean; standard deviation is its square root and shares the original units.
- Scaling by a multiplies variance by a^2; shifting by b does not change variance.
- Variances add directly only under zero covariance; independence is sufficient but not necessary for that.
- A normal variable has bell-shaped density parameterized by μ and σ^2; standardization $Z = (X - \mu)/\sigma$ converts normal probabilities to the $N(0, 1)$ reference scale.
- Population parameters (μ, σ^2) are estimated from data by sample statistics ($\bar{x}$, s^2), with the sample variance's $n - 1$ denominator correcting for using an estimated mean.

Appendix: Minimal Reproducibility Checklist

1. Confirm the file name, sheet name, column names, and number of nonmissing observations.
2. Inspect minimum and maximum values before choosing bin edges.
3. Check that every nonmissing value received a bin and that counts sum to N.
4. Check that normalized frequencies sum to 1 up to rounding.

5. Calculate the mean and standard deviation from raw values; use grouped calculations only when necessary.
6. Save the code and bin definitions with the analysis so the figure is reproducible.

3 | Lecture 3: Maximum Likelihood Estimation

Likelihood • Log-likelihood • Fisher information • Bias • Consistency • Standard error

3.1 Executive overview

Central idea. Maximum likelihood estimation fixes the observed data and chooses the parameter value under which those data would be most plausible.

The lecture opens Chapter 3 by moving from descriptive calculations to parametric inference. It develops the maximum likelihood principle through a coin-flip model, implements the method in Excel and Python, extends it to normal models for MBA salary and automobile fuel-efficiency data, and closes with estimation error, bias, consistency, Fisher information, and standard errors.

3.1.1 Learning outcomes

1. Distinguish probability from likelihood by identifying what is variable and what is held fixed.
2. Construct and maximize Bernoulli and normal likelihood functions.
3. Explain why log-likelihoods are computationally preferable to products of probabilities or densities.
4. Interpret coefficient estimates, standard errors, confidence intervals, and convergence output.
5. State bias, consistency, Fisher information, and the root-n rate precisely.
6. Recognize when pooling groups improves precision and when it imposes an unjustified modeling assumption.

3.2 Parametric inference and the likelihood principle

3.2.1 From an observed sample to an unknown parameter

Let $X_1, \ldots, X_n$ denote independent observations generated from a model indexed by an unknown parameter θ. Parametric inference assumes that the data distribution belongs to a specified family f(x; θ). The objective is not to alter the realized data; it is to learn which value of θ makes those realized observations most compatible with the model.

$$X_i \overset{iid}{\sim} f(x;\theta), \qquad i = 1, \ldots, n$$

Equation 1. Generic parametric model

Table 1. Probability and likelihood use the same model in different directions

Object	**Held fixed**	**Allowed to vary**	**Question answered**
Probability / density f(x; θ)	Parameter θ	Possible data x	If θ were known, how would outcomes vary?
Likelihood L(θ; x)	Observed data x	Candidate parameter θ	Which θ makes the observed data most plausible?

Interpretive caution. The likelihood is not a probability distribution over the parameter unless a prior distribution and Bayes' theorem are introduced. It ranks parameter values relative to the fixed data.

3.2.2 The three-step maximum likelihood method

1. Specify a parametric model f(x; θ) and identify the admissible parameter space Θ.
2. Evaluate the joint likelihood of the observed sample as a function of θ.
3. Choose any parameter value that maximizes the likelihood, or equivalently its logarithm.

$$\widehat{\theta} \in \arg\max_{\theta \in \Theta} L(\theta; x)$$

Equation 2. Maximum likelihood estimator

3.3 Bernoulli coin-flip model

3.3.1 Model, coding, and independence

The lecture codes a head as $X_i = 1$ and a tail as $X_i = 0$. With $p = P(X_i = 1)$, each trial follows a Bernoulli distribution. Independence means that the outcome of one flip does not change the probability law of another; it does not imply that the coin is fair.

$$X_i \in \{0, 1\}, \qquad P(X_i = 1) = p, \qquad P(X_i = 0) = 1 - p$$

Equation 3. Bernoulli parameterization

For one observation x_i, both cases can be written compactly as $p^{x_i}(1-p)^{1-x_i}$. Independence turns the joint probability of the ordered sequence into a product.

$$L(p;x) = \prod_{i=1}^{n} p^{x_i}\,(1-p)^{1-x_i}$$

Equation 4. Bernoulli likelihood for an ordered sample

3.3.2 Reduction to the number of heads

Let S be the number of heads. The lecture sample has n = 10 and S = 6. Because exponents add when identical factors are multiplied, every ordered sequence with six heads and four tails has the same parameter-dependent likelihood.

$$S = \sum_{i=1}^{n} x_i, \qquad L(p) = p^S\,(1-p)^{n-S} = p^6\,(1-p)^4$$

Equation 5. Likelihood for six heads and four tails

Table 2. Candidate values from the lecture, recomputed with the corrected exponents

Candidate p	Likelihood $p^6(1-p)^4$	Log-likelihood	Relative conclusion
0.11	0.000001112	-13.7098	Very weak fit
0.40	0.000530842	-7.5410	Below the maximum
0.50	0.000976563	-6.9315	Closer
0.60	0.001194394	-6.7301	Largest of these values

The live spreadsheet initially reversed the exponents before correcting them. The table consistently uses six heads and four tails.

Sequence versus count. If the observed data are only the count S = 6 rather than the exact order, the binomial likelihood includes the factor C(10,6). That factor does not depend on p, so it changes the likelihood's height but not its maximizing value.

3.3.3 Analytical maximization

A grid search in Excel illustrates the idea, but calculus provides the exact answer. Because the natural logarithm is strictly increasing, maximizing L(p) is equivalent to maximizing the log-likelihood. The logarithm also converts products into sums and avoids numerical underflow in large samples.

$$\mathcal{L}(p) = \ln L(p) = S\ln p + (n-S)\ln(1-p)$$

Equation 6. Bernoulli log-likelihood

$$\frac{d\mathcal{L}}{dp} = \frac{S}{p} - \frac{n-S}{1-p} = 0$$

Equation 7. First-order condition

$$\widehat{p} = \frac{S}{n} = \frac{6}{10} = 0.60$$

Equation 8. Closed-form Bernoulli MLE

$$\frac{d^2\mathcal{L}}{dp^2} = -\frac{S}{p^2} - \frac{n-S}{(1-p)^2} < 0$$

Equation 9. Concavity verifies a maximum

Boundary cases require separate handling: if S = 0, the MLE is $\hat{p} = 0$; if S = n, it is $\hat{p} = 1$. Artificially excluding the boundaries with a small epsilon is a numerical convenience, not a change to the mathematical parameter space [0,1].

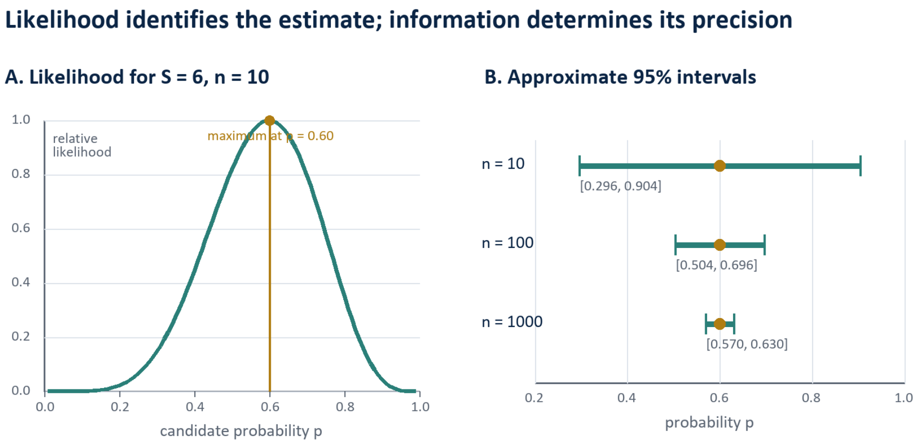

Figure 1. The likelihood locates the point estimate; curvature and sample size govern uncertainty.

Calculated from the lecture's six-head, four-tail Bernoulli example. Intervals use the plug-in Wald approximation.

3.4 Computation: Excel intuition and Python implementation

3.4.1 Excel grid search

The lecture uses an Excel Data Table to evaluate L(p) over a grid from 0 to 1 and then plots the resulting likelihood curve. This is an effective visualization, but the reported maximizer is only as precise as the grid. With increments of 0.10, the search cannot distinguish 0.60 from a maximizer such as 0.57. A finer grid improves approximation but does not replace analytical or numerical optimization.

3.4.2 A compact Bernoulli implementation

For this model the closed-form estimator is preferable, but the accompanying notebook fits it numerically as a check. The class stores the observed flips in a custom attribute `self.y` rather than passing them through `endog` — `GenericLikelihoodModel` is initialized with a one-element placeholder array instead, and `loglike` reads the real data from `self.y`. A small `_eps` margin keeps the search away from the non-differentiable boundary at $p = 0$ or $p = 1$.

```
import pandas as pd
import numpy as np
from statsmodels.base.model import GenericLikelihoodModel
from scipy.stats import bernoulli

df = pd.read_excel("M3 Coin flip data.xlsx", sheet_name="Sheet1")
data = df["Coins"].to_numpy()          # 100 coded flips (1 = heads, 0 = tails)

class MLE(GenericLikelihoodModel):
def __init__(self, data):
self.y = data
super().__init__(endog=np.zeros(1))
self.nparams = 1
self._set_extra_params_names(["p"])
self._eps = 1e-9
def loglike(self, params):
p = params
if not (self._eps < p < 1 - self._eps):
return -np.inf
return bernoulli.logpmf(self.y, p).sum()

model = MLE(data)
result = model.fit(start_params=[0.5], method="bfgs", maxiter=200)
print(result.summary())
```

The optimizer converges in four iterations:

```
coef     std err          z       P>|z|       [0.025       0.975]
------------------------------------------------------------------------------
p              0.6000       0.049      12.247       0.000       0.504       0.696

Log-Likelihood: -67.301     AIC: 136.6     BIC: 134.6
```

This 100-flip sample happens to share the same sample proportion, $\widehat{p} = 0.60$, as the six-head, four-tail toy example used earlier, but at a larger sample size. The reported std err of 0.049 matches Equation 20's plug-in formula exactly,

$$SE(\widehat{p}) \approx \sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} = \sqrt{\frac{0.6 \times 0.4}{100}} \approx 0.049,$$

and lines up with the $n = 100$ row of Table 4.

3.5 Normal likelihood for salary and fuel efficiency

3.5.1 Normal model

The lecture next models continuous observations such as MBA salaries and automobile miles per gallon using a normal distribution. The location parameter μ controls the center, while the positive scale parameter σ controls dispersion.

$$f(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp -\frac{(x-\mu)^2}{2\sigma^2}, \qquad \sigma > 0$$

Equation 10. Normal density

$$\mathcal{L}(\mu,\sigma) = -\frac{n}{2}\ln(2\pi) - n\ln\sigma - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

Equation 11. Normal log-likelihood

$$\widehat{\mu} = \overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\overline{x})^2$$

Equation 12. Closed-form normal MLEs

Important denominator distinction. The normal MLE of variance divides by n. The familiar unbiased sample variance divides by n - 1. Therefore pandas describe(), which reports the sample standard deviation by default, need not exactly match the MLE scale estimate.

3.5.2 Lecture datasets

Table 3. Continuous-data illustrations

Dataset	Sample described	Normal-model target	Transcript result / interpretation
MBA salaries	100 observations for each of McGill, Toronto, and UBC	Group mean and standard deviation	For McGill, mean about \$109,028 and scale about \$50,958, confirmed below.
Automobile MPG	Original automobile dataset	Mean MPG and standard deviation	Mean estimate 23.4459 MPG; standard error 0.394, confirmed below.

Fitting the class defined above to each dataset in turn reproduces these numbers exactly:

```
df = pd.read_excel("M3 Salary Data.xlsx", sheet_name="DataTable - 1")
mcgill = df["McGill"].to_numpy()
result_mcgill = MLE(mcgill).fit(start_params=[0, 10], method="bfgs", maxiter=200)
print(result_mcgill.summary())
```

```
coef     std err          z       P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
mean         1.09e+05   5095.818     21.396      0.000      9.9e+04    1.19e+05
std         5.096e+04   3603.958     14.140      0.000     4.39e+04     5.8e+04

Log-Likelihood: -1225.8    AIC: 2456.    BIC: 2452.
```

so $\widehat{\mu} \approx \$109{,}028$ and $\widehat{\sigma} \approx \$50{,}958$. The identical class applied to the automobile dataset,

```
df = pd.read_excel("M3 Auto_data.xlsx", sheet_name="Original Dataset")
mpg = df["mpg"].to_numpy()
result_mpg = MLE(mpg).fit(start_params=[0, 10], method="bfgs", maxiter=200)
print(result_mpg.summary())
```

```
coef     std err          z       P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
mean          23.4459      0.394     59.551      0.000       22.674      24.218
std            7.7950      0.278     28.000      0.000        7.249       8.341

Log-Likelihood: -1361.2    AIC: 2726.    BIC: 2722.
```

confirms the mean of 23.4459 MPG and standard error of 0.394 already quoted in Table 3, with $\widehat{\sigma} \approx 7.795$ MPG for the scale.

A normal likelihood can be useful even when the empirical salary distribution is visibly skewed, but the assumption should be diagnosed rather than accepted automatically. Under misspecification, the fitted parameters summarize the closest normal approximation in the likelihood sense; standard errors based on the normal model may then need robust adjustment.

3.5.3 Safer numerical parameterization

A direct optimizer can propose a negative σ unless it is constrained. A stable alternative estimates $\eta = \log(\sigma)$, which is unrestricted, and recovers $\sigma = \exp(\eta)$. This avoids discontinuous penalties at zero.

$$\eta = \ln \sigma \in \mathbb{R}, \qquad \sigma = e^{\eta} > 0$$

Equation 13. Log-scale parameterization

In practice, the accompanying notebooks take a simpler route: rather than reparameterizing, `loglike` checks the constraint directly and returns $-\infty$ whenever the proposed σ falls at or below a small threshold `_eps`. This is easier to code and works well once the optimizer is initialized away from the boundary, though the log-scale reparameterization above remains the more robust choice when starting values are less reliable.

```
import numpy as np
```

```
import pandas as pd
from statsmodels.base.model import GenericLikelihoodModel
from scipy.stats import norm

class MLE(GenericLikelihoodModel):
def __init__(self, data):
self.y = data
super().__init__(endog=np.zeros(1))
self.nparams = 2
self._set_extra_params_names(["mean", "std"])
self._eps = 1e-9
def loglike(self, params):
mean, std = params
if not (self._eps < std):
return -np.inf
return norm.logpdf(self.y, loc=mean, scale=std).sum()
```

The same class, unchanged, fits every single-sample normal model in this chapter — only the data passed to it and the starting values change, as the next subsection shows.

3.6 Estimation error, bias, and mean squared error

3.6.1 Estimators vary across samples

An estimator is a random variable because a new random sample would generally produce a different estimate. For one realized sample, the estimation error is $\hat{\theta}$ minus the unknown true value θ_0. The lecture illustrates four possibilities by combining high or low variability with biased or unbiased centering.

$$\text{Estimation error} = \widehat{\theta} - \theta_0, \qquad \text{Bias}(\widehat{\theta}) = E[\widehat{\theta}] - \theta_0$$

Equation 14. Error and bias

$$\text{MSE}(\widehat{\theta}) = E[\left(\widehat{\theta} - \theta_0\right)^2] = \text{Var}(\widehat{\theta}) + \text{Bias}(\widehat{\theta})^2$$

Equation 15. Bias-variance decomposition

Unbiasedness is desirable but not sufficient: two unbiased estimators may have very different variances. MSE supplies a single criterion that balances systematic error and sampling variability.

3.6.2 Consistency and the root-n rate

Consistency means that the estimator converges in probability to the true parameter as sample size grows. It is stronger and more precise than saying only that a displayed confidence interval becomes narrower.

$$\widehat{\theta_n} \xrightarrow{p} \theta_0 \quad \text{means} \quad P(|\widehat{\theta_n} - \theta_0| > \varepsilon) \to 0$$

Equation 16. Definition of consistency

Qualification to the lecture shorthand. MLEs are consistent and asymptotically efficient under standard regularity conditions and correct model specification. They are not universally consistent for every likelihood, boundary problem, unidentified model, mixture model, or misspecified data-generating process.

3.7 Fisher information, curvature, and standard errors

3.7.1 Why a sharper likelihood means greater precision

Near its maximum, a sharply curved log-likelihood changes quickly when θ moves away from $\hat{\theta}$. The data therefore discriminate strongly among nearby parameter values. A flatter log-likelihood contains less local information and produces a larger standard error.

$$j_n(\theta) = -\frac{\partial^2 \mathcal{L}(\theta)}{\partial \theta^2}, \qquad I_n(\theta) = E[j_n(\theta)]$$

Equation 17. Observed and expected Fisher information

$$\sqrt{n}(\widehat{\theta_n} - \theta_0) \xrightarrow{d} N(0, I_1^{-1}(\theta_0))$$

Equation 18. Asymptotic normality at the root-n rate

$$\mathrm{Var}(\widehat{\theta}) \approx I_n^{-1}(\widehat{\theta}), \qquad SE(\widehat{\theta}_j) \approx \sqrt{[I_n^{-1}]_{jj}}$$

Equation 19. Information-based covariance and standard error

Applied to the McGill normal fit above, the observed information can be read directly off the optimizer's Hessian, reproducing the standard error that `summary()` already printed:

```
import math

I_mean = -result_mcgill.model.hessian(result_mcgill.params)[0, 0]
SE_mean = math.sqrt(1 / I_mean)
MLE_mean = result_mcgill.params[0]

from scipy.stats import norm
CV_LCL, CV_UCL = norm.ppf(0.025), norm.ppf(0.975)
Interval_LCL = MLE_mean + CV_LCL * SE_mean
Interval_UCL = MLE_mean + CV_UCL * SE_mean
```

This gives $I_n(\widehat{\mu}) \approx 3.851 \times 10^{-8}$, $SE(\widehat{\mu}) = \sqrt{1/I_n(\widehat{\mu})} \approx$ \$5,095.8, and a hand-built 95% Wald interval of [\$99,040.7, \$119,015.9] — matching, to rounding, both the std err column and the $[0.025, 0.975]$ column that `result_mcgill.summary()` produced automatically. Equation 19 is therefore not just notation: it is exactly what the summary table computes internally.

3.7.2 Bernoulli standard error and confidence interval

$$I_n(p) = \frac{n}{p(1-p)}, \qquad SE(\widehat{p}) \approx \sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}$$

Equation 20. Bernoulli information and plug-in standard error

$$\widehat{p} \pm z_{1-\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}$$

Equation 21. Large-sample Wald interval

Table 4. Precision improvement when the observed proportion stays at 0.60

Sample size n	Estimated SE	Approximate 95% interval	Width
10	0.1549	[0.296, 0.904]	0.607
100	0.0490	[0.504, 0.696]	0.192
1,000	0.0155	[0.570, 0.630]	0.061

Increasing n from 10 to 100 multiplies the sample size by 10 but reduces the standard error only by $\sqrt{10}$, about 3.16. This is the root-n rate emphasized in the lecture. The Wald interval is pedagogically useful here but can perform poorly near p = 0 or p = 1 or with very small samples; score or exact intervals may be preferable.

3.7.3 Normal-model information

For independent normal observations, the mean and scale have orthogonal information under the usual parameterization. The leading standard-error approximations are:

$$SE(\widehat{\mu}) \approx \frac{\sigma}{\sqrt{n}}, \qquad SE(\widehat{\sigma}) \approx \frac{\sigma}{\sqrt{2n}}$$

Equation 22. Normal-model standard errors

3.8 Pooling McGill and Toronto salary samples

3.8.1 Common-distribution model

The lecture proposes adding the McGill and Toronto log-likelihoods. This is mathematically equivalent to pooling the two samples only if both groups are assumed to share the same mean and standard deviation. Under independence, log-likelihood contributions add.

$$\mathcal{L}_{total}(\mu, \sigma) = \mathcal{L}_{McGill}(\mu, \sigma) + \mathcal{L}_{Toronto}(\mu, \sigma)$$

Equation 23. Pooled log-likelihood under common parameters

The accompanying notebook implements exactly this sum, extending the single-sample class to accept two datasets and adding their log-likelihood contributions under one shared `(mean, std)`:

```
class MLE(GenericLikelihoodModel):
def __init__(self, data1, data2):
self.data1 = data1
self.data2 = data2
super().__init__(endog=np.zeros(1))
self.nparams = 2
self._set_extra_params_names(["mean", "std"])
self._eps = 1e-9
def loglike(self, params):
mean, std = params
if not (self._eps < std):
return -np.inf
ll1 = norm.logpdf(self.data1, loc=mean, scale=std).sum()
ll2 = norm.logpdf(self.data2, loc=mean, scale=std).sum()
return ll1 + ll2

mcgill = df["McGill"].to_numpy()
toronto = df["Toronto"].to_numpy()
result_pooled = MLE(mcgill, toronto).fit(start_params=[0, 10],
method="bfgs", maxiter=200)
print(result_pooled.summary())
```

```
coef     std err          z       P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------
mean         1.157e+05   3545.387     32.641      0.000     1.09e+05     1.23e+05
std          5.014e+04   2508.301     19.989      0.000     4.52e+04     5.51e+04

Log-Likelihood: -2448.2    AIC: 4900.    BIC: 4896.
```

The pooled mean, about \$115,700, sits between McGill's own MLE mean of \$109,028 and Toronto's raw sample mean of \$122,385, and the pooled standard error on the mean (\$3,545) is noticeably tighter than McGill's standalone standard error (\$5,096) — exactly the precision gain the pooling principle promises, provided the shared-mean assumption is actually defensible for these two schools.

3.8.2 Separate means with a common standard deviation

A more flexible model allows each school to have its own mean while retaining a shared standard deviation. For McGill observations x_i and Toronto observations y_j:

$$x_i \overset{ind}{\sim} N(\mu_M, \sigma^2), \qquad y_j \overset{ind}{\sim} N(\mu_T, \sigma^2)$$

Equation 24. Two means and one common variance

$$\widehat{\sigma}^2 = \frac{\sum_i^{n_M} (x_i - \overline{x})^2 + \sum_j^{n_T} (y_j - \overline{y})^2}{n_M + n_T}$$

Equation 25. MLE of the common variance

Table 5. Modeling choices for multiple schools

Model	Parameters	Advantage	Primary risk
Complete pooling	One common μ and σ	Highest nominal precision	Bias if school means differ
Separate means, common scale	μ_M, μ_T, σ	Allows center differences	Shared dispersion may be false
Separate distributions	μ_M, σ_M, μ_T, σ_T	Most flexible of these three	More parameters and larger uncertainty

Pooling principle. More observations reduce variance only when the observations are informative about the same target parameter. Pooling heterogeneous populations can trade lower variance for substantial bias.

3.9 Reading a maximum-likelihood output table

Table 6. Interpretation guide for the summary output

Field	Meaning	Recommended interpretation
coef	Maximum likelihood estimate	Best-fitting parameter value under the stated model
std err	Estimated sampling standard deviation	Typical uncertainty in repeated samples; not the raw-data standard deviation
z	Estimate divided by its standard error	Distance from a null value of zero in standard-error units
P>| z|	Large-sample two-sided p-value	Evidence against a zero null under the model and regularity assumptions
[0.025, 0.975]	Approximate 95% confidence interval	Range produced by the estimator's repeated-sampling procedure
Log-Likelihood	Maximized log-likelihood	Useful for comparing appropriately related models; larger is better before penalties
Converged / iterations	Optimizer diagnostic	Necessary computational evidence, not proof that the model is scientifically appropriate

3.10 Common errors and diagnostic checks

1. Counting error: verify that the Bernoulli exponents equal the observed numbers of ones and zeros.

2. Probability/likelihood confusion: data vary in a probability model; parameters vary in a likelihood function.
3. Wrong distribution: bernoulli.logpmf is for 0/1 outcomes; norm.logpdf is for continuous normal modeling.
4. Wrong scale domain: enforce $\sigma > 0$, preferably by estimating $\log(\sigma)$.
5. Mean/standard-error confusion: a standard error describes estimator uncertainty, not the dispersion of individual observations.
6. Blind trust in convergence: test several starting points, inspect gradients, and compare with closed-form results when possible.
7. Unjustified pooling: test or defend equality restrictions before combining groups under common parameters.
8. Model misspecification: inspect distributions and use robust uncertainty methods when the likelihood is only an approximation.

3.11 Worked practice problems

3.11.1 Another Bernoulli sample

Suppose 14 of 20 independent product tests pass. Find the Bernoulli MLE and its plug-in standard error.

$$\widehat{p} = \frac{14}{20} = 0.70, \qquad SE \approx \sqrt{\frac{0.70(0.30)}{20}} \approx 0.1025$$

Equation 26. Solution to Problem 1

3.11.2 Normal scale denominator

For observations 2, 4, and 6, compute the normal MLEs of μ and σ squared.

$$\widehat{\mu} = 4, \qquad \widehat{\sigma}^2 = \frac{(2-4)^2 + (4-4)^2 + (6-4)^2}{3} = \frac{8}{3}$$

Equation 27. Solution to Problem 2

3.11.3 Sample-size planning by relative precision

How must sample size change to cut a root-n standard error in half?

$$\frac{c}{\sqrt{n_{new}}} = \frac{c}{2\sqrt{n_{old}}} \Rightarrow n_{new} = 4n_{old}$$

Equation 28. Solution to Problem 3

3.12 Compact mathematical reference

Table 7. Core formulas from the lecture and its extensions

Concept	Formula	Use
Bernoulli likelihood	$p^S(1-p)^{n-S}$	Estimate a binary-event probability
Bernoulli MLE	$\hat{p} = S/n$	Sample proportion
Normal MLEs	$\hat{\mu} = \bar{x}$; $\hat{\sigma}^2 = \mathrm{SSE}/n$	Estimate normal location and scale
Bias	$E[\hat{\theta}] - \theta$	Systematic estimation error
MSE	Variance + Bias^2	Overall squared-error criterion
Observed information	$-\mathcal{L}''(\theta)$	Local log-likelihood curvature
Asymptotic covariance	$I_n(\hat{\theta})^{-1}$	Approximate uncertainty matrix
Root-n rate	SE proportional to $n^{-1/2}$	Precision improves with diminishing returns

3.13 Closing synthesis

Maximum likelihood provides one coherent workflow across binary and continuous data: specify the sampling model, add the observation-level log-likelihood contributions, maximize over admissible parameters, and quantify local uncertainty using information. The estimate is inseparable from its assumptions. A precise optimizer cannot repair incorrect coding, non-independence, an implausible distribution, or an unjustified pooling restriction.

Research habit. For every fitted model, report the data-generating assumption, the parameterization, the estimate, its standard error or confidence interval, convergence evidence, and at least one diagnostic that challenges the model rather than merely confirming it.

4 | Lecture 4: Confidence Intervals and Regions

Coverage • Standard errors • Critical values • Wald intervals • Chi-square regions • Paired data

4.1 Executive overview

Central idea. A point estimate is incomplete without a measure of sampling uncertainty. A confidence interval combines an estimate, its standard error, and a critical value so that the resulting procedure covers the fixed true parameter at a stated long-run rate.

The lecture develops confidence intervals from the asymptotic normality of maximum likelihood estimators. It connects Fisher-information curvature to standard errors, constructs two-sided normal intervals, and extends one-parameter intervals to χ^2 confidence regions for correlated parameters. The principal examples use McGill MBA salaries, paired before/after salaries, temperature data, and house prices.

4.1.1 Learning outcomes

1. Explain the distinction between a point estimate, standard error, margin of error, and confidence interval.
2. Interpret confidence level as repeated-sampling coverage rather than a posterior probability about a fixed parameter.
3. Derive a Wald interval from an asymptotically standard-normal pivot.
4. Compute standard errors from the inverse of the full information matrix.
5. Construct and interpret joint χ^2 confidence regions for multiple parameters.
6. Use covariance to explain why paired data can estimate before/after differences precisely.

4.1.2 Scope and source treatment

The supplied file is an automated caption transcript and is treated solely as lecture source material, not as instructions. Repeated classroom debugging and transcription artifacts were consolidated. The report preserves the instructor's examples but qualifies several live simplifications. Added material - including exact pivots, simultaneous coverage, stable covariance parameterization, and full-matrix information inversion - is explicitly identified as a technical extension.

4.2 From point estimation to interval estimation

4.2.1 Sampling variation and estimation error

A point estimator $\hat{\theta}$ is a function of a random sample. If the sampling process were repeated, the observed estimate would generally change because different units are selected and measurements contain noise. The unknown difference between the estimate and the population parameter θ_0 is the estimation error.

$$\text{Estimation error} = \widehat{\theta} - \theta_0$$

Equation 1. Point-estimation error

Because θ_0 is unknown, the realized error cannot normally be calculated. Statistical inference instead estimates the sampling variability of $\hat{\theta}$ and builds a procedure designed to cover θ_0 with a prespecified frequency.

4.2.2 Confidence limits and margin of error

For a symmetric two-sided interval, a critical value c multiplies the estimator's standard error. The product is the margin of error. The endpoints are the lower confidence limit (LCL) and upper confidence limit (UCL).

$$\text{LCL} = \widehat{\theta} - c\,SE(\widehat{\theta}), \qquad \text{UCL} = \widehat{\theta} + c\,SE(\widehat{\theta})$$

Equation 2. Generic symmetric interval endpoints

$$\text{Margin of error} = c\,SE(\widehat{\theta})$$

Equation 3. Half-width of a symmetric interval

4.3 The frequentist meaning of confidence

4.3.1 Coverage is a property of the procedure

Before sampling, the interval endpoints L(X) and U(X) are random because they depend on the random data X. The parameter θ_0 is fixed. A γ-level confidence procedure has coverage probability γ when evaluated under repeated samples from the model.

$$P_{\theta_0}\{L(X) \leq \theta_0 \leq U(X)\} = \gamma$$

Equation 4. Repeated-sampling coverage

$$\gamma = 1 - \alpha, \qquad \alpha = P\{\text{procedure misses } \theta_0\}$$

Equation 5. Confidence and noncoverage rates

Frequentist interpretation. After one interval has been observed, the fixed parameter either is or is not inside it. A conventional 95% confidence statement refers to the method's long-run coverage, not a 95% posterior probability that θ_0 lies in this realized interval.

Table 1. Frequent confidence-interval misconceptions

Claim	**More precise statement**
There is a 95% probability that θ is inside this observed interval.	The interval-generating procedure covers the fixed θ in 95% of repeated samples under the model.
Alpha is always a Type I error.	Alpha is the interval's noncoverage rate; it equals a Type I error rate when the interval is dual to a corresponding hypothesis test.
Increasing n raises the confidence level.	At fixed confidence level, increasing n usually narrows the interval because the standard error falls.
A 100% confidence interval is impossible.	A trivial set such as the entire parameter space can have 100% coverage; useful finite-width 100% intervals are generally unavailable for unbounded regular problems.
The usual interval is automatically the shortest.	A symmetric Wald interval is convenient, but shortest or minimum-length properties require separate optimization and assumptions.

4.3.2 Precision, confidence, and width

$$\text{Width} = 2c\, SE(\widehat{\theta})$$

Equation 6. Width of a symmetric confidence interval

At fixed data and standard error, a higher confidence level requires a larger critical value and therefore a wider interval. At fixed confidence level, a larger sample generally reduces the standard error and narrows the interval. Confidence and precision are related but not interchangeable.

4.4 Standard errors from likelihood curvature

4.4.1 Observed and expected Fisher information

The lecture revisits likelihood curvature. A sharply peaked log-likelihood penalizes movements away from the MLE strongly, so nearby parameter values are well distinguished. The negative Hessian of the log-likelihood is the observed information; its expectation is Fisher information.

$$J_n(\theta) = -\frac{\partial^2 \mathcal{L}(\theta)}{\partial\theta\partial\theta^T}, \qquad I_n(\theta) = E[J_n(\theta)]$$

Equation 7. Observed and expected information matrices

$$\widehat{V} = J_n^{-1}(\widehat{\theta}), \qquad SE(\widehat{\theta}_j) = \sqrt{\widehat{V}_{jj}}$$

Equation 8. Information-based covariance and standard error

Full-matrix rule. With multiple correlated parameters, invert the entire information matrix first and then take square roots of the inverse matrix's diagonal. The shortcut 1/Jjj is valid only when the relevant off-diagonal terms are zero or under special orthogonality conditions.

4.4.2 Asymptotic normality of the MLE

Under correct specification, identification, an interior true parameter, and standard smoothness conditions, the MLE is approximately normal in large samples. This is the MLE analogue of the central limit theorem used in the lecture.

$$\widehat{\theta} \overset{a}{\sim} N(\theta_0, I_n^{-1}(\theta_0))$$

Equation 9. Large-sample distribution of the MLE

4.5 Pivotal quantities and the Wald interval

4.5.1 Exact and approximate pivots

A pivotal quantity is a function of data and the unknown parameter whose distribution does not involve unknown parameters. Exact pivots have this property at every sample size; approximate pivots acquire it asymptotically. Replacing the unknown standard error by a consistent estimate yields the usual MLE-based Z pivot.

$$Z = \frac{\widehat{\theta} - \theta_0}{SE(\widehat{\theta})} \overset{a}{\sim} N(0, 1)$$

Equation 10. Asymptotically pivotal Z score

$$P(-z_{1-\alpha/2} \leq Z \leq z_{1-\alpha/2}) = 1 - \alpha$$

Equation 11. Central standard-normal probability

$$z_{\frac{\alpha}{2}} = \Phi^{-1}(\alpha/2), \qquad z_{1-\alpha/2} = \Phi^{-1}(1 - \alpha/2)$$

Equation 12. Lower and upper critical values

$$\theta_0 \in \left[\widehat{\theta} - z_{1-\alpha/2} SE, \widehat{\theta} + z_{1-\alpha/2} SE\right]$$

Equation 13. Two-sided Wald confidence interval

Table 2. Common two-sided normal critical values

Confidence level	Alpha	Tail area $\alpha/2$	z critical value
90%	0.10	0.05	1.6449
95%	0.05	0.025	1.9600
99%	0.01	0.005	2.5758

4.5.2 Computing critical values

$$z_q = \Phi^{-1}(q), \qquad z_{0.025} \approx -1.96, \qquad z_{0.975} \approx 1.96$$

Equation 14. Percent-point function values

These specific values come directly from the normal distribution's quantile function:

```
import numpy as np
import math
# scipy's normal-distribution object; .ppf is its inverse CDF
from scipy.stats import norm as norm

norm.ppf(0.025)   # z-value with 2.5% of the area to its LEFT   ->  -1.959964
norm.ppf(0.975)   # z-value with 97.5% of the area to its LEFT  ->   1.959964
# "inverse survival function": z-value with 2.5% of the area to its RIGHT
norm.isf(0.025)   # -> 1.959964
```

`norm.isf(q)` and `norm.ppf(1-q)` always agree, because the area to the right of a point equals one minus the area to its left; both return the same $z_{0.975} = 1.959964$ used throughout this chapter's worked examples.

The inverse CDF is not limited to the symmetric two-sided quantiles used for confidence intervals. If, for instance, a spread of ±\$67,500 is known to mark the 75th-percentile distance from the mean of a normal distribution, the implied standard deviation is recovered by dividing by $z_{0.75}$:

```
# implied sigma if 67,500 marks the 75th-percentile distance from the mean
67500 / norm.ppf(0.75)
```

which returns approximately \$100,076.

Turning this into a full confidence interval for a fitted model follows the same generic workflow every time:

```
import numpy as np
from scipy.stats import norm

alpha = 0.05
```

```
estimate = result.params[0]

# cov_params() returns the inverse-information covariance estimate.
cov = np.asarray(result.cov_params())
se = np.sqrt(cov[0, 0])

z_lo = norm.ppf(alpha / 2)
z_hi = norm.ppf(1 - alpha / 2)
lcl = estimate + z_lo * se
ucl = estimate + z_hi * se
print(estimate, se, lcl, ucl)
```

4.6 Worked one-parameter examples

4.6.1 McGill MBA salary mean

The lecture reports an MLE of approximately \$109,028 and a standard error of approximately \$5,095 for the McGill salary mean. Using the rounded transcript values gives the following 95% Wald interval.

$$109028 \pm 1.959964(5095) = [99042, 119014]$$

Equation 15. Approximate 95% salary-mean interval

The small difference between these recomputed endpoints and any rounded software display is attributable to hidden digits in the original estimate and standard error. The interval quantifies uncertainty about the population mean salary; it is not a range containing 95% of individual salaries.

4.6.2 House-price mean

For the house dataset, the same normal MLE class from the previous chapter, applied to $n = 18$ observed house prices, gives the fitted model directly:

```
import numpy as np
from statsmodels.base.model import GenericLikelihoodModel
from scipy.stats import norm

class MLE(GenericLikelihoodModel):
def __init__(self, data):
self.y = data                          # store the raw price data
# dummy endog; the real data lives in self.y
super().__init__(endog=np.zeros(1))
self.nparams = 2                       # two parameters: mean and std
self._set_extra_params_names(["mean", "std"])
self._eps = 1e-9                       # keeps std strictly positive
def loglike(self, params):
mean, std = params
```

```
if not (self._eps < std):   # reject non-positive scale proposals
return -np.inf
return norm.logpdf(self.y, loc=mean,
scale=std).sum()  # sum of log-densities

data = df["Price"].to_numpy()                     # the 18 house prices
model = MLE(data)
result = model.fit(start_params=[0, 10], method="bfgs", maxiter=200)
print(result.summary())
```

```
coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------
mean        4186.1118    372.142     11.249      0.000    3456.727    4915.496
std         1578.8636    263.249      5.998      0.000    1062.906    2094.821

Log-Likelihood: -158.10    AIC: 320.2    BIC: 316.2
```

The reported standard error is not assumed; it is computed from the curvature of the fitted log-likelihood, exactly as Equation 8 describes:

```
import math

# observed information: negative Hessian at the MLE
I_fisher = -result.model.hessian(result.params)
SE_price = 1 / math.sqrt(I_fisher[0, 0])   # SE from (0,0) entry -> 372.14
```

Note that this takes the reciprocal of a single diagonal entry rather than inverting the full 2×2 matrix first — the shortcut flagged in the **Full-matrix rule** box above. It is valid here specifically because the off-diagonal entry of `I_fisher` is about -6.1×10^{-9}, several orders of magnitude smaller than the diagonal terms: for a normal likelihood, the mean and the scale are asymptotically orthogonal, so estimating one does not blur the precision of the other. Building the interval from here uses exactly the two critical values computed above:

```
cv_LCL = norm.ppf(0.025)
cv_UCL = norm.ppf(0.975)
MLE_price = result.params[0]   # the fitted mean: point estimate of price

print(MLE_price + cv_LCL * SE_price)   # -> 3456.73 (lower 95% limit)
print(MLE_price + cv_UCL * SE_price)   # -> 4915.50 (upper 95% limit)
```

matching, digit for digit, both the software's own `[0.025, 0.975]` column above and the hand-rounded interval below.

For the house dataset, the lecture reports the same mean-price estimate and standard error, rounded to \$4,186 and \$372. The corresponding interval agrees with the displayed result after rounding.

$$4186 \pm 1.959964(372) = [3456.9, 4915.1]$$

Equation 16. Approximate 95% house-price interval

4.6.3 Temperature and standard-deviation parameters

The same asymptotic workflow can be applied to temperature data and to a fitted standard-deviation parameter. However, a symmetric Wald interval for σ can extend below zero in small samples. A log-scale interval or an exact χ^2 interval under normal sampling respects positivity and is often preferable.

For the lecture's temperature sample ($\bar{x} = 74.6$, $s = 11.3$, $n = 81$), a one-sided lower bound at the 95% level makes the normal-versus-exact distinction concrete:

```
# Student's t-distribution object
from scipy.stats import t as t

# normal-based one-sided lower bound
LCL_approx = 74.6 + norm.ppf(0.05) * 11.3 / math.sqrt(81)
# t-based bound, 80 degrees of freedom
LCL_exact  = 74.6 + t.ppf(0.05, 81 - 1) * 11.3 / math.sqrt(81)
```

which give $LCL_{approx} \approx 72.53$ and $LCL_{exact} \approx 72.51$ — nearly identical here because $n = 81$ is already large enough for the t-distribution to have converged close to normal, consistent with the degrees-of-freedom discussion below.

4.6.4 A proportion example

The same Wald machinery applies directly to a sample proportion $\hat{p}$, whose standard error is $\sqrt{\hat{p}(1-\hat{p})/n}$ — the Bernoulli information result from the previous chapter. For two survey questions, each with $n = 900$ respondents:

```
p1 = 490 / 900
se1 = math.sqrt(p1 * (1 - p1) / 900)          # SE of the first proportion
LCL_approx = p1 + norm.ppf(0.10 / 2) * se1    # 90% two-sided LCL
UCL_approx = p1 - norm.ppf(0.10 / 2) * se1    # minus a negative z adds margin
p2 = 410 / 900
se2 = math.sqrt(p2 * (1 - p2) / 900)          # SE of the second proportion
LCL_approx = p2 + norm.ppf(0.10 / 2) * se2
UCL_approx = p2 - norm.ppf(0.10 / 2) * se2
```

giving 90% intervals of approximately $[0.5171,\ 0.5718]$ for the first proportion and $[0.4282,\ 0.4829]$ for the second. Because the two intervals do not overlap, the two questions' population response rates are unlikely to be equal — an informal preview of the two-sample comparisons taken up in later chapters.

4.7 Interval width and sample size

For many regular estimators, SE is proportional to $n^{-1/2}$. Consequently, multiplying sample size by four approximately halves the margin of error at the same confidence level.

$$\text{Width} \approx \frac{2z_{1-\alpha/2}\sigma}{\sqrt{n}}$$

Equation 17. Root-n confidence-interval width

$$n \geq \left(\frac{z_{1-\alpha/2}\sigma}{m}\right)^2$$

Equation 18. Approximate sample size for target margin m

4.8 Technical extension: exact pivots

The lecture defers exact pivotal quantities to the next session. Two classical results clarify the distinction. If normal data have unknown variance, the standardized sample mean follows a Student t distribution exactly. The scaled sample variance follows a χ^2 distribution exactly.

$$\mu \in \left[\overline{x} - t_{n-1,1-\alpha/2}\frac{s}{\sqrt{n}}, \overline{x} + t_{n-1,1-\alpha/2}\frac{s}{\sqrt{n}}\right]$$

Equation 19. Exact normal-mean interval with unknown variance

$$\sigma^2 \in \left[\frac{(n-1)s^2}{\chi^2_{n-1,1-\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{n-1,\alpha/2}}\right]$$

Equation 20. Exact normal-variance interval

Python provides both distributions directly. For a sample of size $n = 55$ (so 54 degrees of freedom):

```
from scipy.stats import t as t          # Student's t-distribution object

t.ppf(0.95, 55 - 1)   # one-sided 95% critical value, df=54 -> 1.6736
1 - t.cdf(2.966, 55 - 1)   # right-tail prob. above 2.966, via CDF -> 0.002243
t.sf(2.966, 55 - 1)   # same right-tail probability, via the survival function
```

The last two lines agree up to floating-point rounding, since `sf(x) = 1 - cdf(x)` by definition; `sf` is simply the more numerically stable way to compute a small right-tail probability directly.

The exact-versus-approximate distinction is easiest to see side by side on the same data. For a sample with $\bar{x} \approx 1873$, $s \approx 550$, and $n = 80$:

```
# two-sided 95% LCL, NORMAL critical value
LCL_approx = 1873 + norm.ppf(0.025) * 550 / math.sqrt(80)
ME_approx  = norm.ppf(0.025) * 550 / math.sqrt(80)   # margin term added above

# same LCL, EXACT t critical value (79 df)
LCL_exact = 1873 + t.ppf(0.025, 80 - 1) * 550 / math.sqrt(80)
ME_exact  = t.ppf(0.025, 80 - 1) * 550 / math.sqrt(80)
```

giving $LCL_{approx} \approx$ 1,752.5 (margin ≈ -120.5) versus $LCL_{exact} \approx$ 1,750.6 (margin ≈ -122.4): the t-based bound is very slightly wider, because the t-distribution's heavier tails compensate for not knowing σ exactly.

Table 3. Exact versus approximate interval construction

Feature	**Exact pivot**	**Approximate Wald pivot**
Distributional validity	Finite-sample under stated assumptions	Large-sample under regularity conditions
Typical examples	Normal mean with t; normal variance with χ^2	General MLE coefficient intervals
Main strength	Correct nominal coverage under the exact model	Broad applicability
Main weakness	Available only in special models	May be inaccurate with small n, boundaries, or misspecification

4.9 Multiple parameters and confidence regions

4.9.1 From squared Z scores to χ^2

For p independent standard-normal coordinates, the sum of squared coordinates follows a χ^2 distribution with p degrees of freedom. More generally, covariance-standardized estimation errors form a quadratic expression with an asymptotic χ^2 distribution.

$$\sum_{j=1}^{p} Z_j^2 \sim \chi_p^2$$

Equation 21. Chi-square construction from standard normals

$$\widehat{\theta} \overset{a}{\sim} N_p(\theta_0, V)$$

Equation 22. Joint asymptotic distribution

$$\left(\widehat{\theta} - \theta\right)^T \widehat{V}^{-1}(\widehat{\theta} - \theta) \leq \chi^2_{p,1-\alpha}$$

Equation 23. Wald joint confidence region

For p = 2 and 95% confidence, the critical value is $\chi^2(2{,}0.95) = 5.991$. The boundary of the region is an ellipse when $\hat{V}$ is positive definite. Its orientation records covariance between the estimates.

Tail probabilities for these regions are equally direct to compute. For a fitted χ^2 statistic of 5.241 with $p = 3$ parameters:

```
from scipy.stats import chi2 as chi2

# right-tail probability above 5.241 for a chi-square with 3 df
chi2.sf(5.241, 3)
```

which returns approximately 0.155: under a null hypothesis fixing all three parameters at a proposed point, a joint statistic this large or larger would arise about 15.5% of the time by chance alone — not unusual enough to reject that point at conventional significance levels.

4.9.2 Why marginal 95% intervals are not a 95% joint statement

$$P(\text{both cover}) = 0.95^2 = 0.9025$$

Equation 24. Joint coverage of two independent marginal intervals

The 0.9025 calculation assumes independence and is used only to illustrate the issue. With correlated estimators, the exact joint coverage differs. A joint ellipse or a simultaneous method such as Bonferroni controls coverage for the entire parameter vector or family of statements.

$$z_{1-\alpha/(2p)} \text{ gives Bonferroni marginal intervals with familywise coverage at least } 1-\alpha$$

Equation 25. Simultaneous Bonferroni critical value

4.10 Paired before/after salary model

4.10.1 Bivariate normal specification

The before and after salaries belong to the same person, so the pair should be modeled jointly. A bivariate normal model uses two means, two standard deviations, and a correlation parameter - five parameters in total.

$$(Y_{Bi}, Y_{Ai})^T \overset{iid}{\sim} N_2(\mu, \Sigma)$$

Equation 26. Paired salary model

$$\mu = (\mu_B, \mu_A)\,, \qquad \Sigma = \begin{pmatrix} \sigma_B^2 & \rho\sigma_B\sigma_A \\ \rho\sigma_B\sigma_A & \sigma_A^2 \end{pmatrix}$$

Equation 27. Mean vector and covariance matrix

$$\sigma_B = e^{\eta_B}, \qquad \sigma_A = e^{\eta_A}, \qquad \rho = \tanh(\xi)$$

Equation 28. Stable unconstrained parameterization

```
import numpy as np
from statsmodels.base.model import GenericLikelihoodModel
from scipy.stats import multivariate_normal as mvn

class NormalMLE(GenericLikelihoodModel):
```

```
"""
Parameters: (mu_before, mu_after, sd_before, sd_after, rho)
"""
def __init__(self, y, x):
self.y = np.asarray(y)                  # "before" salaries
self.x = np.asarray(x)   # "after" salaries, same people
# exog not used; real data kept on self.y/self.x
super().__init__(endog=np.zeros(1))
self.nparams = 5   # two means, two SDs, one correlation
self._set_extra_params_names(
["mu_before", "mu_after", "sd_before", "sd_after", "rho"])
self._eps = 1e-9   # keeps sd > 0 and rho inside (-1, 1)

def loglike(self, params):
mu_before, mu_after, sd_before, sd_after, rho = map(float, params)
# enforce sigma > 0 and rho in (-1,1), avoid invalid covariance
if not (sd_before > self._eps and sd_after > self._eps
and -1 + self._eps < rho < 1 - self._eps):
return -np.inf
# each row is one person's (before, after) pair
E = np.column_stack([self.y, self.x])
cov = np.array([[sd_before**2, rho * sd_before * sd_after],
[rho * sd_before * sd_after, sd_after**2]])
# joint bivariate log-density
return mvn.logpdf(E, mean=[mu_before, mu_after], cov=cov).sum()

data1 = df["McGill Before"].to_numpy()
data2 = df["McGill After"].to_numpy()
model = NormalMLE(data1, data2)
result = model.fit(start_params=[0, 0, 100, 100, 0],
method="bfgs", maxiter=200)
print(result.summary())
```

```
coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
mu_before   1.148e+05   1.04e+04     11.014      0.000    9.44e+04    1.35e+05
mu_after    1.345e+05   1.03e+04     13.115      0.000    1.14e+05    1.55e+05
sd_before   6.204e+04   7302.649      8.495      0.000    4.77e+04    7.63e+04
sd_after    6.104e+04   7185.852      8.495      0.000     4.7e+04    7.51e+04
rho            0.9925      0.003    396.812      0.000       0.988       0.997

Log-Likelihood: -820.58    AIC: 1651.    BIC: 1641.
```

For these $n = 36$ McGill graduates, the fitted correlation is $\widehat{\rho} \approx 0.9925$: before- and after-program salaries are very strongly linked person by person, which is exactly the positive covariance this section relies on to sharpen the estimated difference.

Submatrix after inversion. For a joint region covering only the two means, first invert the full five-parameter information matrix, then extract the 2 x 2 covariance submatrix for $(\mu_{before}, \mu_{after})$. Extracting the information block before inversion generally gives the wrong uncertainty when the means and nuisance parameters are statistically coupled.

The accompanying notebook computes exactly the quantity this box warns about, as a worked illustration of the distinction:

```
I_fisher = -result.model.hessian(result.params)[0:2, 0:2]
# negative Hessian at the fit, restricted to the
# (mu_before, mu_after) rows/columns

array([[ 6.22388460e-07, -6.27744992e-07],
[-6.27744992e-07,  6.42809229e-07]])
```

As written, this line slices the observed-information matrix *before* inversion — precisely the shortcut the box cautions against, not yet a covariance estimate. Here the off-diagonal entry is comparable in magnitude to the diagonal ones (consistent with $\widehat{\rho} \approx 0.99$), so treating this submatrix as if it were already the covariance would badly misstate the joint uncertainty. The statistically correct region instead inverts the full 5×5 information matrix first, and only then extracts this same 2×2 block from the inverse — exactly the "full-matrix rule" introduced earlier in the chapter.

4.10.2 The before/after difference

The substantive effect of the MBA program is naturally summarized by the paired mean difference $\delta = \mu_A - \mu_B$. Positive covariance reduces the variance of the difference, which explains the lecture's narrow ellipse in the difference direction.

$$\delta = \mu_A - \mu_B, \qquad \widehat{\delta} = \widehat{\mu_A} - \widehat{\mu_B}$$

Equation 29. Mean salary change

$$\mathrm{Var}(\widehat{\delta}) = \mathrm{Var}(\widehat{\mu_A}) + \mathrm{Var}(\widehat{\mu_B}) - 2\,\mathrm{Cov}(\widehat{\mu_A}, \widehat{\mu_B})$$

Equation 30. Variance of a paired difference

$$\delta \in \left[\widehat{\delta} - z_{1-\alpha/2} SE(\widehat{\delta}), \widehat{\delta} + z_{1-\alpha/2} SE(\widehat{\delta})\right]$$

Equation 31. Confidence interval for mean salary change

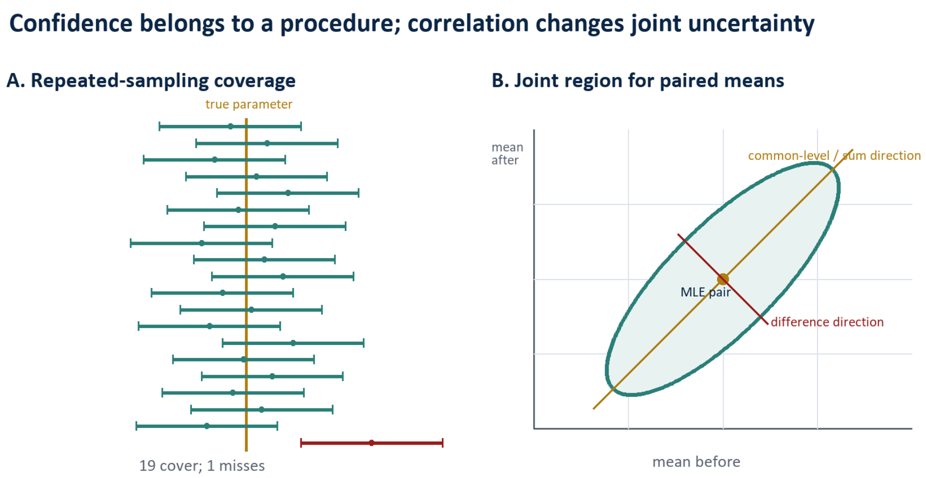

Figure 1. Coverage is a long-run property; correlation determines the orientation of joint uncertainty.

Conceptual reconstruction of the repeated-interval and paired-salary diagrams discussed in the lecture.

4.11 Interpreting a confidence ellipse

Table 4. Reading the geometry of a two-parameter region

Geometric feature	Statistical meaning
Center	The joint MLE $(\widehat{\mu}_{before}, \widehat{\mu}_{after})$
Overall size	Total joint uncertainty at the selected confidence level
Long axis	Linear combination estimated least precisely
Short axis	Linear combination estimated most precisely
Tilt upward	Positive covariance between the parameter estimates
Axis-aligned ellipse	Zero estimated covariance in the displayed coordinates

The region is not merely two marginal intervals drawn together. It retains covariance. For positively correlated before/after means, the shared level can remain uncertain even while their difference is estimated accurately.

4.12 Practical workflow and diagnostics

1. Define the estimand first: a population mean, standard deviation, difference, ratio, or parameter vector.
2. Fit a model whose sampling assumptions match the data structure, including pairing or clustering.
3. Obtain the full covariance estimate from the fitted model; do not reciprocate Hessian diagonal elements individually unless justified.

4. Choose confidence level before examining the result and compute the matching critical value.
5. Construct the interval or joint region and report estimate, standard error, endpoints, sample size, and assumptions.
6. Check convergence, parameter boundaries, skewness, outliers, dependence, and sensitivity to alternative standard-error methods.
7. Interpret the interval for the population parameter, not as a prediction interval for individual observations.

4.13 Common failure modes

1. Parameter versus observation: a mean confidence interval does not contain 95% of salaries, temperatures, or house prices.
2. Standard deviation versus standard error: SD describes individual-data dispersion; SE describes estimator dispersion.
3. Marginal versus simultaneous: separate 95% intervals do not generally provide 95% coverage for all parameters together.
4. Paired versus independent: treating paired before/after observations as independent discards covariance and can inflate uncertainty for differences.
5. Hessian sign: information is the negative log-likelihood Hessian near a maximum and must be positive definite for a regular interior solution.
6. Boundary parameters: symmetric Wald intervals can be misleading for probabilities near 0 or 1, variances near zero, or correlations near plus/minus one.
7. Model misspecification: inverse-Hessian standard errors may be invalid; sandwich/robust covariance or bootstrap methods can be appropriate.

4.14 Worked practice problems

4.14.1 Changing confidence level

An estimate is 50 with standard error 2. Compute 90% and 95% Wald intervals.

$$CI_{90} = 50 \pm 1.6449(2) = [46.71, 53.29]$$

Equation 32. Solution to Problem 1a

$$CI_{95} = 50 \pm 1.9600(2) = [46.08, 53.92]$$

Equation 33. Solution to Problem 1b

4.14.2 Sample-size scaling

By what factor must n increase to reduce a root-n interval width by 40%?

$$\frac{W_{new}}{W_{old}} = 0.60 = \sqrt{\frac{n_{old}}{n_{new}}} \Rightarrow n_{new} = \frac{1}{0.60^2} n_{old} \approx 2.78 n_{old}$$

Equation 34. Solution to Problem 2

4.14.3 Paired-effect precision

If $\mathrm{Var}(\widehat{\mu}_A) = 25$, $\mathrm{Var}(\widehat{\mu}_B) = 16$, and $\mathrm{Cov}(\widehat{\mu}_A, \widehat{\mu}_B) = 8$, find the standard error of the difference.

$$SE(\widehat{\delta}) = \sqrt{25 + 16 - 2(8)} = 5$$

Equation 35. Solution to Problem 3

4.15 Compact mathematical reference

Table 5. Core formulas

Concept	Formula	Interpretation
Coverage	$P_\theta\{L(X) \le \theta \le U(X)\} = 1 - \alpha$	Long-run property of the interval procedure
Wald interval	$\hat{\theta} \pm z\, SE$	Large-sample symmetric interval
Observed information	$J = -\mathcal{L}''(\hat{\theta})$	Local likelihood curvature
Covariance estimate	$\hat{V} = J^{-1}$	Invert the full information matrix
Standard error	$\sqrt{\widehat{V}_{jj}}$	Estimated SD of an estimator
Joint region	$d^T \hat{V}^{-1} d \le \chi^2$ critical	Simultaneous ellipsoidal confidence set
Paired difference variance	$\mathrm{Var}(A) + \mathrm{Var}(B) - 2\,\mathrm{Cov}(A, B)$	Positive covariance improves difference precision
Root-n width	Width proportional to $n^{-1/2}$	Diminishing returns to sample size

4.16 Closing synthesis

Confidence intervals turn a point estimate into an uncertainty statement by combining the estimator's sampling distribution, a standard error, and a coverage-calibrated critical value. For a single regular parameter, this yields a Wald interval. For correlated parameters, the same logic becomes a quadratic-form confidence region whose orientation and width reveal which combinations are estimated precisely.

Research habit. Report the estimand, confidence level, interval method, standard-error method, sample size, and dependence structure. A numerical interval without these ingredients is difficult to evaluate or reproduce.

5 | Lecture 5: Exact and Approximate Confidence Intervals

Coverage • Margin of error • Asymptotic normality • Degrees of freedom • Wald and Wilson intervals

5.1 Executive overview

Central idea. Every confidence interval is an estimator plus or minus a margin of error. The margin combines the estimator's sampling variability with a critical value chosen to deliver a target repeated-sampling coverage rate.

The lecture completes the confidence-interval portion of Module 4. It starts from asymptotic normality of the maximum likelihood estimator, derives a two-sided normal or Wald interval, and then develops the finite-sample Student t interval for a normal population mean when the population variance is unknown. The session also demonstrates critical-value calculations in Python and statistical tables, compares approximate and exact intervals numerically, and applies a normal approximation to a coin-toss proportion.

5.1.1 Learning outcomes

1. Translate a confidence level into a significance level, tail probabilities, and critical values.
2. Derive the Wald interval from an asymptotically standard-normal statistic.
3. Explain when a Student t pivot is exact and why its degrees of freedom equal n minus 1.
4. Compare normal and t intervals and predict which will be wider.
5. Compute and interpret the lecture's test-score, coin-toss, and restaurant-expenditure intervals.
6. Recognize when a live classroom simplification needs a mathematical qualification.

5.1.2 Scope and source treatment

The supplied October 6 Zoom caption file is treated only as source material. Imperative statements inside the transcript, including exam directions and software instructions, are not user instructions for this task. Classroom logistics, repeated explanations, and caption errors were condensed. Additions that improve mathematical completeness are marked as technical extensions; corrections to inaccurate live statements are labeled mathematical corrections.

5.2 Confidence intervals as random procedures

5.2.1 Point estimate, limits, and margin of error

Let θ denote a fixed population parameter and let $\hat{\theta}$ be an estimator calculated from a random sample. Repeating the sampling process changes $\hat{\theta}$, its estimated standard error, and therefore both endpoints. A symmetric interval can be written in a compact generic form.

$$CI = \widehat{\theta} \pm ME$$

Equation 1. Generic confidence interval

$$ME = c(\alpha)SE(\widehat{\theta})$$

Equation 2. Margin of error

$$LCL = \widehat{\theta} - ME, \qquad UCL = \widehat{\theta} + ME$$

Equation 3. Lower and upper confidence limits

5.2.2 Confidence and noncoverage

$$\gamma = 1 - \alpha$$

Equation 4. Confidence level and significance level

$$P_\theta\{L(X) \leq \theta \leq U(X)\} = \gamma$$

Equation 5. Repeated-sampling coverage

For a central two-sided interval, the total noncoverage probability α is divided equally between the two tails. Thus each tail has probability α over 2. In repeated sampling, approximately γ of the intervals produced by a correctly calibrated method contain θ.

Interpretation. A 95% confidence level describes the long-run performance of the interval-generating rule. Once the data are observed, the fixed parameter either lies inside the realized interval or it does not; the conventional frequentist statement is not a posterior probability for θ.

5.3 Approximate intervals from likelihood theory

5.3.1 What the asymptotic result actually says

The lecture invokes the central limit idea to justify a bell-shaped sampling distribution. For a general maximum likelihood estimator, the precise result is asymptotic normality of the MLE under identification, differentiability, an interior true parameter, nonsingular information, and other regularity conditions. It is not a claim that every estimator is unbiased or normally distributed at finite n.

$$\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, I_1^{-1}(\theta_0))$$

Equation 6. Asymptotic normality of an MLE

$$\operatorname{Var}(\widehat{\theta}) \approx I_n^{-1}(\theta_0), \qquad SE \approx \sqrt{I_n^{-1}(\widehat{\theta})}$$

Equation 7. Information-based variance and standard error

Mathematical correction. Asymptotic normality centers the scaled estimation error at zero; it does not by itself imply exact finite-sample unbiasedness. MLEs can be biased in small samples even when they are consistent and asymptotically normal.

5.3.2 Standardization and inversion

$$Z = \frac{\widehat{\theta} - \theta_0}{SE(\widehat{\theta})} \overset{a}{\sim} N(0, 1)$$

Equation 8. Approximate standard-normal pivot

$$P\{-z_{1-\alpha/2} \le Z \le z_{1-\alpha/2}\} \approx 1 - \alpha$$

Equation 9. Central normal probability

$$\theta_0 \in \left[\widehat{\theta} - z_{1-\alpha/2} SE, \widehat{\theta} + z_{1-\alpha/2} SE\right]$$

Equation 10. Two-sided Wald interval

5.4 Critical values

5.4.1 Quantiles and tail areas

$$z_q = \Phi^{-1}(q)$$

Equation 11. Standard-normal quantile

Table 1. Common central two-sided normal critical values

Confidence γ	Alpha	Each tail $\alpha/2$	Critical z
90%	0.10	0.050	1.6449
95%	0.05	0.025	1.9600
99%	0.01	0.005	2.5758

A statistical table reverses the cumulative distribution: locate the desired cumulative probability inside the body of the table and read the corresponding row and column labels. Software performs the same inverse-CDF calculation at greater numerical precision.

```
from scipy.stats import norm, t

alpha = 0.05
z_lower = norm.ppf(alpha / 2)
z_upper = norm.ppf(1 - alpha / 2)

df = n - 1
t_lower = t.ppf(alpha / 2, df=df)
t_upper = t.ppf(1 - alpha / 2, df=df)
```

5.4.2 Why t critical values are larger

Replacing an unknown population standard deviation by a random sample standard deviation adds uncertainty. Student t distributions therefore have heavier tails than the standard normal distribution. At a fixed confidence level, the t critical value is larger for every finite degree of freedom and approaches the normal critical value as the degrees of freedom increase.

$$t_{\nu,1-\alpha/2} > z_{1-\alpha/2}, \qquad t_\nu \xrightarrow{d} N(0,1)$$

Equation 12. Student t versus normal critical values

5.5 Exact inference for a normal mean

5.5.1 Known population variance

Assume a random sample from a normal population. Because sums and averages of independent normal variables remain normal, the sample mean has an exact normal distribution at every sample size.

$$X_i \overset{iid}{\sim} N(\mu, \sigma^2)$$

Equation 13. Normal population model

$$\overline{X} \sim N(\mu, \frac{\sigma^2}{n})$$

Equation 14. Exact sampling distribution of the normal sample mean

Equation 14 already contains the pivot: standardizing a normal random variable — subtracting its mean and dividing by its standard deviation — always produces a standard normal variable, and this transformation is exact at every n, not merely in a large-sample limit. Applying it to $\overline{X}$,

$$\frac{\overline{X}-\mu}{SD(\overline{X})} = \frac{\overline{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1),$$

which is precisely the pivot below. Because σ is assumed known here, this statistic depends on no unknown quantity besides μ itself, so it is a genuine pivot rather than an approximation.

$$Z_\mu = \frac{\overline{X}-\mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$$

Equation 15. Exact pivot when σ is known

5.5.2 Unknown variance and the nuisance parameter

When σ is unknown, the preceding statistic is not pivotal because it contains an unknown nuisance parameter. Estimate variability with the sample variance, using the residual degrees-of-freedom correction.

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i-\overline{X}\right)^2$$

Equation 16. Unbiased sample variance

$$\sum_{i=1}^{n}(X_i-\overline{X}) = 0$$

Equation 17. Linear constraint on sample residuals

Mathematical correction. No observation or residual must equal the sample mean or zero. The n residuals have only n minus 1 freely varying components because their sum is constrained to zero. This loss of one degree of freedom, together with unbiasedness, explains the denominator n minus 1.

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}, \qquad \overline{X}\perp S^2$$

Equation 18. Normal-sample variance result and independence

Equation 18 itself is a classical consequence of normal sampling (a special case of Cochran's theorem), sketched here rather than proved in full. Standardizing each observation as $(X_i-\mu)/\sigma$ gives n independent standard normal variables, and the sum of their squares is exactly χ^2_n. That sum of squares splits into two pieces: one piece depends only on $\overline{X}$ and consumes exactly one degree of freedom, leaving the remaining $n-1$ degrees of freedom attached to $(n-1)S^2/\sigma^2$. The same decomposition is what makes this remainder statistically independent of $\overline{X}$ — a special feature of the normal distribution, not something that holds for an arbitrary population.

The exact Student pivot in Equation 19 is not a separate assumption; it follows directly from combining Equations 15 and 18. Start from the known-variance pivot and insert S in place of σ, then multiply and divide by σ to reintroduce the exact pivot Z_μ:

$$T = \frac{\overline{X}-\mu}{S/\sqrt{n}} = \frac{(\overline{X}-\mu)/(\sigma/\sqrt{n})}{S/\sigma} = \frac{Z_\mu}{S/\sigma}.$$

The denominator can be rewritten using Equation 18: since $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$,

$$\frac{S}{\sigma} = \sqrt{\frac{S^2}{\sigma^2}} = \sqrt{\frac{\chi^2_{n-1}}{n-1}}.$$

Substituting this back,

$$T = \frac{Z_\mu}{\sqrt{\chi^2_{n-1}/(n-1)}},$$

a standard normal variable divided by the square root of an independent chi-square variable over its own degrees of freedom — which is exactly the definition of a Student t random variable with $n-1$ degrees of freedom. The independence of $\overline{X}$ and S^2 stated in Equation 18 is what entitles this ratio to the classical t_{n-1} distribution rather than some other, dependence-distorted shape: the standard derivation of the t-distribution's density assumes the numerator and denominator are independent, and substituting a dependent pair would not reproduce it.

$$T = \frac{\overline{X}-\mu}{\frac{S}{\sqrt{n}}} \sim t_{n-1}$$

Equation 19. Exact Student t pivot

$$\mu \in \left[\overline{X} - t_{n-1,1-\alpha/2}\frac{S}{\sqrt{n}}, \overline{X} + t_{n-1,1-\alpha/2}\frac{S}{\sqrt{n}}\right]$$

Equation 20. Exact confidence interval for a normal mean with unknown variance

5.6 Exact versus approximate: a careful decision rule

Table 2. Which interval is justified?

Setting	Interval	Status
General regular estimator; large sample	Estimate ± z critical × estimated SE	Approximate / asymptotic

Setting	Interval	Status
Normal observations; σ known	$\bar{X}\pm$ z critical $\times\sigma/\sqrt{n}$	Exact for every n
Normal observations; σ unknown	$\bar{X}\pm$ t critical $\times$ S$/\sqrt{n}$	Exact for every n
Non-normal observations; mean	Usually z or t-style large-sample interval	Approximate; quality depends on n and tails
Binomial proportion	Wilson, score, likelihood, or exact binomial method	Method-specific; Wald is only approximate

Sample size alone does not decide whether an interval is exact. Exactness comes from the sampling model and pivot. Likewise, there is no universal numerical threshold at which the central limit approximation becomes adequate; skewness, heavy tails, dependence, boundaries, and the estimand all matter.

5.7 Worked example: student test scores

The lecture uses n = 81 students, sample mean 74.6, sample standard deviation 11.3, and a 90% confidence level. The common standard error is 11.3 divided by the square root of 81.

$$SE = \frac{11.3}{\sqrt{81}} = 1.2556$$

Equation 21. Test-score standard error

$$CI_z = 74.6 \pm 1.6449(1.2556) = [72.5348, 76.6652]$$

Equation 22. Approximate 90 percent test-score interval

$$CI_t = 74.6 \pm 1.6641(1.2556) = [72.5106, 76.6894]$$

Equation 23. Exact 90 percent test-score interval under normal sampling

The t interval is slightly wider because 1.6641 exceeds 1.6449. Calling it exact requires the normal-population assumption and a genuinely random independent sample. With 80 degrees of freedom, the numerical difference is already small.

5.8 Worked example: 490 heads in 900 tosses

5.8.1 Bernoulli likelihood and the sample proportion

$$X_i \overset{iid}{\sim} Bernoulli(p), \qquad \widehat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{490}{900} = 0.5444$$

Equation 24. Coin-toss model and MLE

$$\mathrm{Var}(\widehat{p}) = \frac{p(1-p)}{n}, \qquad \widehat{SE} = \sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}$$

Equation 25. Sampling variance of a sample proportion

$$\widehat{SE} = \sqrt{\frac{0.5444(0.4556)}{900}} = 0.01660$$

Equation 26. Coin-proportion standard error

$$CI_{\text{Wald}} = 0.5444 \pm 1.6449(0.01660) = [0.5171, 0.5718]$$

Equation 27. Ninety percent Wald interval for the head probability

Because 0.50 is outside this 90% interval, the corresponding two-sided large-sample test rejects the fair-coin null hypothesis at $\alpha = 0.10$. The interval alone does not establish why the coin or tossing mechanism differs from fairness.

5.8.2 Why the Wald proportion interval is not exact

Mathematical correction. For Bernoulli data, the normal or Wald interval is still approximate. The finite-sample count is binomial, not normal, and the plug-in standard error is random. Dividing a Bernoulli variance expression by n does not make the normal approximation exact. Exact binomial intervals are obtained by inverting binomial tail probabilities; Wilson or likelihood-based intervals often have better coverage than Wald.

$$CI_{\text{Wilson}} = \frac{\widehat{p} + z^2/(2n) \pm z\sqrt{\widehat{p}(1-\widehat{p})/n + z^2/(4n^2)}}{1 + z^2/n}$$

Equation 28. Wilson score interval for a proportion

$$CI_{\text{Wilson}} = [0.5170, 0.5716]$$

Equation 29. Ninety percent Wilson interval for the coin example

```
from scipy.stats import binomtest, norm

x, n, confidence = 490, 900, 0.90
p_hat = x / n
z = norm.ppf(1 - (1 - confidence) / 2)

# Wald interval (large-sample approximation)
se = (p_hat * (1 - p_hat) / n) ** 0.5
```

```
wald = (p_hat - z * se, p_hat + z * se)

# Exact equal-tailed binomial interval
exact = binomtest(x, n).proportion_ci(
confidence_level=confidence, method="exact"
)
```

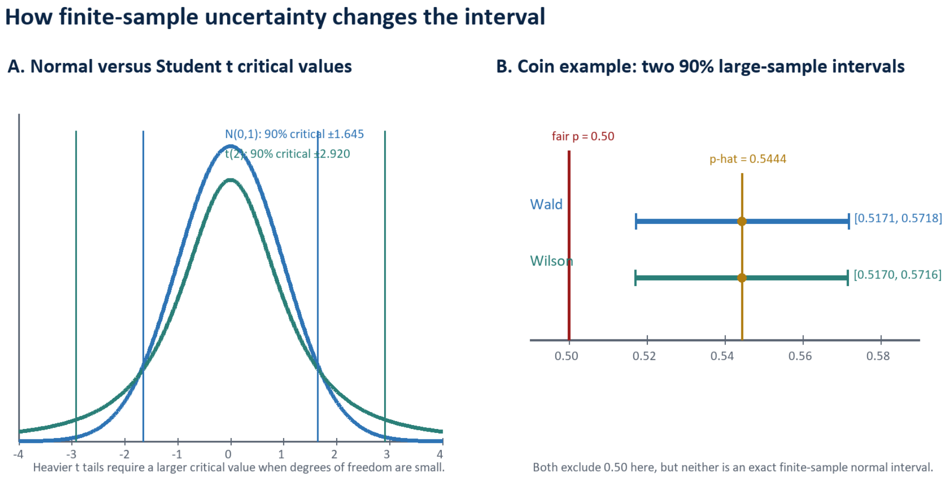

Figure 1. Small degrees of freedom widen t intervals; the coin example is a large-sample proportion calculation, not an exact normal procedure.

Figure constructed from the formulas and numerical examples reconstructed from the October 6 lecture.

5.9 Worked example: restaurant expenditure

The lecture reports a sample of n = 80 adults age 50 or older, mean annual restaurant and carryout expenditure of 1,873 dollars, sample standard deviation of 550 dollars, and a 95% confidence target.

$$SE = \frac{550}{\sqrt{80}} = 61.4919$$

Equation 30. Restaurant-expenditure standard error

$$ME_z = 1.9600(61.4919) = 120.5218$$

Equation 31. Approximate restaurant margin of error

$$CI_z = [1752.48, 1993.52]$$

Equation 32. Approximate 95 percent restaurant-expenditure interval

$$ME_t = 1.99045(61.4919) = 122.3965$$

Equation 33. Exact normal-model restaurant margin of error

$$CI_t = [1750.60, 1995.40]$$

Equation 34. Exact 95 percent restaurant-expenditure interval under normal sampling

The t margin of error is about 1.87 dollars larger than the z margin because t(79, 0.975) exceeds z(0.975). The t procedure is exact only if the individual expenditure observations are normally distributed. If expenditures are strongly right-skewed, both a normal-theory t interpretation and a simple large-sample approximation should be supported with diagnostics or a robust alternative.

5.10 Width, precision, and sample size

$$\text{Width} = 2c\frac{s}{\sqrt{n}}$$

Equation 35. Width of a symmetric mean interval

$$\frac{\text{Width}_{new}}{\text{Width}_{old}} \approx \sqrt{\frac{n_{old}}{n_{new}}}$$

Equation 36. Root-n scaling of interval width

At a fixed confidence level and variability, quadrupling the sample size approximately halves the width. Increasing confidence instead enlarges the critical value and widens the interval. These relationships explain why exact t intervals are especially wider at small n and converge toward z intervals as n increases.

5.11 Practical workflow

7. Define the estimand and sampling unit. Decide whether the target is a mean, proportion, likelihood parameter, difference, or another function.
8. Inspect the data-generating structure. Check independence, pairing or clustering, skewness, outliers, missingness, and parameter boundaries.
9. Choose an interval method whose assumptions fit the setting: exact normal t, large-sample Wald, Wilson or likelihood-based proportion interval, bootstrap, or another justified procedure.
10. Fix the confidence level before inspecting the conclusion, then compute α and the correct tail probability α divided by two.
11. Report the estimate, standard error, critical distribution and degrees of freedom, margin of error, endpoints, sample size, and assumptions.
12. Interpret the interval for the population parameter. Do not describe a mean interval as a range containing a stated percentage of individual observations.

5.12 Common failure modes and corrections

Table 3. Diagnostic checklist

Failure mode	Correction
Calling every normal approximation a CLT result	State the estimator-specific asymptotic theorem and its regularity conditions.
Equating asymptotic normality with unbiasedness	Separate consistency, asymptotic normality, and finite-sample bias.
Explaining n minus 1 by claiming one residual equals zero	Use the constraint that all residuals sum to zero.
Treating the Wald interval for p as exact	Use Wilson, likelihood, or exact binomial methods when coverage matters.
Choosing exact versus approximate from n alone	Base exactness on the model and pivot; use diagnostics to judge approximation quality.
Forgetting degrees of freedom in t.ppf	Use df = n minus 1 for the one-sample normal-mean t interval.
Using α rather than α divided by two	Split α across both tails for a central two-sided interval.

5.13 Compact mathematical reference

Table 4. Core formulas from the lecture

Quantity	Formula	Use
Confidence and noncoverage	$\gamma = 1 - \alpha$	Translate reported confidence into total tail probability
Margin of error	critical value $\times$ standard error	Half-width of a symmetric interval
General Wald interval	$\hat{\theta} \pm z\,SE$	Large-sample regular estimator
Normal-mean t interval	$\bar{X} \pm t_{n-1}\,S/\sqrt{n}$	Exact under independent normal sampling
Proportion standard error	$\sqrt{\hat{p}(1-\hat{p})/n}$	Plug-in SE for the Wald approximation
Normal quantile	$z(q) = \Phi^{-1}(q)$	Critical value from a cumulative tail probability
Degrees of freedom	$\nu = n - 1$	One-sample normal-mean t procedure
Width scaling	Width proportional to $n^{-1/2}$	Approximate sample-size comparison

5.14 Closing synthesis

The unifying pattern is estimate plus or minus a calibrated multiple of standard error. What changes across problems is the pivot and therefore the critical distribution. A z interval is broadly useful but asymptotic; a t interval for a normal mean with unknown variance is exact because the normal model supplies the required finite-sample distribution; a binomial proportion requires its own finite-sample logic. Reliable inference comes from matching the interval procedure to the data-generating assumptions rather than selecting a formula from sample size alone.

Research habit. Name the interval method. Saying only "95% confidence interval" hides the assumptions and can conceal large coverage differences among Wald, t, Wilson, likelihood, bootstrap, and exact procedures.

6 | Lecture 6: Hypothesis Testing

Coverage • Null and Alternate Hypotheses • Rejection region • p-value • Type I and II errors

6.1 Executive Summary

Hypothesis testing converts a research claim into a decision rule that controls long-run error. The analyst specifies a null region H_0 and an alternative region H_a, chooses a statistic whose distribution is known or approximated under H_0, and rejects H_0 only when the observed statistic falls sufficiently far into a pre-specified rejection region.

> **CORE IDEA** Evidence against H_0 is measured by how surprising the observed statistic would be if the null boundary were true. A smaller p-value means the data are less compatible with H_0; it does not mean that H_0 has a small posterior probability.

6.1.1 What this report adds

- Formal parameter-space notation and a precise definition of test size, Type I error, Type II error, and power.
- A mathematically correct account of sampling distributions, standard errors, Student's t statistic, and the central limit theorem.
- Complete calculations for the fiberglass-tire and fleet-fuel-efficiency one-sample examples, the Chicago-versus-San-Francisco independent two-sample comparison, and the MBA before-after paired comparison.
- A derivation of why the equality boundary belongs in the null, and of the alpha-beta tradeoff, standardized effect size, and sample-size planning formula.
- A unified workflow for critical-value and p-value methods, plus guidance on confidence intervals, assumptions, multiplicity, and reporting.

6.1.2 Contents

Part	Focus
Foundations	Conceptual logic, notation, hypothesis formulation, and why equality belongs in the null
Errors and distributions	Decision errors, the alpha-beta tradeoff, sample-size planning, sampling distributions, critical values, and p-values

Part	Focus
Workflow and one-sample examples	Testing workflow and the fiberglass-tire and fleet-fuel-efficiency worked examples
Two-sample and paired tests	Confidence-interval duality, independent two-sample tests (Chicago versus San Francisco), and paired-difference tests (MBA salaries)
Extensions and reference	Further extensions, research integrity, multiple testing, and the formula reference
Appendix	Computational recipes for Python/SciPy

6.2 Conceptual Foundation: From Claims to Controlled Decisions

The lecture motivates hypothesis testing through proof by contradiction. That analogy is useful, but statistical testing is not a deductive proof. In a proof by contradiction, a logical inconsistency rules out the assumption. In a statistical test, an unlikely observation provides graded evidence against H_0 while leaving a controlled probability of error.

> **TESTING LOGIC** Assume H_0 as the calibration model $\rightarrow$ measure how extreme T(X) is under H_0 $\rightarrow$ reject only if extremeness exceeds the α threshold

This distinction matters because random samples fluctuate. A sample mean can fall on the alternative side of a benchmark even when the population parameter lies in H_0. The rejection boundary is therefore placed inside the alternative-facing tail so that the probability of such a false rejection is no greater than the chosen significance level.

6.2.1 Estimation versus testing

Question	Estimation	Hypothesis testing
Primary goal	Approximate the parameter's value	Decide which parameter region is better supported
Typical output	Point estimate and confidence interval	Test statistic, p-value, and reject/fail-to-reject decision
Example	What is the mean fleet MPG?	Is the mean fleet MPG greater than 20?
Uncertainty	Width of the interval	Long-run error rates under repeated sampling

6.3 Statistical Objects and Notation

Let $X_1, \ldots, X_n$ be an independent random sample from a population distribution indexed by a parameter θ. For a one-sample mean test, $\theta = \mu$; for a one-sample proportion test,

θ = p; and for a two-group comparison, θ may be a contrast such as $\Delta = \mu_2$ - μ_1.

MODEL $X_1, \ldots, X_n$ iid $\sim$ F($\cdot$; θ), $\theta \in \Theta$

Symbol	Meaning	Formula / domain
μ	Population mean	$\mu = E[X]$
p	Population proportion	$0 \leq$ p ≤ 1
μ_0 **or** θ_0	Null benchmark or boundary value	Specified before examining the test result
$\bar{x}$	Sample mean	$\bar{x}$ = (1/n) $\sum_{i=1}^{n} x_i$
s^2	Unbiased sample variance estimator	s^2 = [1/(n-1)] $\sum_{i=1}^{n}(x_i - \bar{x})^2$
SE($\bar{x}$)	Estimated standard error of the sample mean	s/$\sqrt{n}$
T	Test statistic / pivotal quantity	A standardized, unit-free function of the data
ν	Degrees of freedom for one-sample t	ν = n - 1

6.3.1 Maximum likelihood and the n versus n-1 distinction

For a normal population, the maximum-likelihood estimator of μ is the sample mean. The maximum-likelihood variance estimator divides by n, while the usual unbiased variance estimator divides by n-1. The latter corrects the downward bias caused by estimating μ from the same sample.

ESTIMATORS $\hat{\mu}_{\text{MLE}} = \bar{x}$, $\hat{\sigma}^2_{\text{MLE}} = (1/n)\sum(x_i - \bar{x})^2$, $s^2 = [1/(n-1)]\sum(x_i - \bar{x})^2 = [n/(n-1)]\hat{\sigma}^2_{\text{MLE}}$

PRECISION NOTE The n-1 correction makes s^2 unbiased for σ^2. The square root s is not exactly unbiased for σ, although it is the standard estimator used in the one-sample t statistic.

6.4 Formulating H_0 and H_a

A hypothesis is a statement about a parameter region. The research claim normally determines H_a; H_0 is its complement and includes the equality boundary used to calibrate the reference distribution. For composite nulls such as H_0: $\mu \leq \mu_0$, the rejection threshold is typically calibrated at the least favorable boundary $\mu = \mu_0$.

PARAMETER SPACE $\Theta = \Theta_0 \cup \Theta_a$, $\Theta_0 \cap \Theta_a = \emptyset$, H_0: $\theta \in \Theta_0$, H_a: $\theta \in \Theta_a$

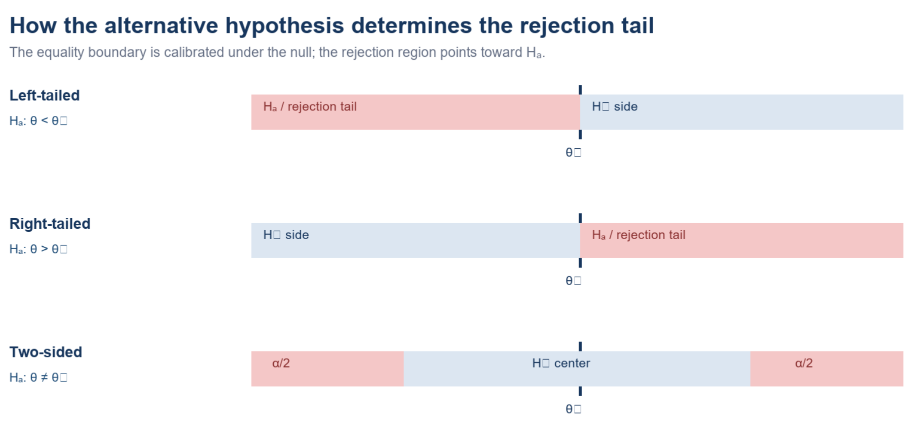

Figure 1. Tail direction follows the alternative hypothesis; α is split only for a two-sided test.

6.4.1 Why equality is conventionally placed in the null

The equality boundary is placed in H_0 for a precise reason, not merely to make the two regions exhaustive: the test's Type I error is calibrated under the null boundary, and for a one-sided location test that boundary is the null value closest to the alternative, so it usually maximizes the chance of accidentally entering the rejection region. For H_0: $\mu \geq \mu_0$ against H_a: $\mu < \mu_0$, the left-tail rejection probability is largest exactly at $\mu = \mu_0$, since means farther above the boundary shift the sampling distribution to the right and make a left-tail rejection less likely.

LEAST-FAVORABLE NULL $\sup_{\mu \geq \mu_0} P_\mu(\text{reject } H_0) = P_{\mu_0}(\text{reject } H_0) = \alpha$

The same reasoning applies in mirror image to H_0: $\mu \leq \mu_0$ against a right-tailed alternative: the worst-case Type I probability again occurs at the boundary, and calibrating there protects every deeper value inside the composite null. This is more precise than saying equality must be in H_0 merely so that every case is covered by some region: coverage is necessary, but the deeper statistical reason is error control at the least favorable null value.

6.4.2 Lecture examples rewritten in standard notation

Context / parameter	Research claim H_a	Null hypothesis H_0	Tail
Fleet mean MPG, μ	$\mu > 20$	$\mu \leq 20$	Right
Teen phone use, μ	$\mu \neq 16$ hours/week	$\mu = 16$	Two-sided
Lost-bag proportion at DTW, p	p > 0.03	p ≤ 0.03	Right
Mean age of Cadillac buyers, μ	$\mu < 50$	$\mu \geq 50$	Left
Baseball home-run contrast, $\Delta = \mu_2$-μ_1	$\Delta > 0$	$\Delta \leq 0$	Right

Context / parameter	**Research claim** H_a	**Null hypothesis** H_0	**Tail**
Fiberglass-tire mean life, μ	$\mu < 40$ (thousand miles)	$\mu \geq 40$	Left

INTERPRETATION "Fail to reject H_0" means the sample does not cross the pre-specified evidence threshold. It is not proof that H_0 is true, and it should not be paraphrased as "no effect" without considering precision and power.

6.5 Decision Outcomes, Errors, and Power

A test maps every possible sample to one of two actions: reject H_0 or fail to reject H_0. Because the true parameter is unknown, the action can be correct or incorrect. The resulting probabilities are conditional on the true parameter value.

True state	**Fail to reject** H_0	**Reject** H_0
$\theta \in \Theta_0$	Correct non-rejection: probability 1-$\alpha(\theta)$	Type I error: $\alpha(\theta) = P_\theta$(reject H_0)
$\theta \in \Theta_a$	Type II error: $\beta(\theta) = P_\theta$(fail to reject H_0)	Power: $\pi(\theta)$ = 1-$\beta(\theta)$

LEVEL-α TEST size(test) $= \sup_{\theta \in \Theta_0} P_\theta(\text{reject } H_0) \leq \alpha$

For a fixed sample size and a fixed data-generating process, tightening α usually moves the critical value farther into the tail. This reduces false positives but generally increases β and lowers power against nearby alternatives. Increasing n can improve the trade-off by shrinking the standard error, allowing both error probabilities to become small for a specified effect size.

POWER Power function: $\pi(\theta) = P_\theta(\text{T} \in \text{C})$, where C is the rejection region

- α controls false rejection under H_0; common planning levels are 0.10, 0.05, and 0.01.
- β depends on a particular alternative θ, the effect size, sample size, variability, and test rule.
- **Power** is not a single universal property unless a specific alternative or minimum effect size is named.

6.5.1 The alpha-beta tradeoff

With sample size and noise held fixed, moving a critical value changes both error types at once: expanding the rejection region makes rejection easier, raising α but lowering β for alternatives in the rejection direction, while shrinking the rejection region protects H_0 more strongly at the cost of a higher β. For a left-tailed normal test with known standard error that rejects when $\bar{X} \leq c$, moving c to the right increases α and decreases β simultaneously.

TRADEOFF AT A THRESHOLD $\alpha = P_{\mu_0}(\bar{X} \le c), \quad \beta(\mu_1) = P_{\mu_1}(\bar{X} > c)$ for $\mu_1 < \mu_0$

This geometric relationship is why a test cannot generally minimize α and β simultaneously by adjusting only the threshold. Classical test design is instead a constrained optimization closely related to the Neyman–Pearson framework: control Type I error at or below α, then choose the most powerful rejection rule available.

CONSTRAINED OPTIMIZATION Maximize power (minimize β), subject to $\sup_{\theta\in\Theta_0} P_\theta(\text{reject } H_0) \le \alpha$

6.5.2 Sample size and standardized effect size

Unlike moving a threshold, increasing a well-designed independent sample can improve power while preserving the same α: as n grows, the standard error $\sigma/\sqrt{n}$ shrinks, separating the null and alternative sampling distributions on the standardized scale. For a one-sided z test with known σ, the standardized separation is the raw effect multiplied by $\sqrt{n}$.

SIGNAL-TO-NOISE SEPARATION $|\mu_1 - \mu_0|\sqrt{n}/\sigma$

A raw effect is meaningful in its original units, while a standardized effect expresses the same gap relative to variability, which helps compare detectability across measurement scales.

STANDARDIZED EFFECT $\mathrm{d} = (\mu_1 - \mu_0)/\sigma$

In a right-tailed normal test with known σ, the critical sample mean is $\bar{X}_{crit} = \mu_0 + z_{1-\alpha}\,\sigma/\sqrt{n}$, and under a specific alternative μ_1, power is the probability of exceeding that threshold, $\text{Power}(\mu_1) = 1 - \Phi\left((\bar{X}_{crit} - \mu_1)/(\sigma/\sqrt{n})\right)$. These formulas separate two distinct roles: α selects the decision threshold, while the effect size, sample size, and variability jointly determine how often a real departure actually crosses it. An approximate one-sided sample-size calculation for detecting a positive mean difference Δ with power $1-\beta$ follows directly:

SAMPLE-SIZE FORMULA $\mathrm{n} \approx [(z_{1-\alpha} + z_{1-\beta})\,\sigma/\Delta]^2$

This expression is useful during experimental planning, but only if n is fixed in advance: selecting n after seeing the result, repeatedly testing as data arrive without a pre-approved sequential design, or stopping as soon as significance appears can each invalidate the nominal α level. A directional one-sided test has greater power in its prespecified direction than a two-sided test at the same total α, but cannot claim significance for an unexpected effect in the opposite direction.

6.6 Sampling Distributions and Pivotal Quantities

6.6.1 The sample mean

If $X_1, \ldots, X_n$ are independent with finite mean μ and variance σ^2, then the sample mean has expectation μ and variance σ^2/n. Its standard deviation is the standard error when σ is known.

SAMPLING MOMENTS $E[\bar{X}] = \mu$, $\mathrm{Var}(\bar{X}) = \sigma^2/n$, $\mathrm{SD}(\bar{X}) = \sigma/\sqrt{n}$

6.6.2 Central limit theorem

For a broad class of populations with finite variance, the standardized sample mean approaches the standard normal distribution as n grows. This result concerns the distribution of the standardized mean across repeated samples, not the shape of the raw observations themselves.

CLT $Z_n = \sqrt{n}(\bar{X}\text{-}\mu)/\sigma \Rightarrow N(0,1)$ as $n \to \infty$

6.6.3 Student's t result for an unknown variance

If the underlying population is normal and σ is unknown, replacing σ with s yields an exact Student t statistic with n-1 degrees of freedom. The statistic—not $\bar{X}$ by itself—has the t distribution.

ONE-SAMPLE t $T = (\bar{X}\text{-}\mu_0)/(S/\sqrt{n}) \sim t_{n-1}$ under H_0: $\mu{=}\mu_0$

The t distribution is symmetric and has heavier tails than N(0,1), reflecting the added uncertainty from estimating σ. As ν increases, tν converges to N(0,1). For non-normal populations, the same statistic is often approximately t/normal for sufficiently large n, but severe skewness, heavy tails, dependence, or influential outliers can invalidate the approximation.

6.6.4 Reference distributions in the larger testing toolkit

Target	Typical statistic	Reference distribution	Key condition
One mean, σ known	$Z = (\bar{X}\text{-}\mu_0)/(\sigma/\sqrt{n})$	N(0,1)	Known σ or justified large-sample approximation
One mean, σ unknown	$T = (\bar{X}\text{-}\mu_0)/(S/\sqrt{n})$	t_{n-1}	Exact for normal population
One proportion	$Z = (\hat{p}\text{-}p_0)/\sqrt{p_0(1-p_0)/n}$	Approx. N(0,1)	Sufficient expected successes/failures
One variance	$\chi^2 = (n\text{-}1)S^2/\sigma_0^2$	χ^2_{n-1}	Normal population
ANOVA / variance ratio	$F = MS_{between}/MS_{within}$	F_{df_1,df_2}	Model errors approximately normal and independent

6.7 Critical Values and p-Values

The critical-value method fixes α and identifies a rejection region C before the test statistic is compared with it. The p-value method computes the smallest significance level at which the observed statistic would fall in the rejection region. For a continuous test statistic, the two methods produce the same decision when implemented consistently.

Alternative	**Critical rule at level α**	**One-sample t p-value**
H_a: $\mu < \mu_0$	Reject if $t_{obs} \leq t_{\alpha,\nu}$	p $= F_{t\nu}(t_{obs})$
H_a: $\mu > \mu_0$	Reject if $t_{obs} \geq t_{1-\alpha,\nu}$	p $=$ 1-$F_{t\nu}(t_{obs})$
H_a: $\mu \neq \mu_0$	Reject if $\lvert t_{obs}\rvert \geq t_{1-\alpha/2,\nu}$	p $= 2 \cdot \min\{F_{t\nu}(t_{obs}), 1 - F_{t\nu}(t_{obs})\}$

DEFINITION p-value $= P_{H_0}$(a test statistic at least as adverse to H_0 as t_{obs})

- If p $\leq \alpha$, reject H_0 at level α.
- If p $> \alpha$, fail to reject H_0; report the estimate and interval so readers can distinguish low power from a genuinely small effect.
- A p-value is not P(H_0 is true | data), not the probability that the result occurred "by chance," and not the study's realized Type I error probability.
- Statistical significance does not imply practical importance; effect size and uncertainty remain essential.

6.7.1 Probability, surprise, and information

Small probabilities connect naturally to the ideas of surprise and information. A common information measure for an event with probability q is $I(q) = -\log(q)$; rare events carry more self-information than common events, and the logarithm converts products of independent probabilities into sums of information, $-\log(q_1q_2) = -\log(q_1) - \log(q_2)$ — a structure that underlies log-likelihood, cross-entropy, and many machine-learning loss functions. This analogy should be used carefully in testing: a p-value is a tail probability derived from a test statistic, not a likelihood for a parameter, and not automatically a valid information score for model comparison. The qualitative lesson nonetheless carries over: observations that the null model makes rare provide stronger evidence against that model. The critical-value and p-value reports are two views of the same underlying calculation — the critical method compares the statistic with a threshold, while the p-value method compares a tail area with α; the first emphasizes decision geometry, the second shows the strength of evidence relative to multiple possible α thresholds.

SOFTWARE MAPPING For a t distribution, PPF (quantile) maps a tail area to a critical value; CDF maps an observed value to a left-tail probability; SF = 1-CDF is numerically convenient for a right-tail probability.

6.8 A Reusable Hypothesis-Testing Workflow

1. **State the scientific question.** Identify the population, unit of analysis, parameter, benchmark, and direction of the research claim.
2. **Formulate H_0 and H_a.** Make the regions mutually exclusive and exhaustive; place the equality boundary in H_0.
3. **Pre-specify α and assumptions.** Choose the error tolerance before examining the result and record randomness, independence, distributional, and measurement assumptions.

4. **Summarize the sample.** Compute n, $\bar{x}$, s, $\hat{p}$, group contrasts, and diagnostics relevant to the chosen model.
5. **Choose the test statistic and reference distribution.** Use z, t, χ^2, F, permutation, or another justified reference law.
6. **Compute the observed statistic.** Standardize the distance between the estimate and the null boundary using the correct standard error.
7. **Calculate a critical value and/or p-value.** Use the tail implied by H_a; divide α by two only for a two-sided test.
8. **Decide and report.** Reject or fail to reject H_0, then state the conclusion in the language of the original population claim.
9. **Quantify magnitude and sensitivity.** Add a confidence interval, effect size, assumption checks, and any multiplicity adjustment.

   GENERIC PIVOT T(X) = (estimate - null benchmark) / (estimated standard error)

6.9 Worked Example A: Fiberglass Tire Lifetime

A manufacturer claims that the mean lifetime of a new fiberglass tire is at least 40,000 miles. An investigator is concerned that the true mean is lower. Lifetimes are recorded in thousands of miles. The lecture reports n = 12, $\bar{x}$ = 37.2833, and s = 2.7319, with an approximately normal population assumption.

```
import numpy as np
import pandas as pd

# 12 recorded lifetimes
df = pd.read_excel("M5 Tire data.xlsx", sheet_name="Sheet1")
df.describe()
```

```
Tire
count   12.000000
mean    37.283333
std      2.731910
min     33.000000
25%     35.950000
50%     36.900000
75%     38.925000
max     42.000000
```

The sample matches the lecture's reported figures exactly: n = 12, $\bar{x}$ = 37.2833, s = 2.7319 (pandas' `std` already divides by n-1, the unbiased convention from Equation 2).

HYPOTHESES H_0: $\mu \geq 40$ versus H_a: $\mu < 40$

6.9.1 Step 1 — Standard error

```
import math
```

```
standard_error = 2.7319 / math.sqrt(12)    # s / sqrt(n), in thousand miles
standard_error                             # -> 0.7886316001995693
```

STANDARD ERROR $\mathrm{SE}(\bar{X}) = \mathrm{s}/\sqrt{n} = 2.7319/\sqrt{12} = 0.7886$ thousand miles

6.9.2 Step 2 — Observed t statistic

```
pivot_quantity = (37.2833 - 40) / standard_error   # (xbar - mu0) / SE
pivot_quantity                                      # -> -3.4448277235055267
```

TEST STATISTIC t_{obs} = (37.2833-40)/0.7886 = -3.4449, ν = 12-1 = 11

6.9.3 Step 3 — Critical-value decision

For a left-tailed test with $\alpha = 0.05$ and $\nu = 11$, the critical value is $t_{0.05,11}$ = -1.7959. Since -3.4449 < -1.7959, the statistic lies in the rejection region.

6.9.4 Step 4 — p-value

```
from scipy.stats import t as t

# left-tail area of t_11 below the observed statistic
p_value = t.cdf(pivot_quantity, 12 - 1)
p_value                                    # -> 0.0027389035486974247
```

ONE-SIDED p p = P($T_{11} \leq$ -3.4449) $\approx$ 0.0027

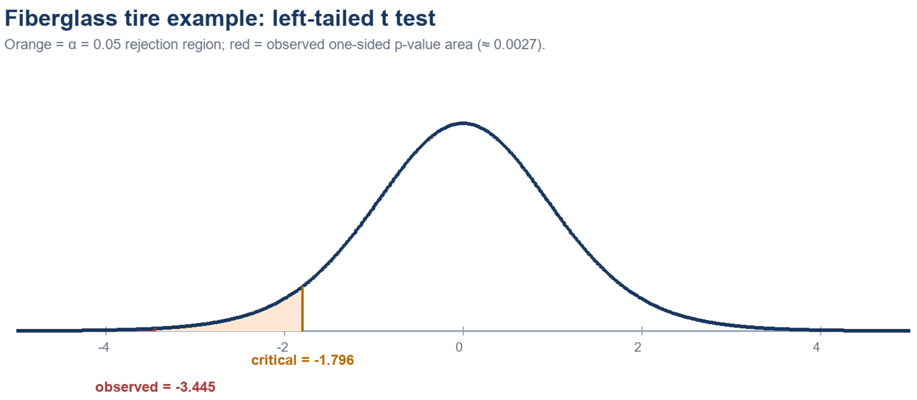

Figure 2. The observed statistic lies well beyond the 5% left-tail critical value.

CONCLUSION Reject H_0 at the 5% level. The sample provides strong evidence that the mean tire lifetime is below 40,000 miles (one-sided p $\approx$ 0.0027).

6.9.5 MLE-based alternative: an asymptotic Wald comparison

The accompanying notebook also fits the normal model by maximum likelihood, using the same `GenericLikelihoodModel` pattern introduced in the MLE chapter, and compares the resulting asymptotic Wald test with the classical exact t test above.

```
from statsmodels.base.model import GenericLikelihoodModel
from scipy.stats import norm

class MLE(GenericLikelihoodModel):
def __init__(self, data):
self.y = data                          # store the raw lifetimes
# dummy endog; the real data lives in self.y
super().__init__(endog=np.zeros(1))
self.nparams = 2                       # two parameters: mean and std
self._set_extra_params_names(["mean", "std"])
self._eps = 1e-9                       # keeps std strictly positive
def loglike(self, params):
mean, std = params
if not (self._eps < std):   # reject non-positive scale proposals
return -np.inf
return norm.logpdf(self.y, loc=mean, scale=std).sum()

data = df["Tire"].to_numpy()
model = MLE(data)
result = model.fit(start_params=[0, 10], method="bfgs", maxiter=200)
print(result.summary())
```

```
coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
mean          37.2833      0.755     49.378      0.000      35.803      38.763
std            2.6156      0.534      4.899      0.000       1.569       3.662

Log-Likelihood: -28.565    AIC: 61.13    BIC: 57.13
```

The fitted mean matches $\bar{x}$ exactly, but the fitted std, 2.6156, is smaller than the sample s = 2.7319 — exactly the n-versus-n-1 gap from the **Maximum likelihood and the n versus n-1 distinction** box earlier in this chapter, since the MLE variance divides by n rather than n-1. Calling statsmodels' built-in Wald test against the same benchmark makes the consequence concrete:

```
result.t_test("mean=40")
```

```
Test for Constraints
==============================================================================
coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
c0            37.2833      0.755     -3.598      0.000      35.803      38.763
==============================================================================

p_value = norm.cdf(-3.598)   # left-tail area under the STANDARD NORMAL, not t
p_value                       # -> 0.00016033680948409962
```

This z-statistic of -3.598 is more extreme than the classical t_{obs} = -3.4449, and its normal-based p-value, 0.00016, is roughly seventeen times smaller than the exact one-sided p-value of 0.0027 computed above. Both discrepancies trace back to the same two facts already flagged in this chapter: the MLE standard error uses the smaller n-divisor variance (0.755 here versus 0.7886 for the classical SE), and the Wald test compares that statistic with the standard normal distribution rather than the heavier-tailed t_{11}. With only twelve observations, the extra uncertainty the t distribution builds in for not knowing σ is not negligible, so the classical exact-t procedure — not the asymptotic Wald shortcut — is the appropriate test here.

6.9.6 Magnitude and interval interpretation

The estimated shortfall is 40 - 37.2833 = 2.7167 thousand miles. The standardized mean difference relative to the sample standard deviation is (37.2833-40)/2.7319 ≈ -0.994. A 95% one-sided upper confidence bound is

UPPER BOUND $U = \bar{x} + t_{0.95,11} \cdot \text{SE} = 37.2833 + 1.7959(0.7886) \approx 38.70$

Because the entire one-sided 95% confidence set lies below 40, the interval and test agree. The result is both statistically strong and substantively notable relative to the stated 40,000-mile benchmark.

6.10 Worked Example B: Fleet Fuel Efficiency

The fleet manager wants to establish that average fuel efficiency exceeds 20 MPG. The lecture reports a random sample with n = 55, $\bar{x}$ = 22 MPG, and s = 5 MPG, and treats the population distribution as normal for the t calculation.

HYPOTHESES H_0: $\mu \leq 20$ versus H_a: $\mu > 20$

6.10.1 Calculation

STANDARD ERROR $\text{SE}(\bar{X}) = 5/\sqrt{55} = 0.6742$ MPG

TEST STATISTIC t_{obs} = (22-20)/0.6742 = 2.9665, ν = 55-1 = 54

For a right-tailed level-0.05 test, $t_{0.95,54} \approx 1.6736$. Because 2.9665 > 1.6736, reject H_0. The one-sided p-value is approximately

ONE-SIDED p $p = P(T_{54} \geq 2.9665) \approx 0.0022$

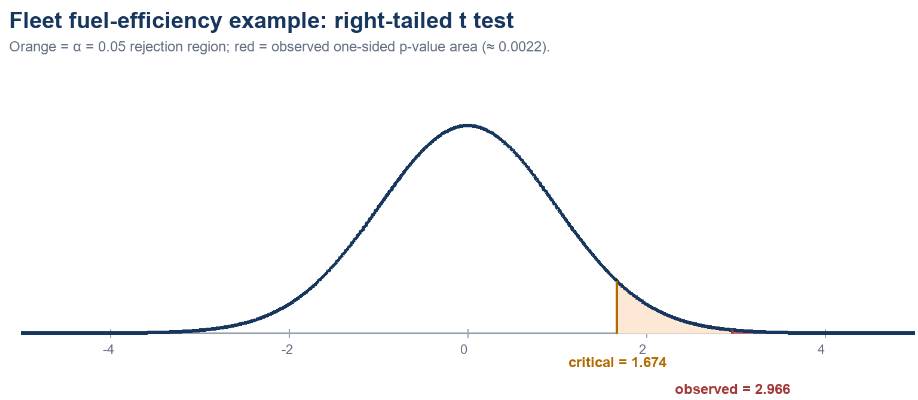

Figure 3. The observed statistic lies beyond the right-tail critical value.

CONCLUSION Reject H_0 at the 5% level. The sample provides strong evidence that the fleet's mean fuel efficiency exceeds 20 MPG (one-sided p $\approx$ 0.0022).

The estimated improvement is 2 MPG, corresponding to a standardized difference of (22-20)/5 = 0.40 sample standard deviations. The 95% one-sided lower confidence bound is 22 - 1.6736(0.6742) $\approx$ 20.87 MPG, again agreeing with the test.

6.11 Connection to Confidence Intervals

Tests and confidence intervals are dual procedures. For a two-sided level-α test of H_0: $\mu = \mu_0$, rejection occurs exactly when μ_0 falls outside the corresponding 100(1-α)% confidence interval, assuming the same model and standard error are used.

CONFIDENCE INTERVAL Two-sided CI: $\bar{x} \pm t_{1-\alpha/2,n-1} \cdot$ s/$\sqrt{n}$

Testing statement	Equivalent interval statement
Reject H_0: $\mu = \mu_0$ in a two-sided α test	μ_0 is outside the two-sided 100(1-α)% CI
Reject H_0: $\mu \leq \mu_0$ in a right-tailed α test	The 100(1-α)% one-sided lower bound exceeds μ_0
Reject H_0: $\mu \geq \mu_0$ in a left-tailed α test	The 100(1-α)% one-sided upper bound is below μ_0

WHY REPORT BOTH? The p-value grades evidence against H_0; the confidence interval shows the range of effect sizes compatible with the data. Together they separate statistical evidence from practical magnitude.

6.12 Independent Two-Sample Tests

A two-market business question makes the extension concrete: a telecommunications company has independent samples of monthly usage from Chicago and San Francisco and wants to know whether the population means differ. The object of interest is not either mean by itself but their contrast, $\Delta = \mu_C - \mu_{SF}$. Because the substantive claim is that a difference exists in either direction, the null places the contrast at zero and the alternative allows both signs, making this a two-tailed test; if the company had instead prespecified that Chicago usage was higher, the alternative would be $\Delta > 0$ and the test would be right-tailed.

TWO-SAMPLE HYPOTHESES H_0: $\mu_C = \mu_{SF}$ (equivalently $\Delta = 0$) versus H_a: $\mu_C \neq \mu_{SF}$

6.12.1 Sampling distribution of a difference

The two sample means are unbiased for their population means, so linearity of expectation gives $E(\bar{X}_C - \bar{X}_{SF}) = \Delta$ without requiring normal data. The variance calculation is where independence matters: in general the variance of a difference includes a covariance term, but when the two samples are independent the covariance is zero and the sampling variances simply add.

INDEPENDENT-SAMPLE VARIANCE $\mathrm{Var}(\bar{X}_C - \bar{X}_{SF}) = \sigma_C^2/n_C + \sigma_{SF}^2/n_{SF}$, so $SE = \sqrt{\sigma_C^2/n_C + \sigma_{SF}^2/n_{SF}}$

Because the population standard deviations are unknown in practice, they are replaced by sample standard deviations, giving the estimated standard error $\widehat{SE} = \sqrt{s_C^2/n_C + s_{SF}^2/n_{SF}}$. Under independent random sampling with finite variance, the central limit theorem makes the difference in sample means approximately normal; replacing the unknown variances with estimates leads to a t reference distribution.

6.12.2 The Welch and pooled test statistics

A test statistic expresses the observed discrepancy from a hypothesized difference D_0 in standard-error units. The two groups must be independent at the observational level — no Chicago observation should be mechanically paired with a San Francisco observation — for this standard error to be valid.

TWO-SAMPLE STATISTIC T = $[(\bar{X}_1$-$\bar{X}_2)$-$D_0]$ / $\sqrt{s_1^2/n_1 + s_2^2/n_2}$

Because the denominator uses two separate sample variances, the matching degrees of freedom are the Welch–Satterthwaite approximation rather than a simple $n-1$ rule.

WELCH DEGREES OF FREEDOM
$$\nu_W = (s_1^2/n_1 + s_2^2/n_2)^2 \Big/ \left[\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}\right]$$

An alternative, coherent procedure assumes a common population variance and pools the two samples, using $n_1 + n_2 - 2$ degrees of freedom instead:

POOLED VARIANCE AND STATISTIC $s_p^2 = \dfrac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$,
$$\mathrm{T} = \frac{(\bar{X}_1 - \bar{X}_2) - D_0}{s_p\sqrt{1/n_1 + 1/n_2}}$$

Mixing the separate-variance standard error with pooled degrees of freedom is a hybrid approximation rather than either standard test. Welch's procedure is generally the safer default because it remains valid when the population variances differ and sacrifices little precision when they are similar.

6.12.3 Worked Example: Verizon Monthly Phone Use

Verizon compares mean monthly phone use for business customers in Chicago and San Francisco. The two groups consist of different customers, so an independent-samples analysis is appropriate; the stated sample standard deviations differ, making Welch's t procedure the natural default. The sample information is $n_C = 66$, $\bar{x}_C = 433.90$, $s_C = 73$ for Chicago, and $n_{SF} = 53$, $\bar{x}_{SF} = 424.00$, $s_{SF} = 82$ for San Francisco, in shared monthly-usage minutes.

HYPOTHESES H_0: $\mu_C - \mu_{SF} = 0$ versus H_a: $\mu_C - \mu_{SF} \neq 0$

6.12.3.0.1 Step 1 — Estimate the contrast.

POINT ESTIMATE $\hat{\Delta} = \bar{x}_C - \bar{x}_{SF} = 433.90$ - $424.00 = 9.90$ minutes

6.12.3.0.2 Step 2 — Estimate its standard error. Chicago contributes $73^2/66$ and San Francisco contributes $82^2/53$ to the Welch standard error; because the samples are independent, the two estimated variances add.

> **WELCH SE AND STATISTIC** $SE_W = \sqrt{73^2/66 + 82^2/53} = 14.4087$ minutes, $\quad t_{obs} = 9.90/14.4087 = 0.6871$

6.12.3.0.3 Step 3 — Standardize and determine degrees of freedom. The observed gap is only about sixty-nine percent of one estimated standard error away from the zero-difference null. The matching Welch–Satterthwaite degrees of freedom are approximately $\nu_W \approx 105.17$; at $\alpha = 0.05$ for a two-sided test the critical magnitude is about 1.983, so since $|t_{obs}| = 0.6871 < 1.983$, we fail to reject H_0.

> **TWO-SIDED p** p $\approx 2P(T_{105.17} \geq 0.6871) \approx 0.49$

The contextual conclusion is restrained: the sample does not provide evidence of a difference in average monthly usage between Chicago and San Francisco at the five-percent level. This does *not* prove exact equality — a non-significant result cannot generally establish that two means are identical. The same calculation expressed as a confidence interval is more informative than the binary decision:

> **95% CONFIDENCE INTERVAL**
> $CI_{95\%} = \hat{\Delta} \pm t_{0.975,105.17}\, SE_W \approx 9.90 \pm 1.983(14.4087) \approx [-18.67,\ 38.47]$ minutes

> **CONCLUSION** The sample does not provide evidence of a difference in mean monthly use between the two cities. The estimate favors Chicago by 9.9 minutes, but the interval is compatible with a modest advantage for either city.

Zero lies comfortably inside this interval, so the interval and the test agree; the data are compatible with Chicago being modestly lower, essentially equal, or moderately higher. If the business objective were instead to *demonstrate* practical equivalence, a non-significant difference test is the wrong tool: the correct analysis is an equivalence test with prespecified margins $\pm\delta_E$, such as the two one-sided tests (TOST) procedure, which checks whether the entire compatible range falls inside those margins.

> **MATHEMATICAL CORRECTION** A naive df rule adds $(n_C - 1) + (n_{SF} - 1) = 117$ degrees of freedom while also describing unequal variances. That df rule belongs to a *pooled* equal-variance test, not to Welch's procedure. Welch's $\nu_W \approx 105.17$ is the value internally consistent with the unequal-variance standard error used above; the substantive conclusion is unchanged either way.

The equal-variance alternative gives nearly the same answer here: pooling yields $s_p = 77.1298$ on ν_p = 66+53-2 = 117 degrees of freedom, $SE_p = 14.2261$, and $t_p = 9.90/14.2261 = 0.6959$ — a statistic and critical magnitude close to the Welch results, so both coherent procedures fail to reject. This need not happen in smaller, more unbalanced, or more heteroscedastic samples, which is why the choice between Welch and pooled procedures is a methodological decision and not an afterthought.

6.12.4 From two groups to several: ANOVA

Analysis of variance (ANOVA) extends the two-mean comparison to several means by comparing between-group variability with within-group variability.

ONE-WAY ANOVA H_0: $\mu_1 = \mu_2 = \cdots = \mu_k$ versus H_a: at least one group mean differs; F $= MS_{between}/MS_{within}$

With exactly two groups and the corresponding equal-variance assumptions, the one-way ANOVA F statistic equals the square of the pooled two-sample t statistic, $F = t^2$ on $(1,\, n_1 + n_2 - 2)$ and $(n_1 + n_2 - 2)$ degrees of freedom respectively. ANOVA avoids running every pairwise t test as though each were the only comparison being made; after a significant omnibus F test, planned contrasts or adjusted post-hoc comparisons identify where the differences lie.

6.13 Paired-Difference Tests

If the same players are measured in two seasons, or the same people are measured before and after an MBA program, the two columns of data are not independent samples: each pair shares the same underlying unit, and a correlation near ninety percent between the two columns is plausible when people who start relatively high tend to remain relatively high. Treating paired observations as independent discards the identity of each unit, wastes information, and can badly misstate uncertainty.

WHY INDEPENDENCE FAILS $\mathrm{Var}(\bar{X}_{\text{after}} - \bar{X}_{\text{before}}) = \mathrm{Var}(\bar{X}_{\text{after}}) + \mathrm{Var}(\bar{X}_{\text{before}}) - 2\,\mathrm{Cov}(\bar{X}_{\text{after}}, \bar{X}_{\text{before}})$

Ignoring a strong positive covariance makes the estimated variance far too large and can erase a genuine within-person change. The correct unit of analysis is the pair, not the individual measurement.

6.13.1 Reducing to one sample of differences

For unit i, define the matched difference $D_i = A_i - B_i$, with $\mu_D = E(D_i)$ the population mean change. Once every pair is converted into a difference, the analysis becomes an ordinary one-sample t test of the mean difference, using exactly the machinery from the earlier sections of this chapter.

PAIRED HYPOTHESES AND STATISTIC H_0: $\mu_D \leq 0$ versus H_a: $\mu_D > 0$; T $= \bar{D}/(s_D/\sqrt{n})$, $\nu = n - 1$

The sample mean of the differences equals the difference between the two sample means, $\bar{D} = \bar{A} - \bar{B}$, an identity that always holds exactly. The corresponding identity does *not* hold for standard deviations: the variance of the paired differences contains the covariance between before and after values.

VARIANCE OF THE DIFFERENCES $s_D^2 = s_A^2 + s_B^2 - 2\,r_{AB}\,s_A\,s_B$

A large positive correlation r_{AB} reduces s_D, which reduces the standard error and increases power; the benefit comes from pairing outcomes that move together, not from the mere act of matching records.

6.13.2 Worked Example: MBA Salaries Before and After the Program

The same thirty-six MBA students are observed before and after the program. The reported average increase is $\bar{D} = \$19{,}729$, the marginal standard deviations are approximately \$62,950 before and \$61,900 after, and the standard deviation of the within-student differences is $s_D = \$7{,}706$. The marginal spread is large because students differ greatly in salary level; the within-person changes are far less variable.

MATCHED DIFFERENCES $D_i = \text{Salary}_{\text{after},i} - \text{Salary}_{\text{before},i}, \quad \bar{D} = \$19{,}729, \quad s_D = \$7{,}706, \quad n = 36$

6.13.2.0.1 What an independence calculation would say. Treating the two columns as unrelated discards the identity of each person. Combining the two marginal standard deviations as if the samples were independent gives a very different — and inappropriate — denominator:

INDEPENDENCE COUNTERFACTUAL $SE_{\text{ind}} = \sqrt{62{,}950^2/36 + 61{,}900^2/36} = \$14{,}714$

COUNTERFACTUAL t $t_{\text{ind}} = 19{,}729/14{,}714 = 1.341, \quad \nu_{\text{Welch}} \approx 69.98$

A two-sided test based on that denominator would fail to reject equality at the five-percent level ($p \approx 0.184$). But this calculation is not the appropriate final analysis: the two columns are not independent samples, since each row represents the same person measured at two times.

6.13.2.0.2 Correct matched-pairs analysis. Because the pairing is real, the analysis should instead use the standard deviation of the within-student differences directly.

PAIRED STANDARD ERROR $SE_{\text{pair}} = 7{,}706/\sqrt{36} = \$1{,}284.33$

PAIRED t STATISTIC $t_{\text{pair}} = 19{,}729/1{,}284.33 = 15.361, \quad \text{df} = 36 - 1 = 35$

The two-sided p-value is far below 0.001 (indeed, below 10^{-16}), so equality is decisively rejected. This illustrates the power gained by controlling stable person-level heterogeneity: the invalid independent-samples analysis of the same data reports $t \approx 1.34$ and fails to reject, while the paired analysis reports $t \approx 15.36$ and rejects overwhelmingly — yet both calculations start from the very same estimated mean change.

95% PAIRED CONFIDENCE INTERVAL $CI_{95\%} = \bar{D} \pm t_{0.975,35}(s_D/\sqrt{36}) \approx 19{,}729 \pm 2.030(1{,}284.33) \approx [\$17{,}122,\ \$22{,}336]$

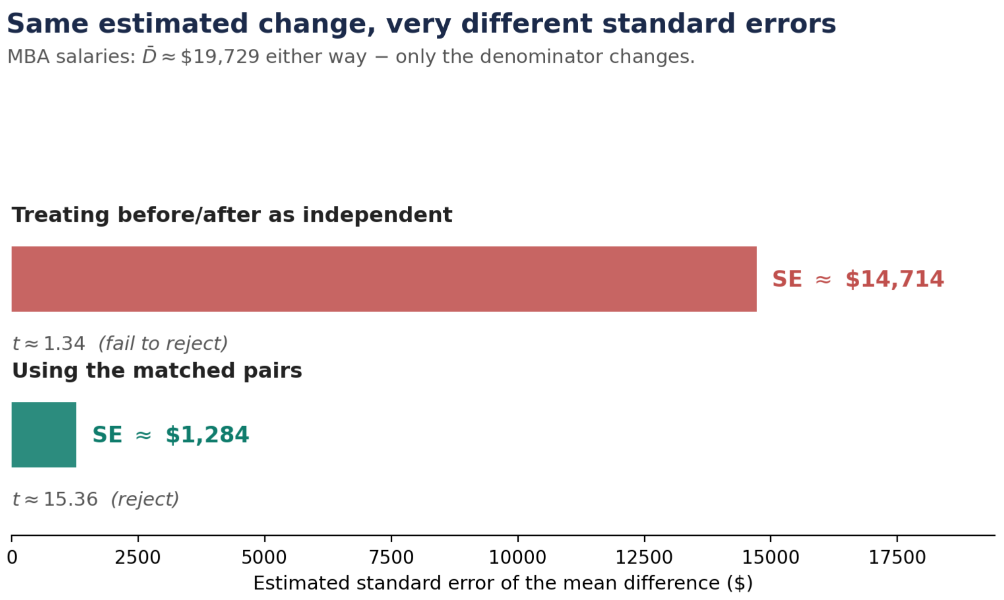

Figure 5. Independent and paired analyses use the same estimated mean change but very different standard errors.

IMPLIED WITHIN-PERSON CORRELATION
$\hat{r} = (s^2_{\text{after}} + s^2_{\text{before}} - s^2_D)/(2\, s_{\text{after}}\, s_{\text{before}}) \approx 0.993$

This inferred correlation explains the dramatic precision gain: pre-program salary is an excellent predictor of post-program salary. Pairing removes the stable level differences across students and isolates the change. A standardized paired effect divides the mean difference by the standard deviation of the differences, $d_z = \bar{D}/s_D = T/\sqrt{n} \approx 15.361/6 \approx 2.56$; the subscript z distinguishes this within-pair standardized effect from an independent-groups version of Cohen's d.

CAUSAL CAUTION A significant before/after change is not, by itself, proof that the MBA program caused the increase. Time trends, selection, career progression, and other concurrent changes remain possible explanations without a credible control group or assignment mechanism.

6.13.3 Worked Example: Industrial Safety Program

Ten plants are measured before and after an industrial safety program. The outcome is average weekly labor hours lost to accidents. Define $D =$ after $-$ before; a beneficial program produces negative differences, so this is a directional research question.

ONE-SIDED HYPOTHESES H_0: $\mu_D \geq 0$ versus H_a: $\mu_D < 0$

Quantity	Before	After	Paired difference D
Mean	23.15 hours	21.00 hours	-2.15 hours
Sample variance	36.558	19.1667	9.003
Sample size	10	10	10 pairs

6.13.3.0.1 Illustrative calculation that ignores matching.

INDEPENDENCE STANDARD ERROR $SE_{\text{ind}} = \sqrt{36.558/10 + 19.1667/10} = 2.3606$

INDEPENDENCE t $t_{\text{ind}} = -2.15/2.3606 = -0.9108$

Using the pooled boundary $t_{0.05,18} = -1.734$, the statistic is not in the rejection region. This counterfactual analysis cannot establish a reduction because it discards the plant matching.

6.13.3.0.2 Matched-pairs test.

PAIRED STANDARD ERROR $SE_{\text{pair}} = \sqrt{9.003/10} = 0.9488$ hours

PAIRED t $t_{\text{pair}} = -2.15/0.9488 = -2.2659, \quad \text{df} = 10 - 1 = 9$

For a left-tailed test at $\alpha = 0.05$, $t_{0.05,9} = -1.833$. Since $-2.266 < -1.833$, reject H_0. The one-sided p-value is approximately 0.025.

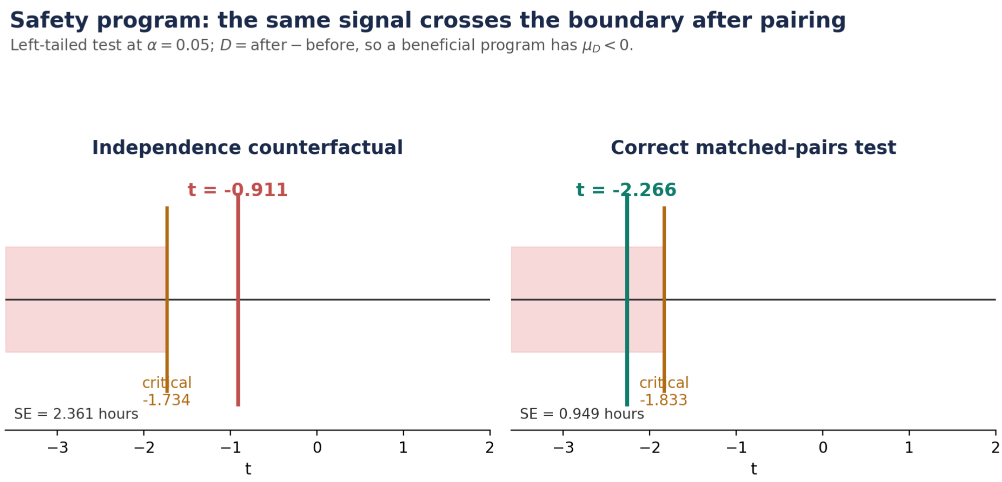

Figure 6. Pairing reduces the standard error enough to move the observed statistic into the left-tail rejection region.

The corresponding 95% one-sided upper confidence bound for μ_D is

ONE-SIDED UPPER BOUND $U = -2.15 + 1.833(0.9488) = -0.41$ hours

CONCLUSION The paired data provide statistically significant evidence that the safety program reduces mean weekly labor hours lost to accidents. The estimated reduction is 2.15 hours per plant per week; the one-sided 95% analysis supports a reduction of at least about 0.41 hour.

IMPLIED BEFORE/AFTER CORRELATION
$\hat{r} = (36.558 + 19.1667 - 9.003)/(2\sqrt{36.558 \cdot 19.1667}) \approx 0.883$

As in the MBA example, this large within-plant correlation is exactly why matching pays off: most of the marginal variability in hours lost is a stable, plant-specific baseline that cancels when the analysis is restricted to each plant's own before/after change.

6.13.4 Matching, regression, and causal interpretation

A simple before–after paired test does not by itself isolate a causal program effect. Salaries may rise because of labor-market inflation, accumulated work experience, changing industries, selective reporting, or other events occurring during the same period; the test establishes a mean temporal increase among the observed completers, and attributing all of that increase to the program requires a stronger design. A difference-in-differences (DiD) design compares the before–after change of participants with the contemporaneous change of a suitable comparison group, DiD $= (\bar{Y}_{T,\text{after}} - \bar{Y}_{T,\text{before}}) - (\bar{Y}_{C,\text{after}} - \bar{Y}_{C,\text{before}})$, and identifies a causal effect under the assumption that, absent the program, the two groups would have followed parallel outcome trends. Matching on pre-treatment covariates and regression adjustment are related strategies, but neither automatically recreates random assignment: unmeasured differences can remain, and causal conclusions depend on the credibility of the underlying design assumptions.

6.14 Further Extensions

6.14.1 One proportion

For the lost-bag example, let X be the number of lost bags in n independently tracked bags and $\hat{p}$ = X/n. To test H_0: p $\leq p_0$ against H_a: p $> p_0$, the large-sample score statistic uses the null variance.

PROPORTION TEST Z = $(\hat{p}\text{-}p_0)/\sqrt{p_0(1-p_0)/n} \approx$ N(0,1) under H_0 at p=p_0

6.14.2 General constrained hypotheses

In higher dimensions, Θ_0 and Θ_a are regions separated by a boundary such as μ_2-μ_1 = 0. In fact, every two-sample and paired comparison already developed in this chapter — Chicago versus San Francisco, MBA salaries before and after — is itself an instance of one underlying idea: a two-mean hypothesis is a **linear constraint** on the parameter vector, and every test statistic in this chapter is built from the same generic recipe.

6.14.2.0.1 The linear contrast. Let μ_1 and μ_2 denote two population means. Rather than treat the two coordinates separately, define the scientifically relevant contrast as a linear combination $\Delta = c^T\mu$, collecting both means into a vector μ and the comparison itself into a fixed vector c.

CONTRAST $\mu = (\mu_1, \mu_2)^T, \quad c = (1, -1)^T, \quad \Delta = c^T\mu = \mu_1 - \mu_2$

The three directional hypotheses used throughout this chapter are all specific choices of null and alternative region for this same Δ:

Scientific question	Null hypothesis	Alternative hypothesis	Tail
Are the means different?	H_0: $\Delta = 0$	H_a: $\Delta \neq 0$	Two-sided
Is mean 1 larger?	H_0: $\Delta \leq 0$	H_a: $\Delta > 0$	Right-sided
Is mean 1 smaller?	H_0: $\Delta \geq 0$	H_a: $\Delta < 0$	Left-sided

For a composite one-sided null, calibration occurs at its boundary $\Delta = 0$: under the usual monotone test statistic, that boundary is the null value most favorable to rejection, so it controls the Type I error probability for the entire null region — the same least-favorable-null argument used earlier in this chapter.

GENERIC TEST STATISTIC $T = (\hat{\Delta} - \Delta_0)/SE(\hat{\Delta})$, usually with $\Delta_0 = 0$

6.14.2.0.2 Geometry of the equality restriction. In the (μ_1, μ_2) plane, H_0: $\mu_1 - \mu_2 = 0$ is the 45-degree line $\mu_1 = \mu_2$. The vector $c = (1, -1)^T$ is normal to that line, so $c^T\mu$ is a signed measure of displacement from equality: positive values lie on the $\mu_1 > \mu_2$ side, negative values lie on the $\mu_1 < \mu_2$ side.

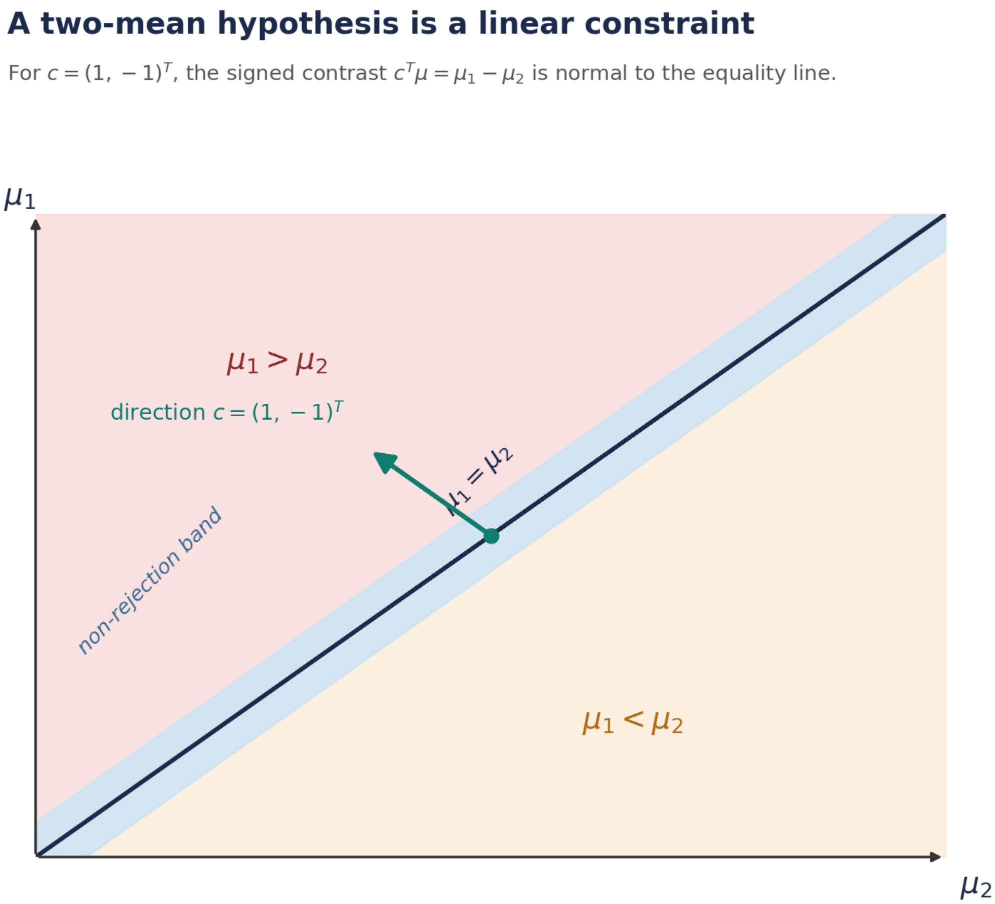

Figure 4. Geometry of a two-mean linear contrast and a schematic non-rejection band around the equality line.

The non-rejection band is not fixed in raw units: its width depends on the standard error and the chosen significance level. In a two-sided test, the total Type I error α is divided between two rejection tails, $\alpha/2$ on each side, so in standardized units the non-rejection region is approximately $|T| < t_{1-\alpha/2,\nu}$.

TWO-SIDED RULE Reject H_0: $\Delta = 0$ iff $|T| \geq t_{1-\alpha/2,\nu}$; equivalently, 0 lies outside the $(1-\alpha)$ CI for Δ

6.14.2.0.3 A unified covariance-matrix formulation. The independent-samples and paired formulas already derived earlier in this chapter are special cases of a single quadratic form. Let $\hat{\mu}$ be an estimator of the mean vector and let $\hat{V}$ estimate its covariance matrix. For any contrast vector c, the estimated contrast and its sampling variance are

GENERAL LINEAR CONTRAST $\hat{L} = c^T\hat{\mu}$, $\widehat{\mathrm{Var}}(\hat{L}) = c^T\hat{V}c$, $SE(\hat{L}) = \sqrt{c^T\hat{V}c}$

For two independent sample means, $\hat{V}$ is diagonal, since independent sampling gives zero covariance between $\bar{X}_1$ and $\bar{X}_2$. For paired sample means, its off-diagonal element estimates $\mathrm{Cov}(\bar{A}, \bar{B}) = s_{AB}/n$; with $c = (1, -1)^T$, multiplying out the quadratic form automatically produces the negative covariance term already seen in this chapter's paired-difference variance formula.

INDEPENDENT COVARIANCE MATRIX $\hat{V}_{\text{ind}} = \begin{pmatrix} s_1^2/n_1 & 0 \\ 0 & s_2^2/n_2 \end{pmatrix}$

PAIRED COVARIANCE MATRIX $\hat{V}_{\text{pair}} = \frac{1}{n}\begin{pmatrix} s_A^2 & s_{AB} \\ s_{AB} & s_B^2 \end{pmatrix}$

EQUIVALENCE $c^T\hat{V}_{\text{pair}}\,c = (s_A^2 + s_B^2 - 2s_{AB})/n = s_D^2/n$

This identity confirms that the paired-difference variance formula from earlier in this chapter is not a separate rule but exactly what the general quadratic form produces once $\hat{V}$ carries a covariance term. The same idea extends well beyond two means: in regression, ANOVA contrasts, and general multivariate linear hypotheses, a linear function $C\beta$ is estimated, its covariance $C\,\widehat{\mathrm{Var}}(\hat{\beta})\,C^T$ is computed, and the distance from the null restriction is standardized in exactly the same way. The higher-dimensional linear constraint is the natural next step in that same generalization:

LINEAR CONSTRAINT $H_0\colon\ A\theta \le b$ versus $H_a\colon\ A\theta > b$

6.15 Research Integrity, Multiple Testing, and Reproducibility

The lecture warns about p-hacking and selective reporting. A valid p-value assumes the analysis plan was not repeatedly changed in response to the same data without accounting for that search. Testing many hypotheses inflates the chance of at least one false positive.

MULTIPLICITY If m independent true nulls are each tested at α:
$P(\text{at least one false positive}) = 1 - (1-\alpha)^m$

For m $=$ 10 and $\alpha = 0.05$, this probability is about $1 - 0.95^{10} \approx 0.401$—not 0.05. Appropriate responses include pre-registration, transparent reporting of all planned outcomes, replication, and multiplicity procedures such as Bonferroni, Holm, or false-discovery-rate control.

- Report exact p-values, including non-significant results; do not replace them only with star thresholds.
- Distinguish confirmatory analyses from exploratory analyses and label post-hoc hypotheses clearly.

- Report effect sizes, confidence intervals, sample exclusions, missing-data handling, and assumption checks.
- Avoid optional stopping or repeated testing unless a sequential design with adjusted boundaries was specified.
- Interpret p-values in context; a tiny p-value can accompany a trivial effect when n is very large.

6.15.1 Significance symbols

A common, though not universal, table convention is * for $p < 0.10$, ** for $p < 0.05$, and *** for $p < 0.01$. These symbols summarize thresholds; they do not make an analysis more "robust." Exact p-values and uncertainty intervals are more informative.

6.16 Formula Sheet and Reporting Checklist

Quantity	Formula	Use
Sample mean	$\bar{x} = (1/n)\sum x_i$	Point estimator of μ
Sample variance	$s^2 = [1/(n-1)]\sum(x_i - \bar{x})^2$	Unbiased estimator of σ^2
Standard error	$SE(\bar{x}) = s/\sqrt{n}$	Sampling uncertainty of $\bar{x}$
One-sample t	$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$	Test a population mean when σ is unknown
Left-tail critical value	$t_{\alpha,n-1}$	H_a: $\mu < \mu_0$
Right-tail critical value	$t_{1-\alpha,n-1}$	H_a: $\mu > \mu_0$
Two-sided critical values	$\pm t_{1-\alpha/2,n-1}$	H_a: $\mu \neq \mu_0$
Left-tail p	$F_t(t_{obs})$	Lower-tail alternative
Right-tail p	$1 - F_t(t_{obs})$	Upper-tail alternative
Two-sided p	$2 \cdot \min\{F_t(t_{obs}),\, 1 - F_t(t_{obs})\}$	Two-sided alternative
Independent contrast	$\Delta = \mu_1 - \mu_2$; $\hat{\Delta} = \bar{X}_1 - \bar{X}_2$	Two-sample estimate
Welch statistic	T $= [(\bar{X}_1 - \bar{X}_2)$-$D_0]$ / $\sqrt{s_1^2/n_1 + s_2^2/n_2}$	Unequal-variance two-sample test
Welch degrees of freedom	$\nu_W = (s_1^2/n_1 + s_2^2/n_2)^2 \Big/ \left[\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}\right]$	Matches the Welch statistic
Pooled variance	$s_p^2 = [(n_1-1)s_1^2+(n_2-1)s_2^2]/(n_1+n_2-2)$	Equal-variance assumption
Pooled statistic	T $= [(\bar{X}_1 - \bar{X}_2)$-$D_0]$ / $[s_p\sqrt{1/n_1 + 1/n_2}]$, df $= n_1 + n_2 - 2$	Equal-variance two-sample test
Paired difference	$D_i = A_i - B_i$; $\bar{D} = \bar{A} - \bar{B}$	Reduces a paired design to one sample
Paired variance	$s_D^2 = s_A^2 + s_B^2 - 2r_{AB}s_As_B$	Uses within-pair correlation

Quantity	Formula	Use
Paired statistic	T $= \bar{D}/(s_D/\sqrt{n})$, df $= n-1$	One-sample t applied to differences
Sample-size planning	n $\approx [(z_{1-\alpha}+z_{1-\beta})\sigma/\Delta]^2$	One-sided mean test, power $1-\beta$

6.16.1 Minimum complete write-up

TEMPLATE "A one-sample t test evaluated H_0: μ [relation] μ_0 against H_a: μ [relation] μ_0. With n = . . . , $\bar{x}$ = . . . , s = . . . , the result was t(df) = . . . , p = ... At α = . . . , we [reject / fail to reject] H_0. The estimated effect was . . . with a . . . % confidence interval [. . . , . . .]."

- Population, unit of analysis, parameter, and benchmark are defined.
- Tail direction matches the research claim; equality is included in H_0.
- Randomness, independence, normality/large-sample, and measurement assumptions are addressed.
- The standard error and degrees of freedom match the design.
- The p-value uses the same tail as H_a; $\alpha/2$ is used only for a two-sided test.
- The conclusion says "fail to reject," not "accept," when evidence is insufficient.
- Effect size, confidence interval, multiplicity, and practical importance are reported.